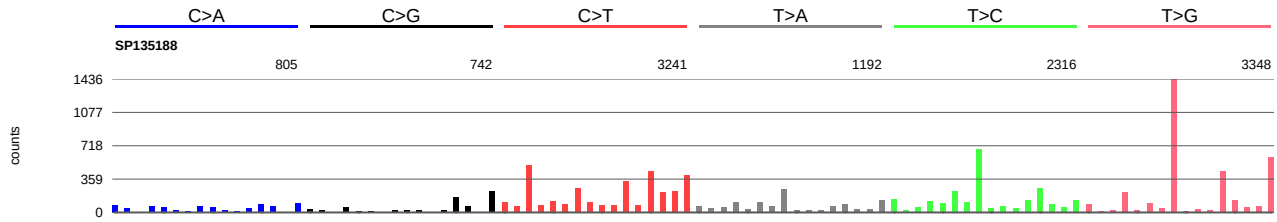
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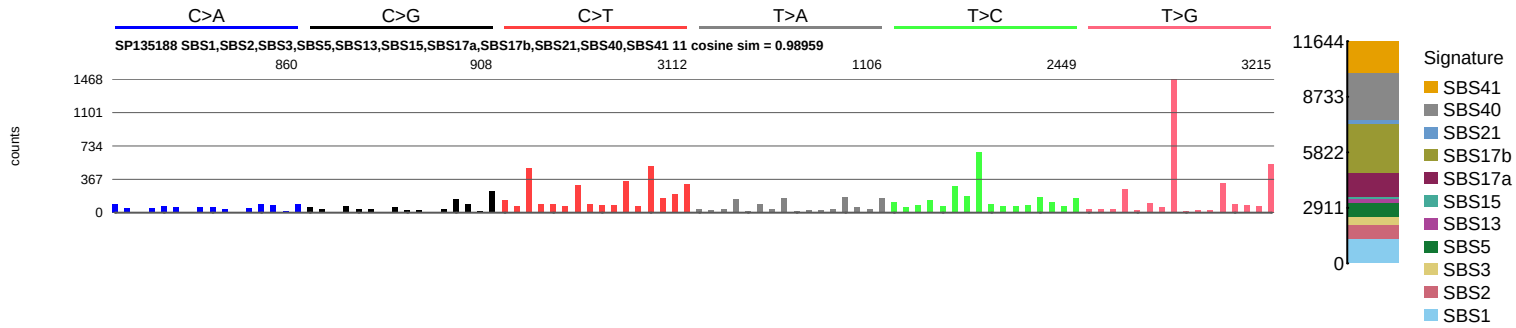
SP135188
805
742
3241
1192
2316
3348
1436
1077
718
359
0
counts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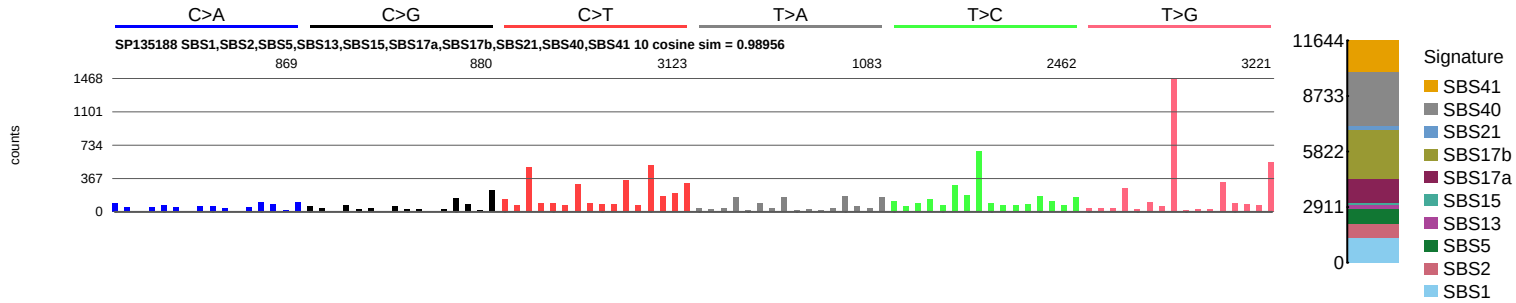
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 11 cosine sim = 0.98959
860
908
3112
1106
2449
3215
1468
1101
734
367
0
counts
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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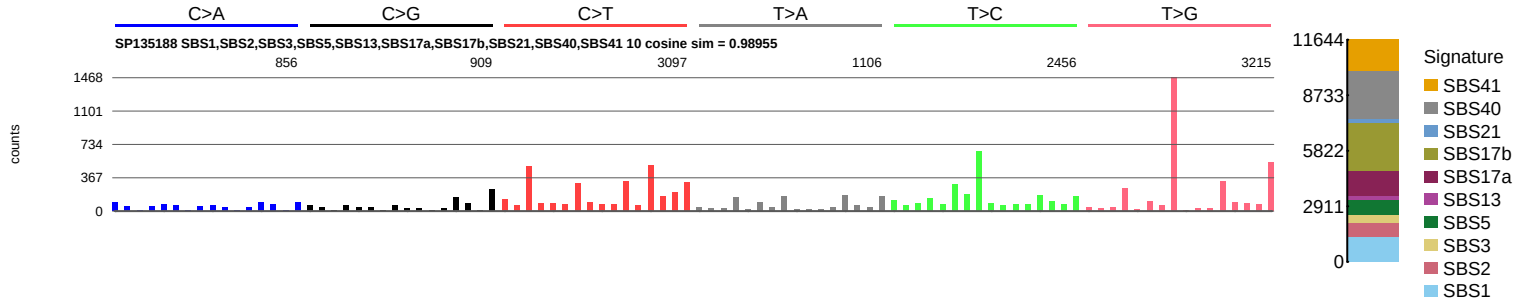
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98956
869
880
3123
1083
2462
3221
1468
1101
734
367
0
counts
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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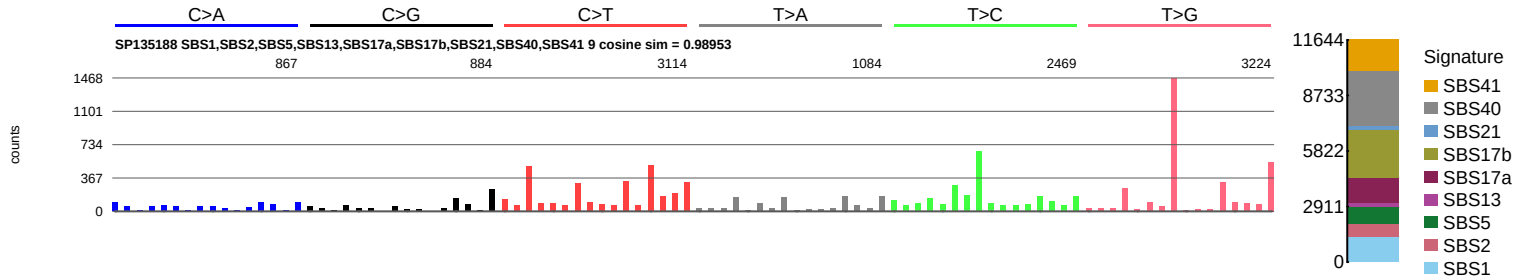
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98955
856
909
3097
1106
2456
3215
1468
1101
734
367
0
counts
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98953
867
884
3114
1084
2469
3224
1468
1101
734
367
0
counts
11644
8733
5822
2911
0
Signature
SBS41
SBS40
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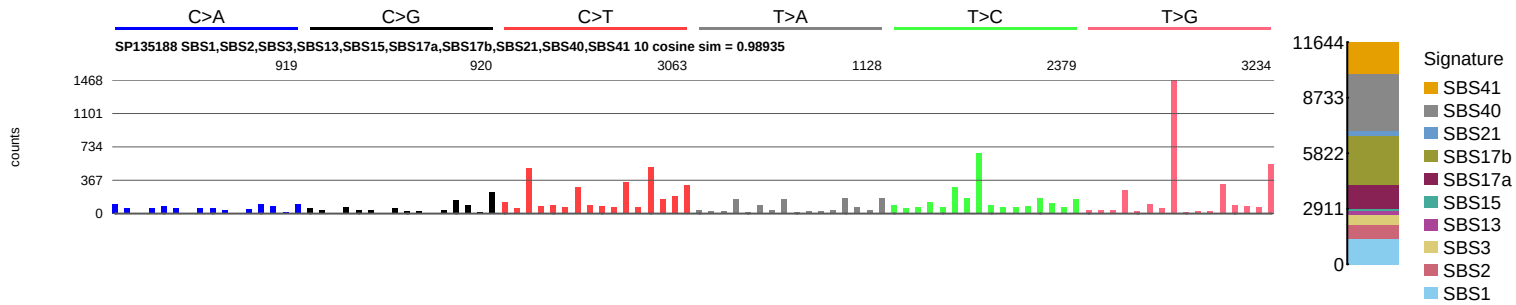
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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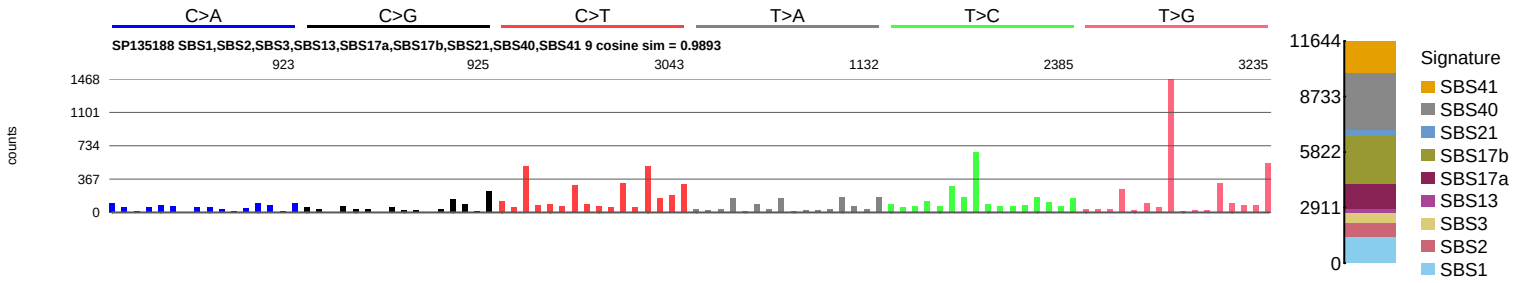
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98935
919
920
3063
1128
2379
3234
1468
1101
734
367
0
counts
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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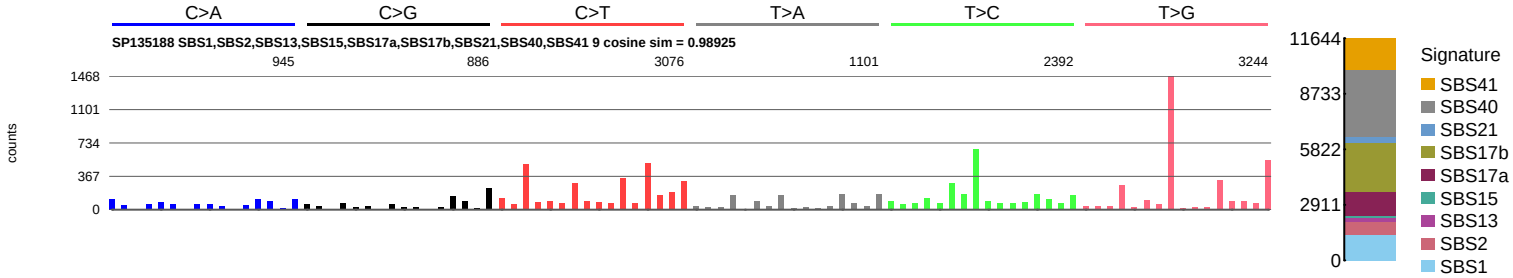
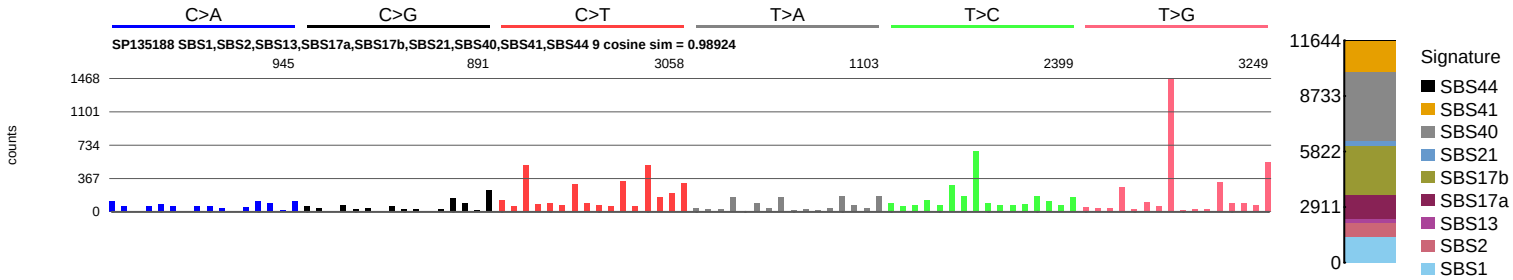
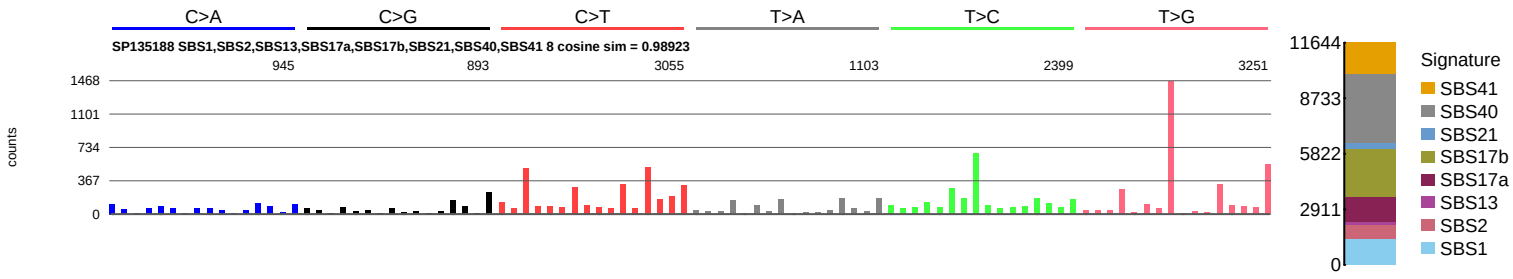
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.9893
923
925
3043
1132
2385
3235
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98925
945
886
3076
1101
2392
3244
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98924
945
891
3058
1103
2399
3249
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98923
945
893
3055
1103
2399
3251
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
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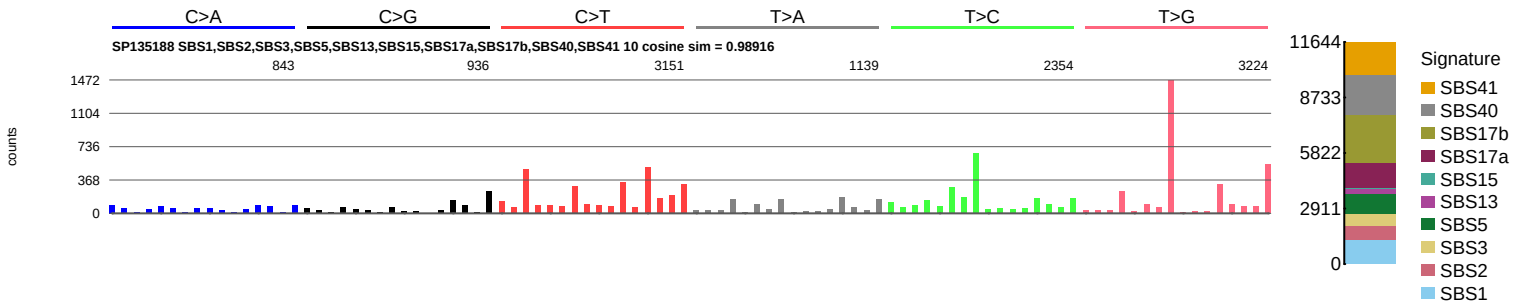
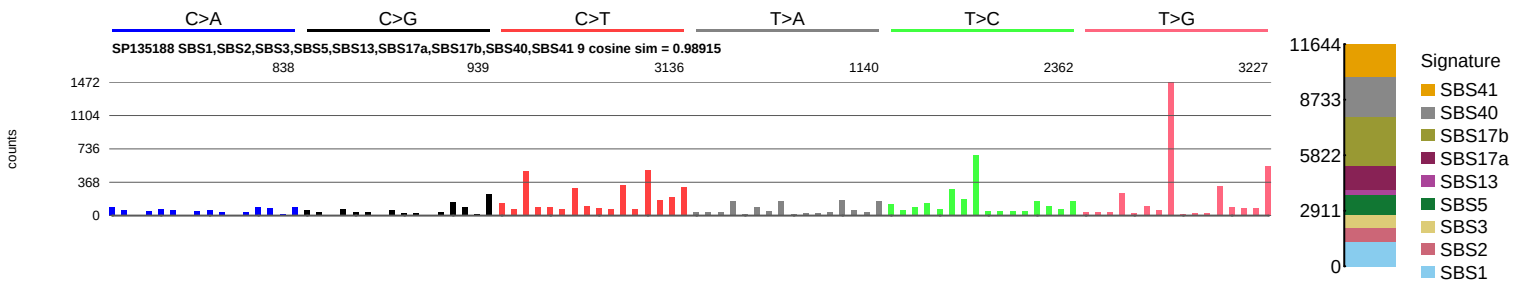
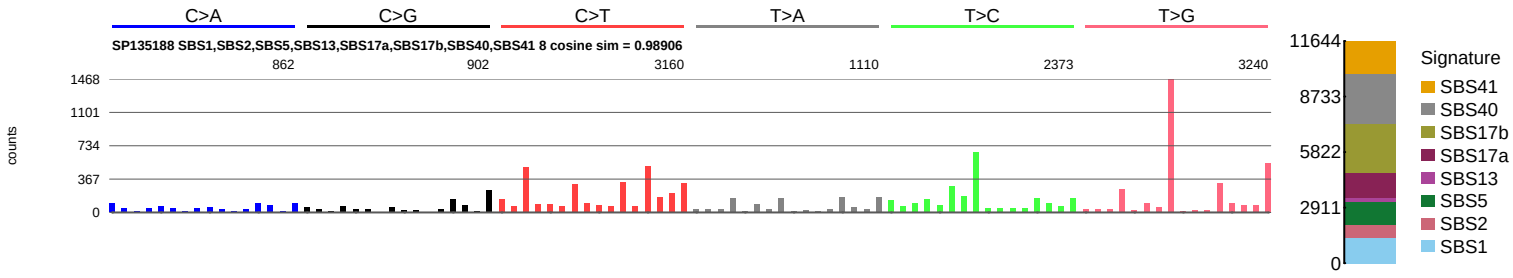
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 10 cosine sim = 0.98916
843
936
3151
1139
2354
3224
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98915
838
939
3136
1140
2362
3227
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98906
862
902
3160
1110
2373
3240
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
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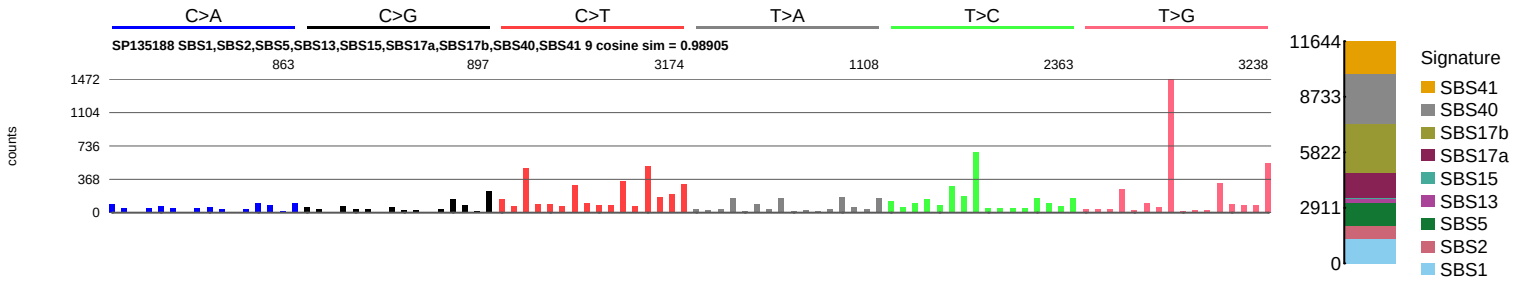
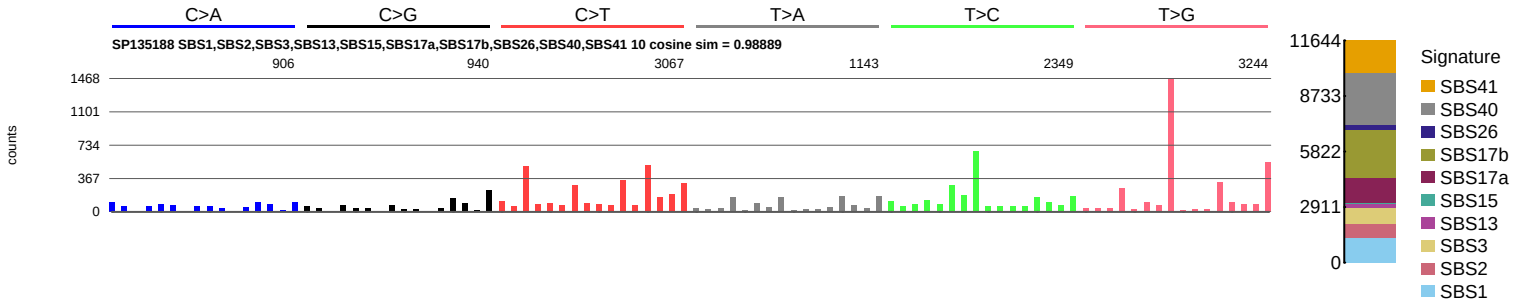
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98905
863
897
3174
1108
2363
3238
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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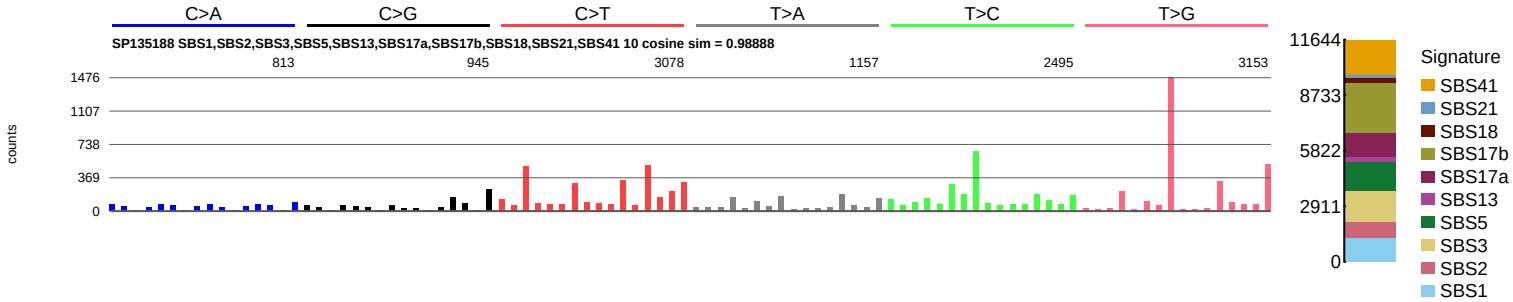
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98889
906
940
3067
1143
2349
3244
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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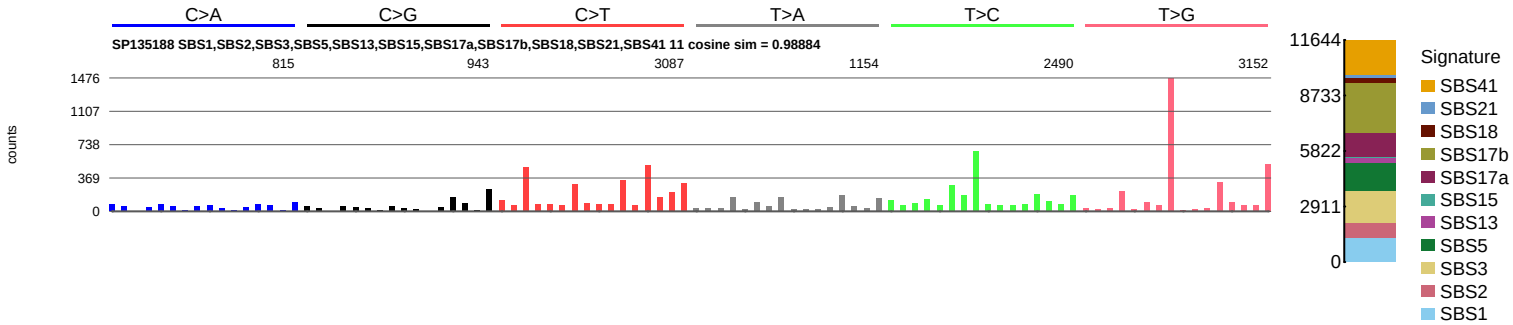
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98888
813
945
3078
1157
2495
3153
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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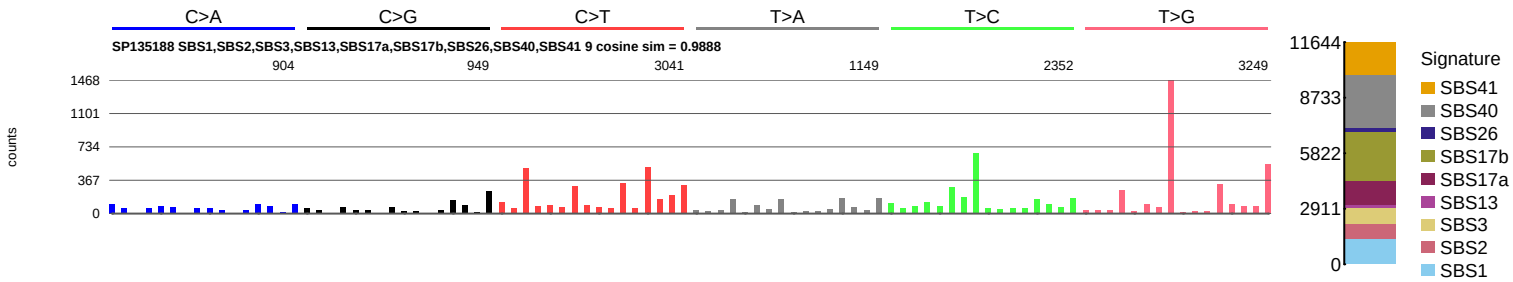
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98884
815
943
3087
1154
2490
3152
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.9888
904
949
3041
1149
2352
3249
counts
1468
1101
734
367
0
11644
8733
5822
2911
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Signature
SBS41
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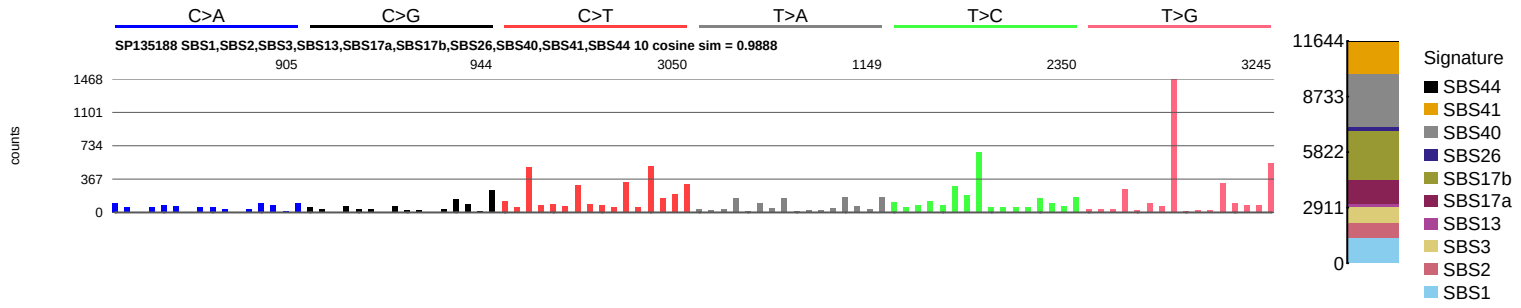
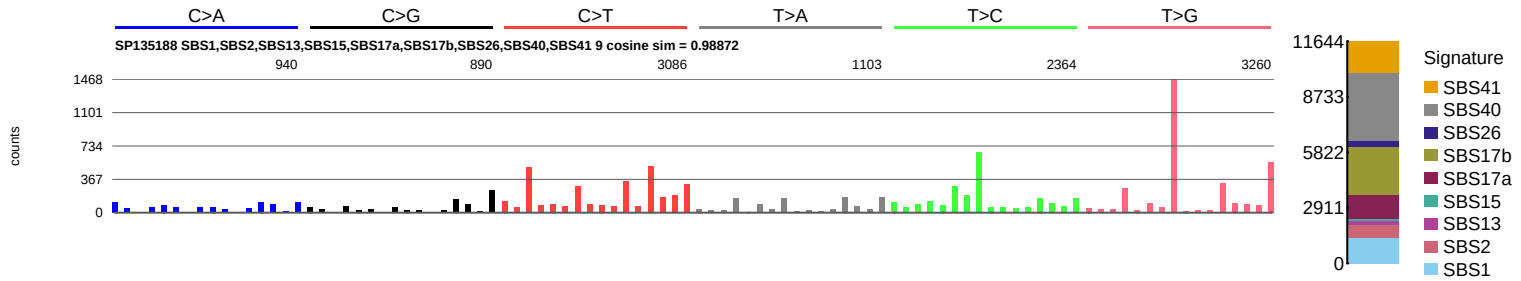
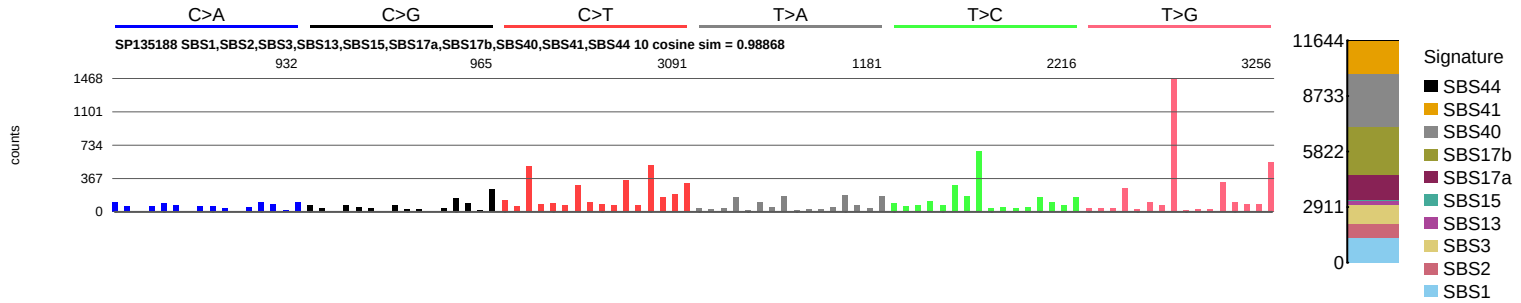
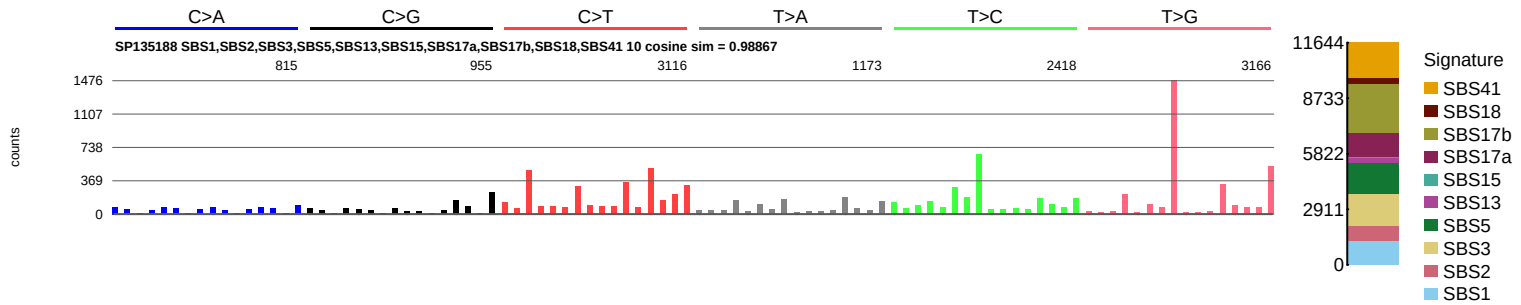
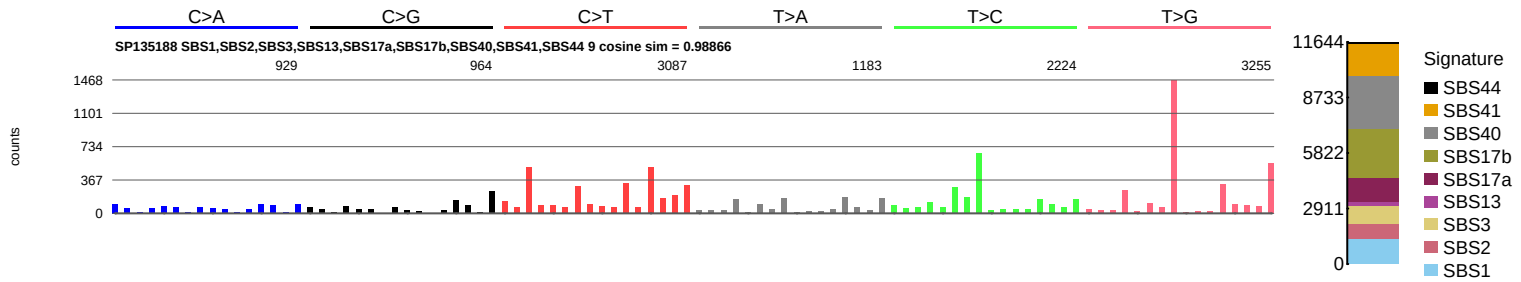
SBS3
SBS2
SBS1

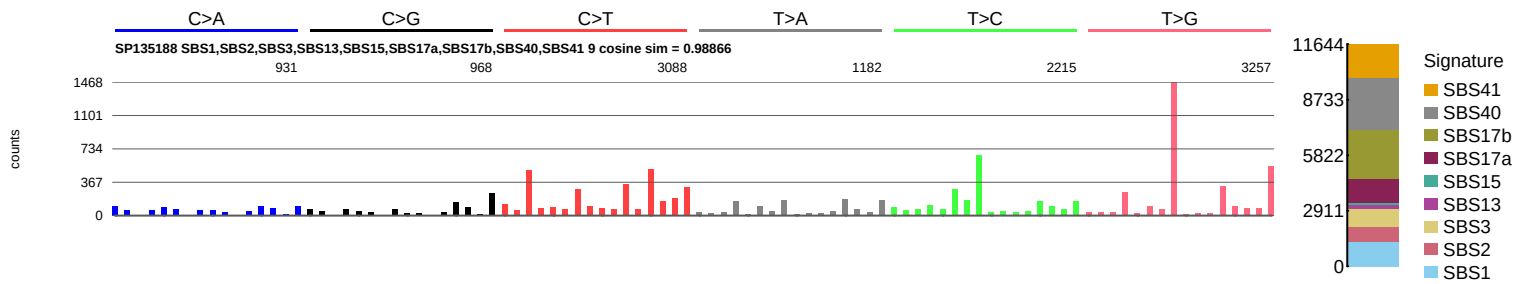

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9888
905
944
3050
1149
2350
3245
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

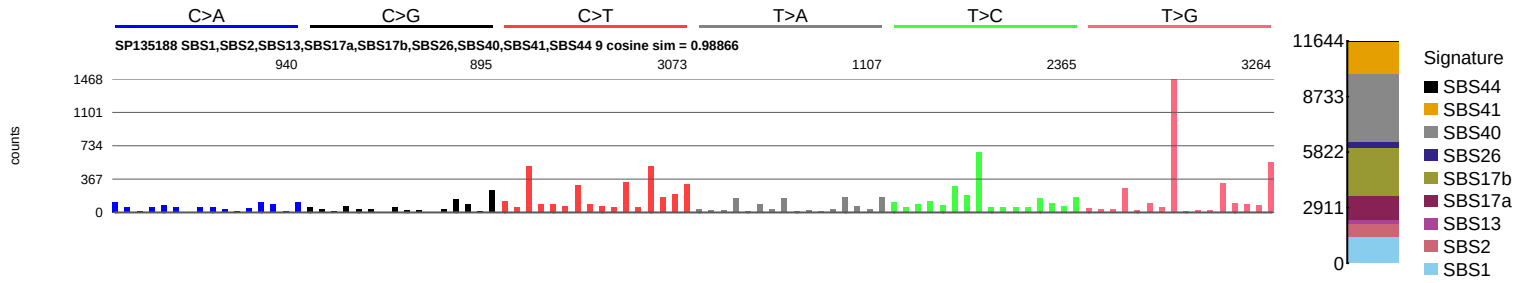

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98872
940
890
3086
1103
2364
3260
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98868
932
965
3091
1181
2216
3256
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98867
815
955
3116
1173
2418
3166
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98866
929
964
3087
1183
2224
3255
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98866
931
968
3088
1182
2215
3257
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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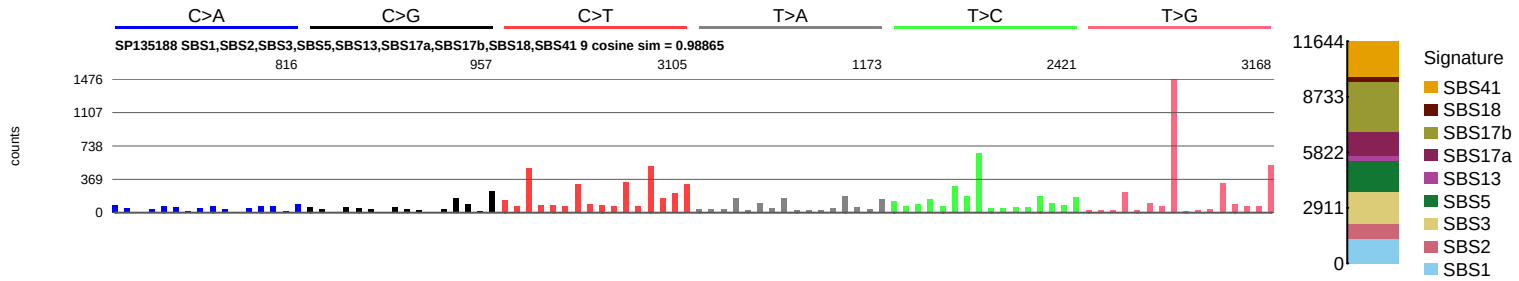
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98866
940
895
3073
1107
2365
3264
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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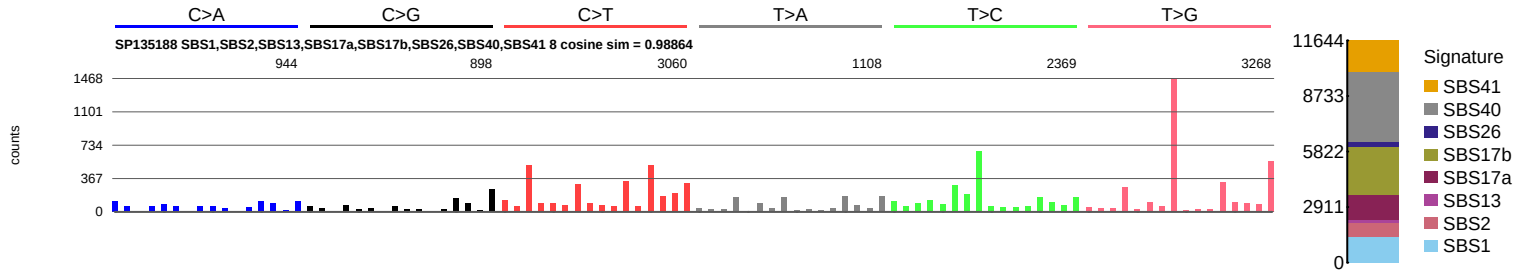
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98865
816
957
3105
1173
2421
3168
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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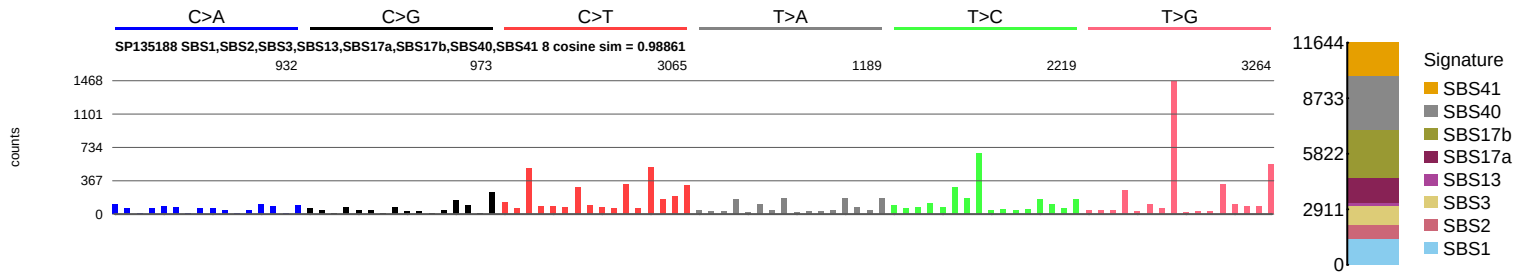
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98864
944
898
3060
1108
2369
3268
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>T
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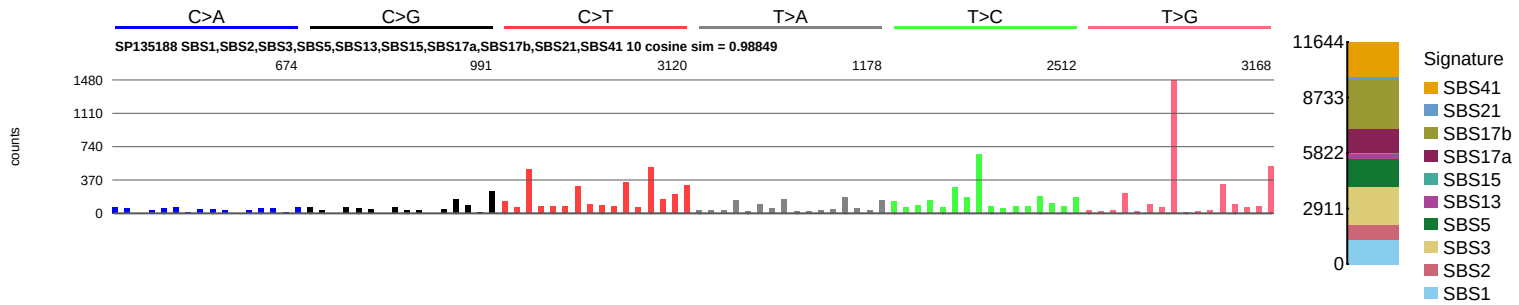
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98861
932
973
3065
1189
2219
3264
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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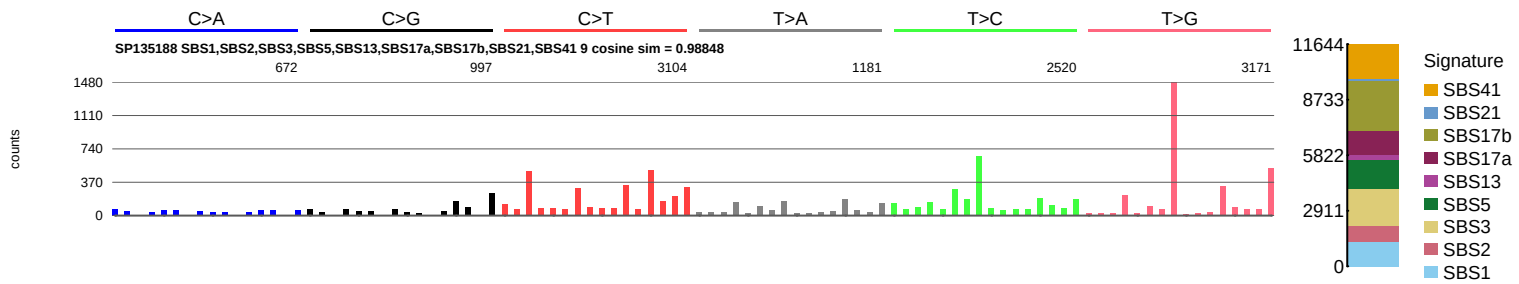
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.98849
674
991
3120
1178
2512
3168
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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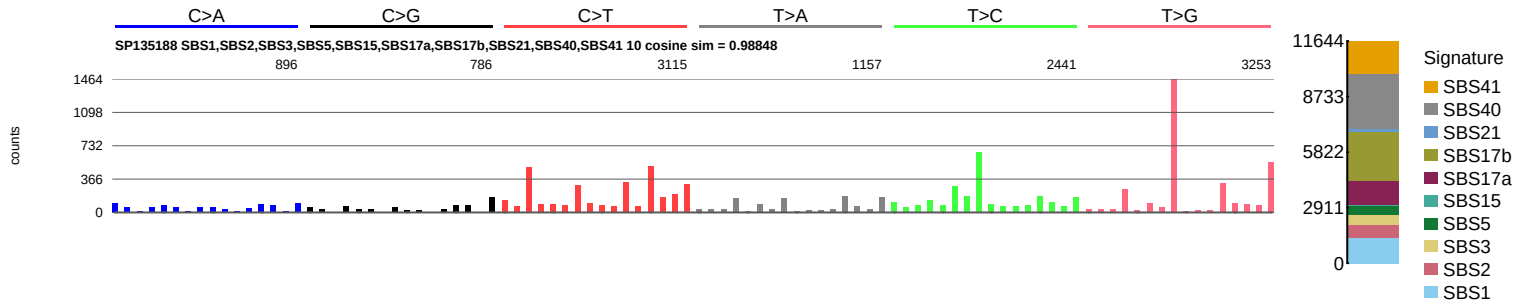
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98848
672
997
3104
1181
2520
3171
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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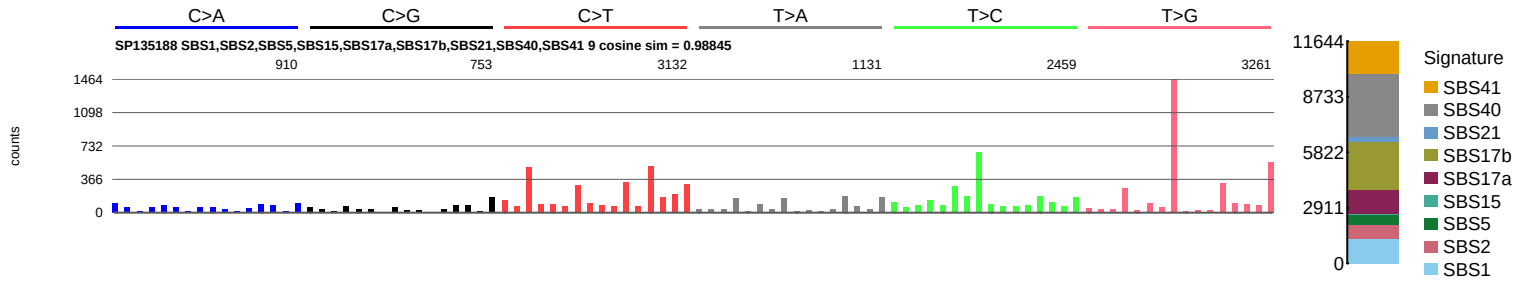
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98848
896
786
3115
1157
2441
3253
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
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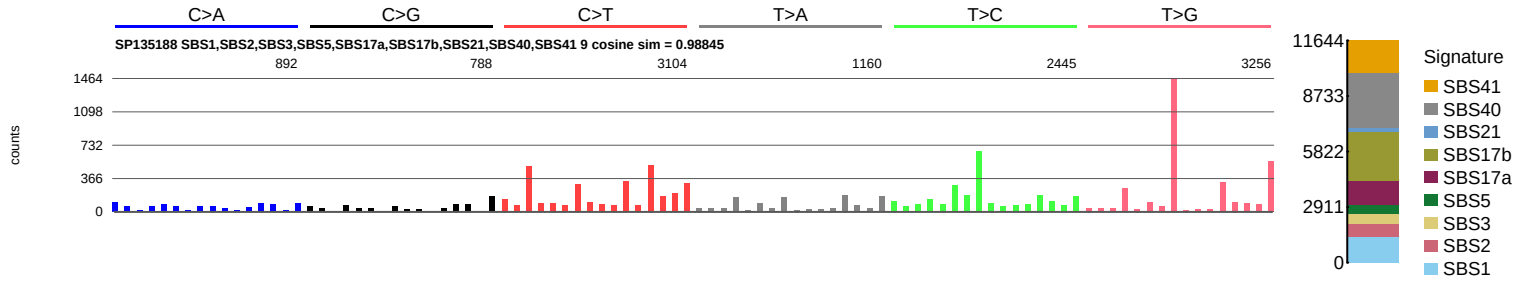
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98845
910
753
3132
1131
2459
3261
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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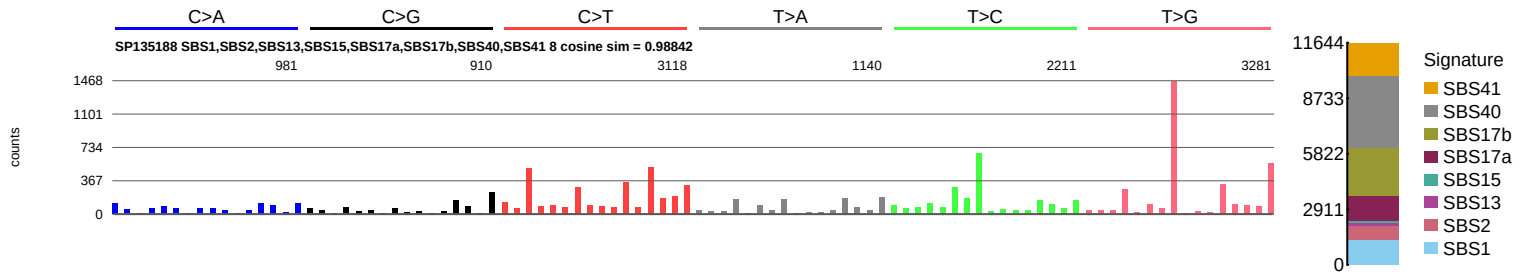
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98845
892
788
3104
1160
2445
3256
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98842
981
910
3118
1140
2211
3281
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
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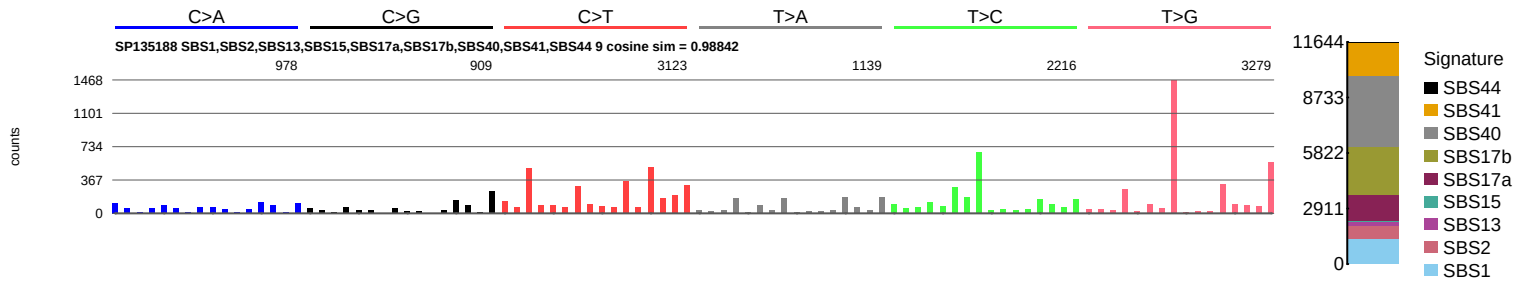
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98842
978
909
3123
1139
2216
3279
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
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SBS41
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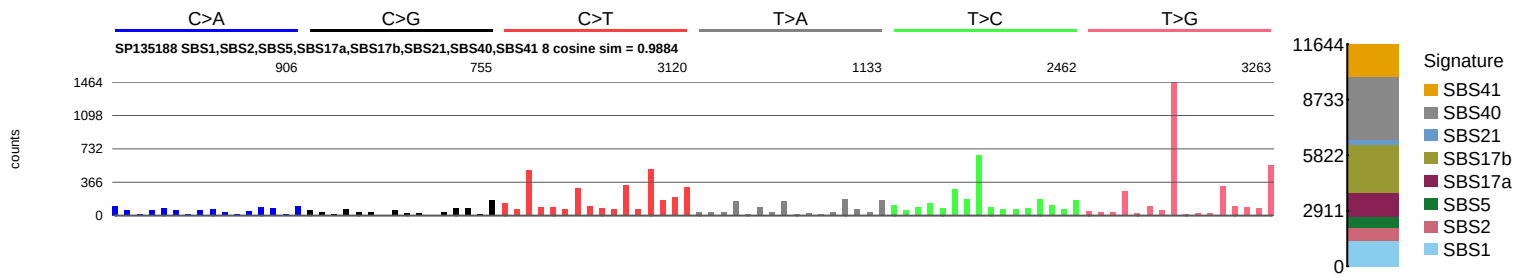
SBS13
SBS2
SBS1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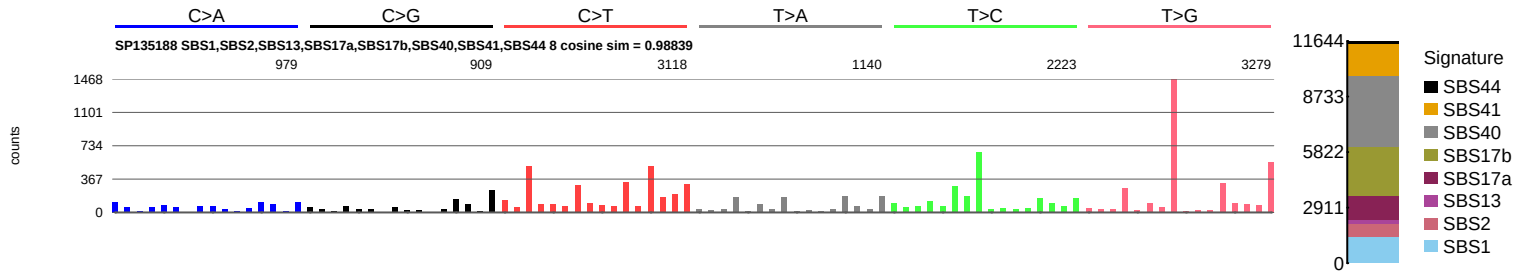
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.9884
906
755
3120
1133
2462
3263
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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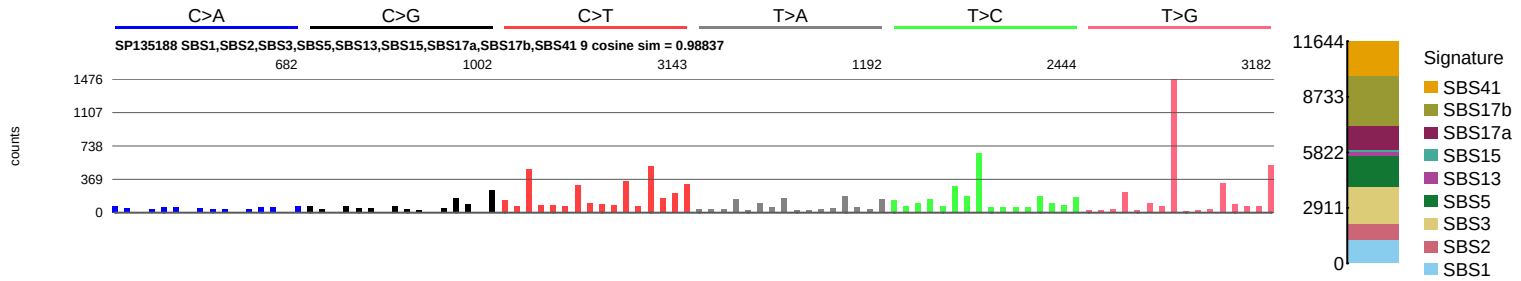
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98839
979
909
3118
1140
2223
3279
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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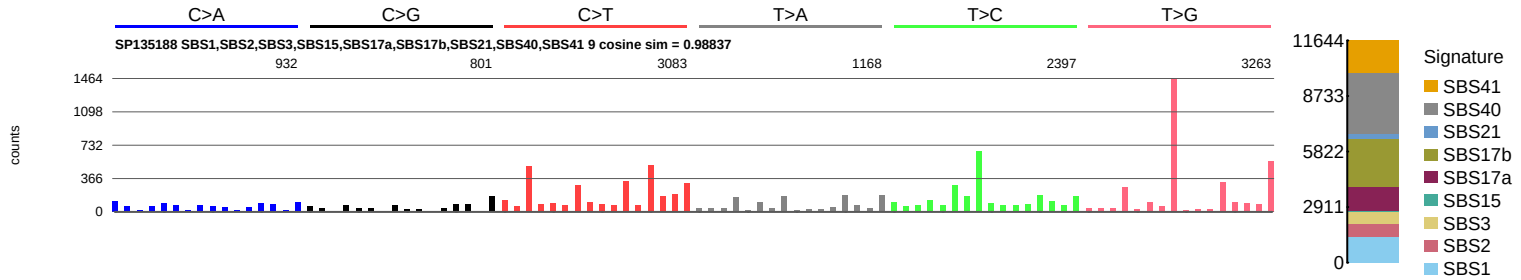
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.98837
682
1002
3143
1192
2444
3182
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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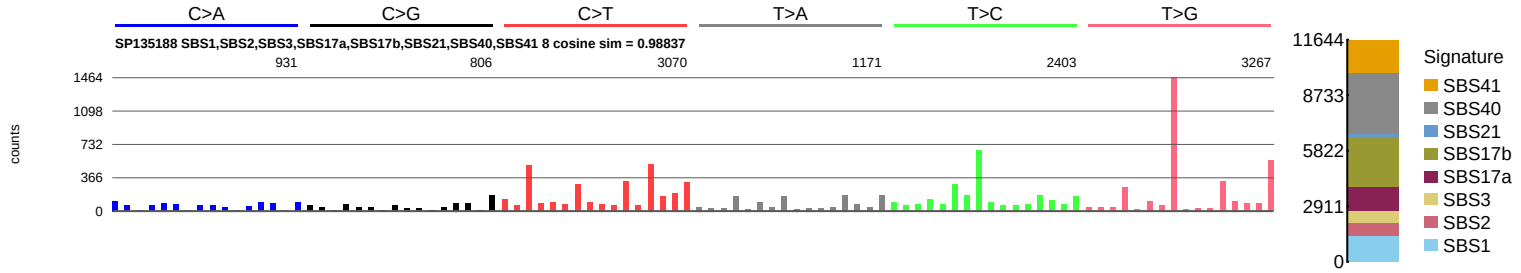
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98837
932
801
3083
1168
2397
3263
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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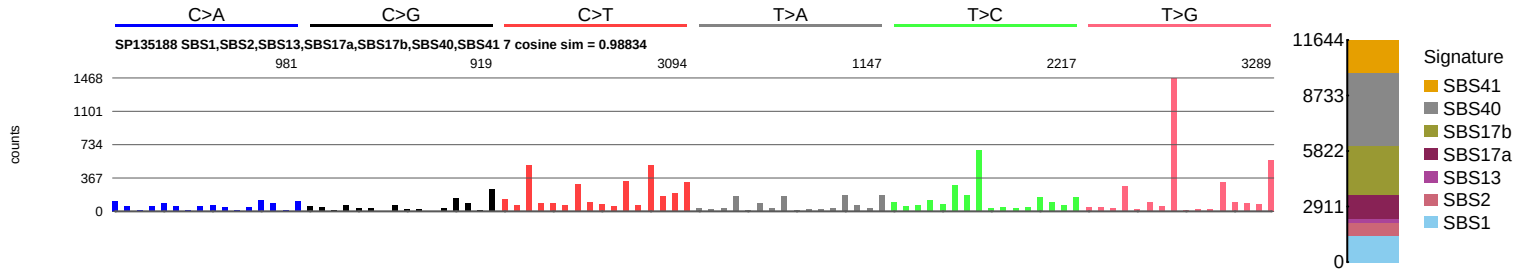
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98837
931
806
3070
1171
2403
3267
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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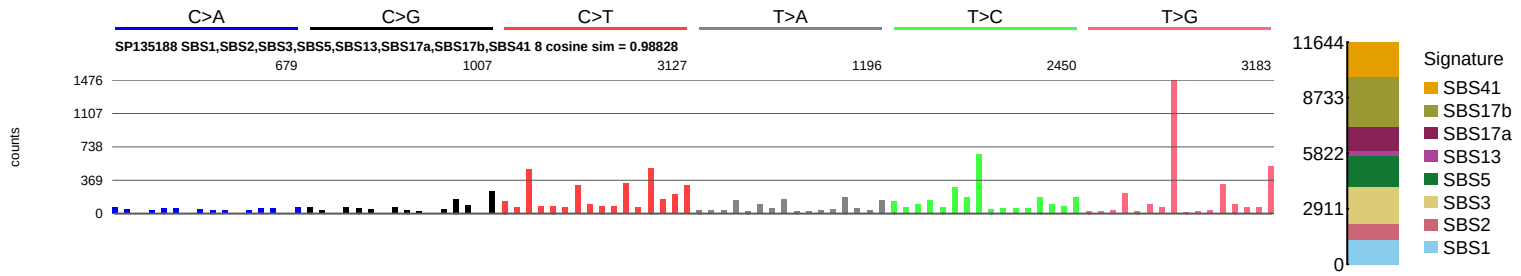
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98834
981
919
3094
1147
2217
3289
counts
1468
1101
734
367
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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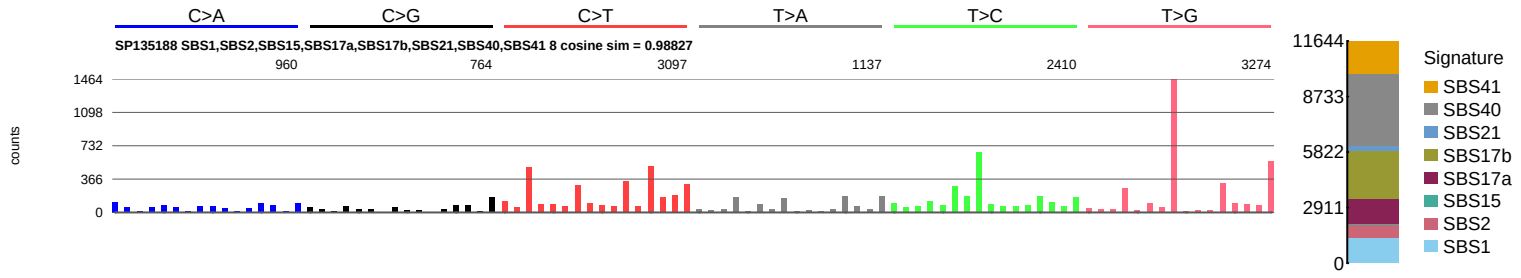
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98828
679
1007
3127
1196
2450
3183
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98827
960
764
3097
1137
2410
3274
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
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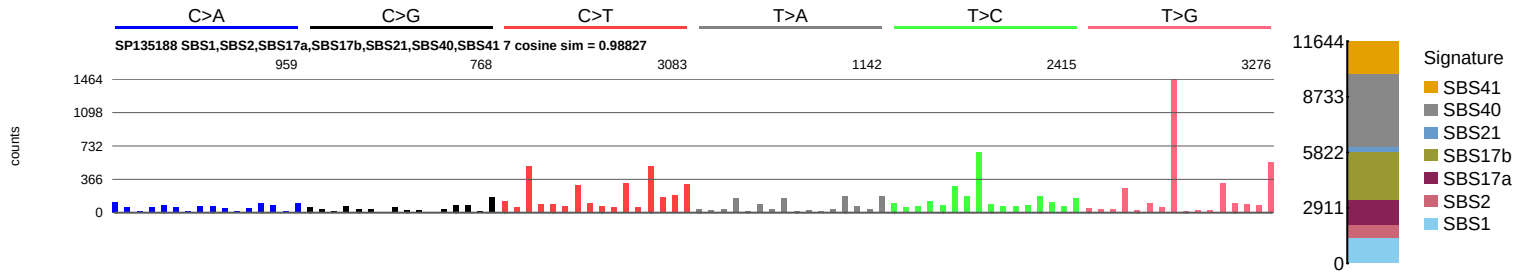
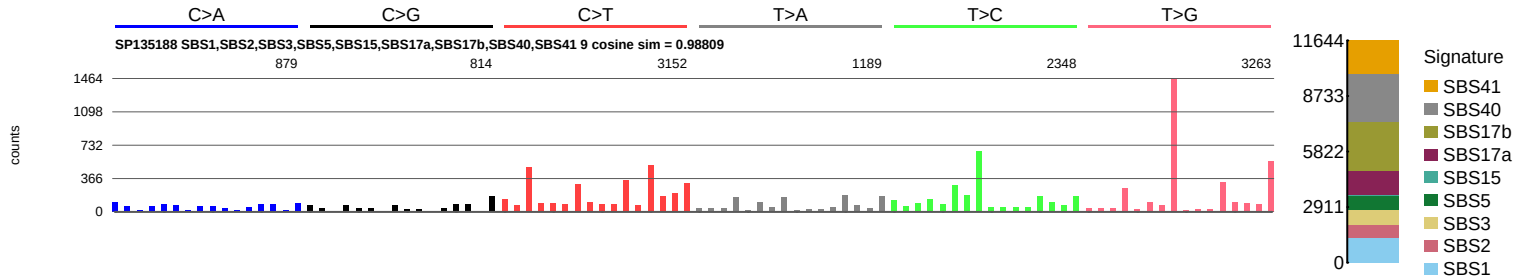
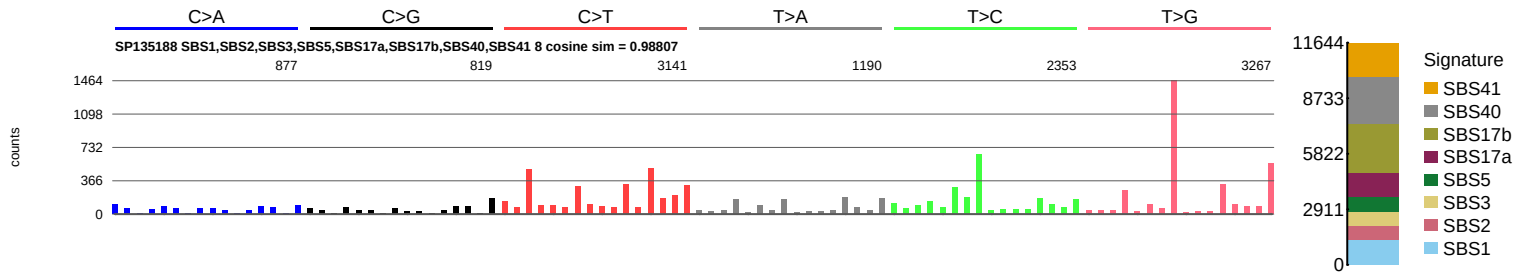
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98827
959
768
3083
1142
2415
3276
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98809
879
814
3152
1189
2348
3263
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98807
877
819
3141
1190
2353
3267
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
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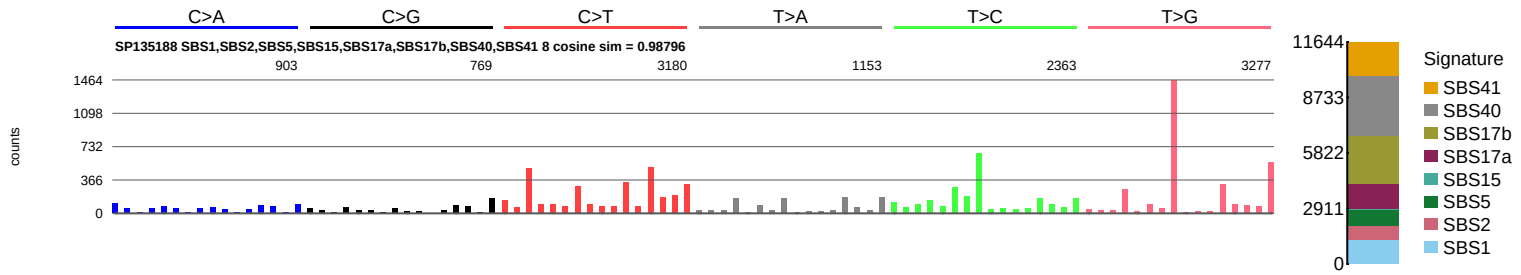
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98796
903
769
3180
1153
2363
3277
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
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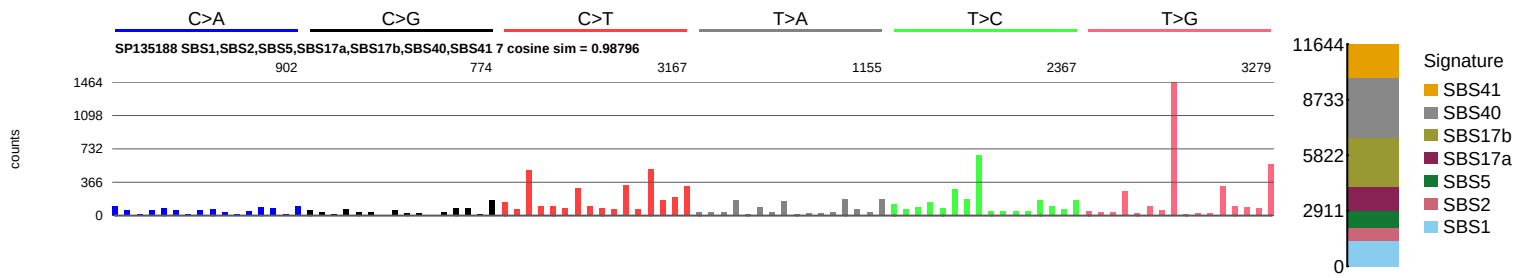
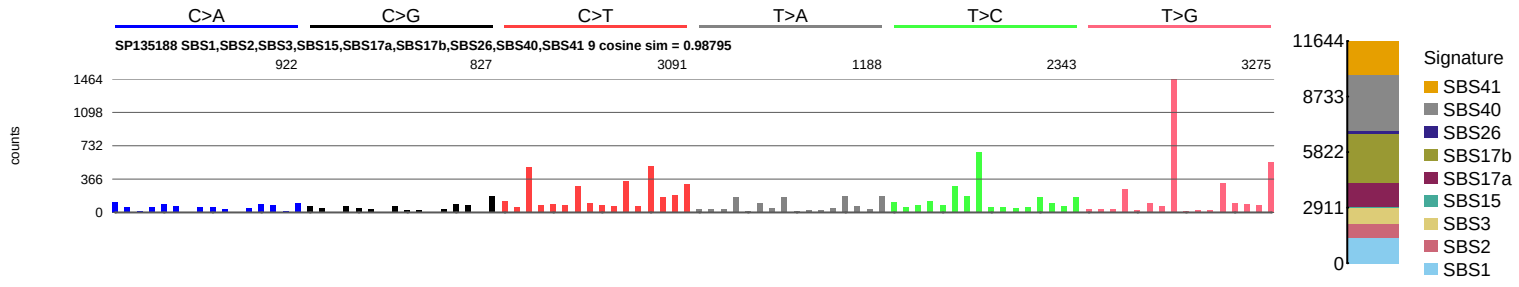
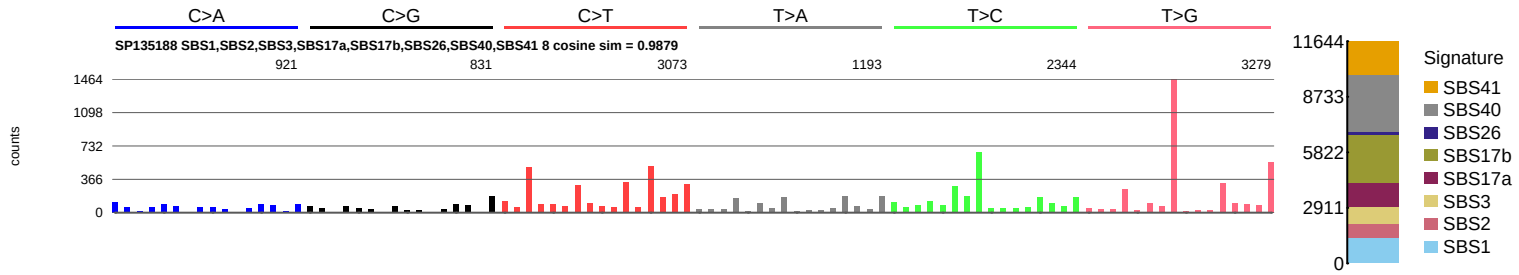
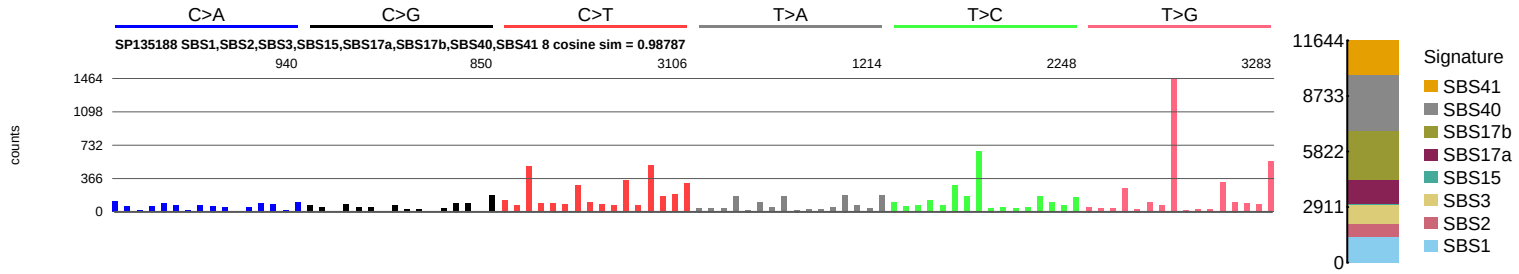
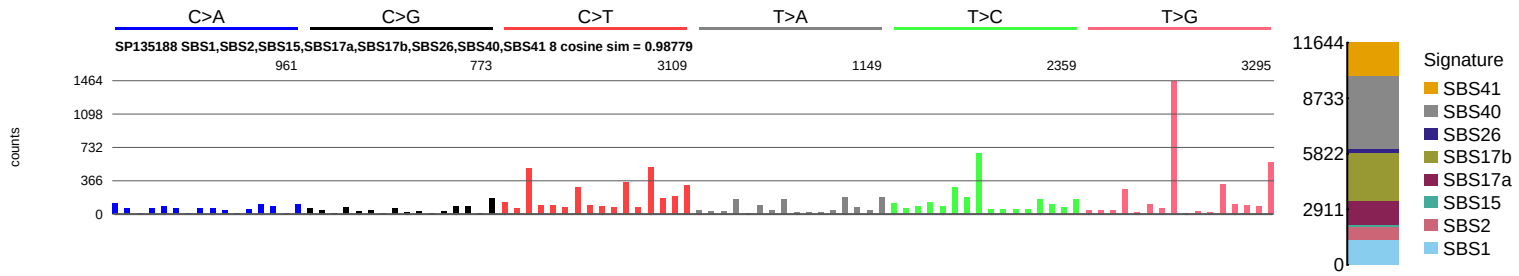
SBS2
SBS1

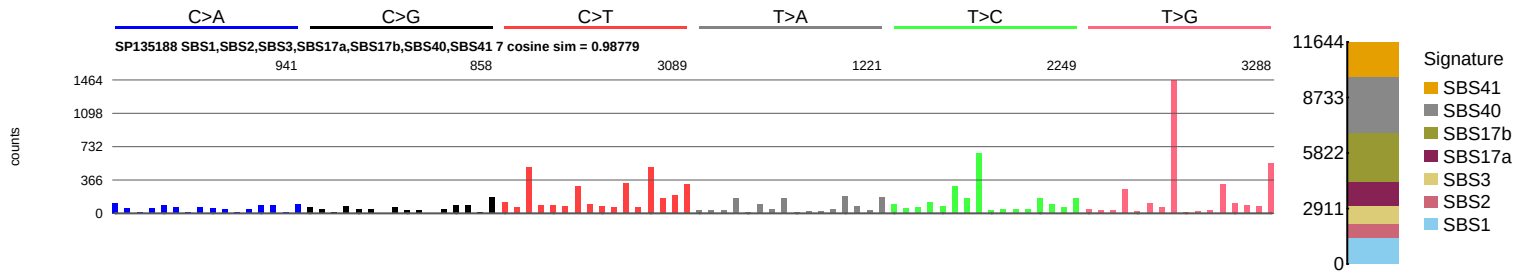

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98796
902
774
3167
1155
2367
3279
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

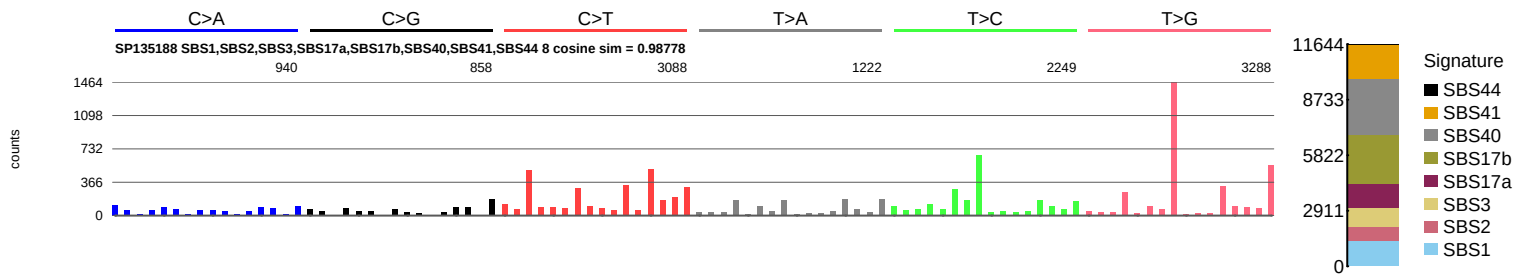

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98795
922
827
3091
1188
2343
3275
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

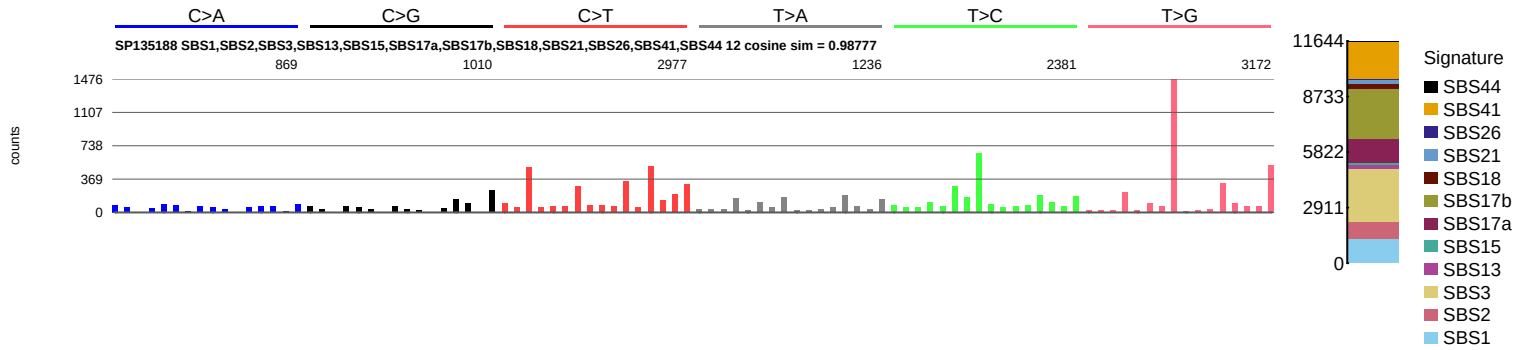

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.9879
921
831
3073
1193
2344
3279
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

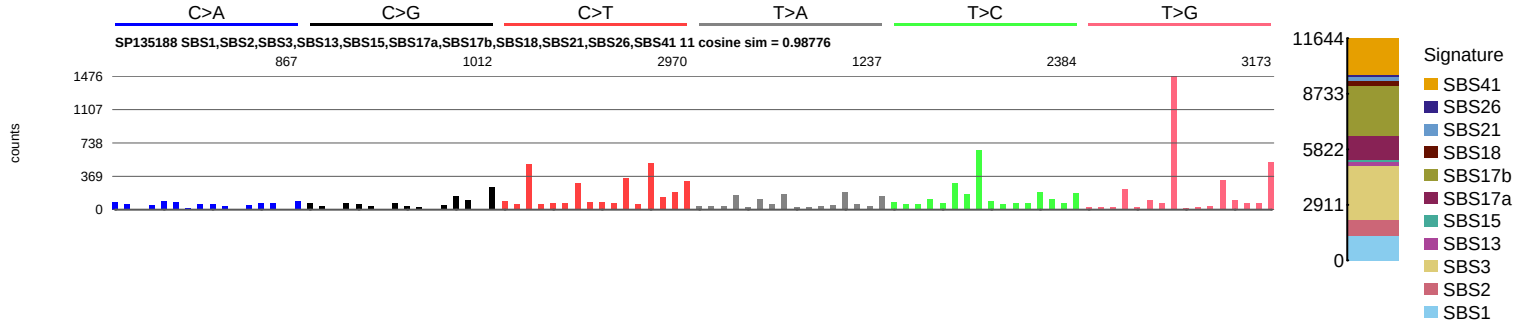

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98787
940
850
3106
1214
2248
3283
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

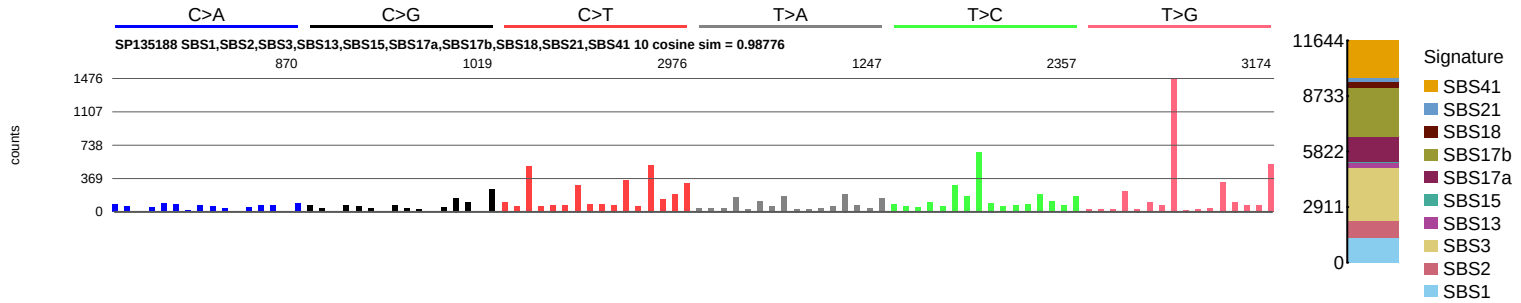

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98779
961
773
3109
1149
2359
3295
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

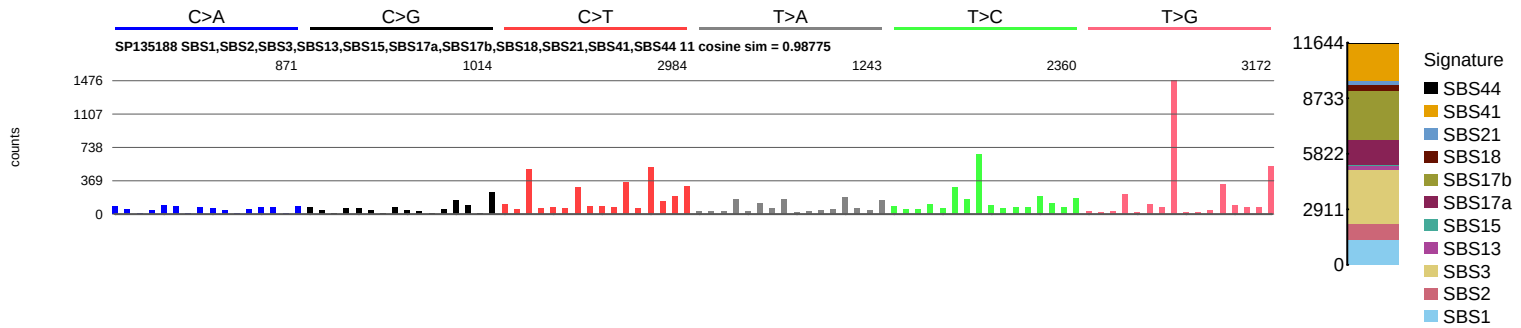

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98779
941
858
3089
1221
2249
3288
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

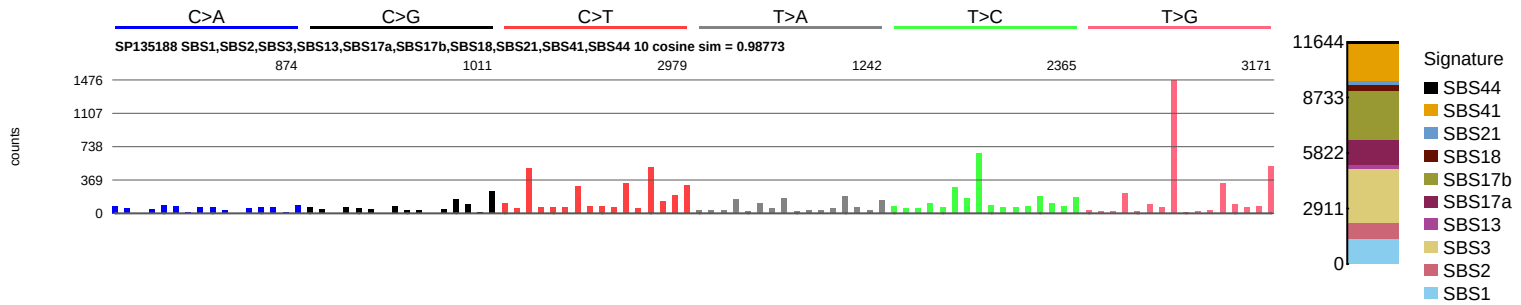

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98778
940
858
3088
1222
2249
3288
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

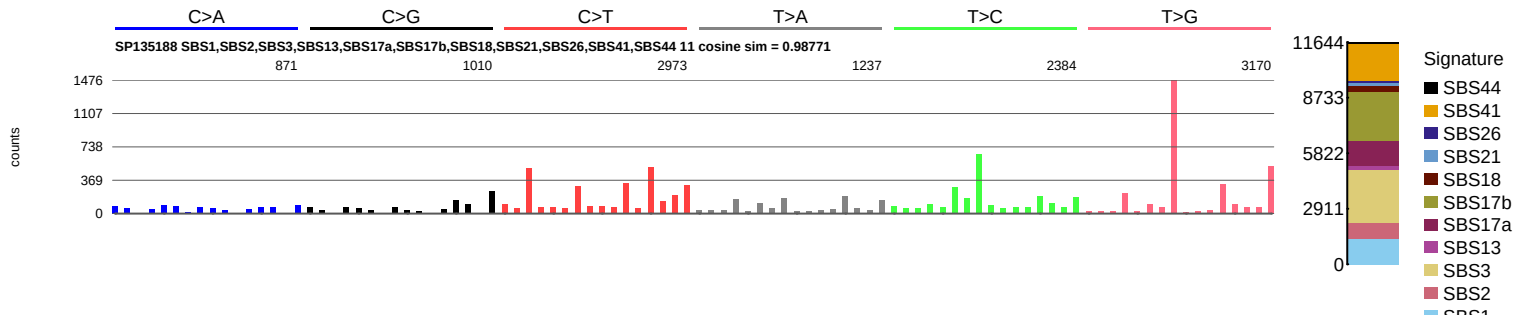

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.98777
869
1010
2977
1236
2381
3172
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

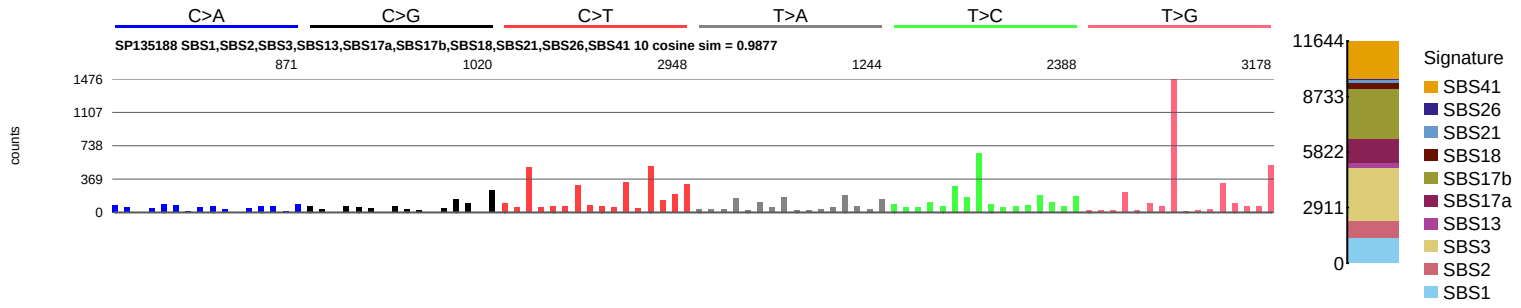

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.98776
867
1012
2970
1237
2384
3173
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

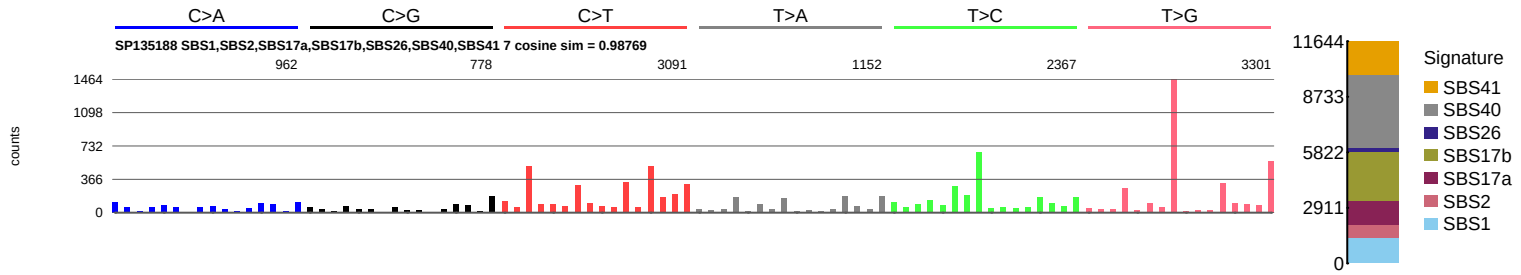

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98776
870
1019
2976
1247
2357
3174
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

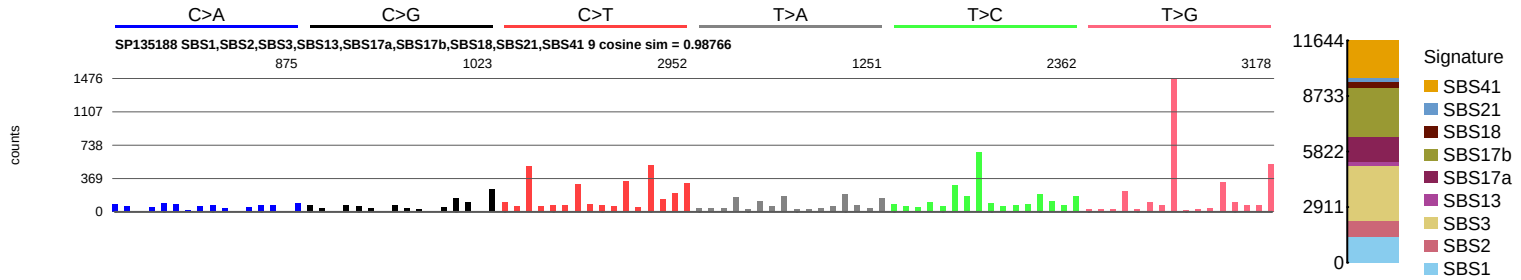

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98775
871
1014
2984
1243
2360
3172
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

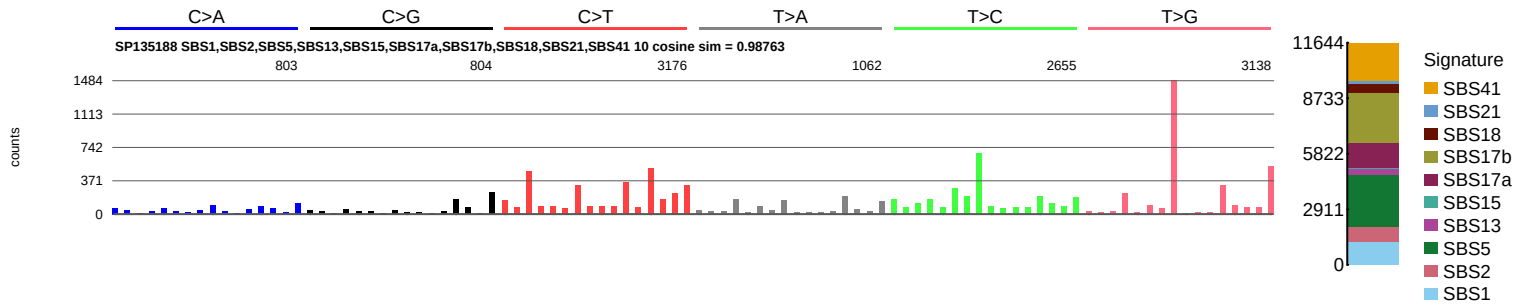

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98773
874
1011
2979
1242
2365
3171
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

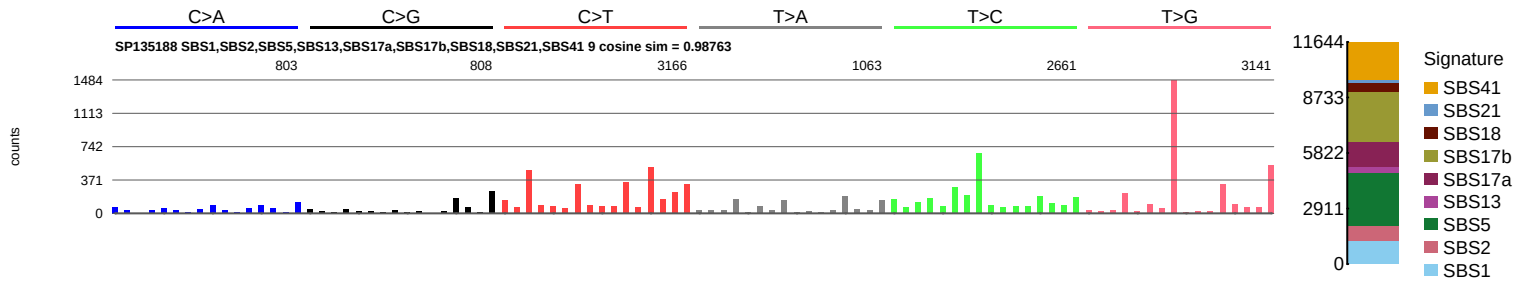

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98771
871
1010
2973
1237
2384
3170
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

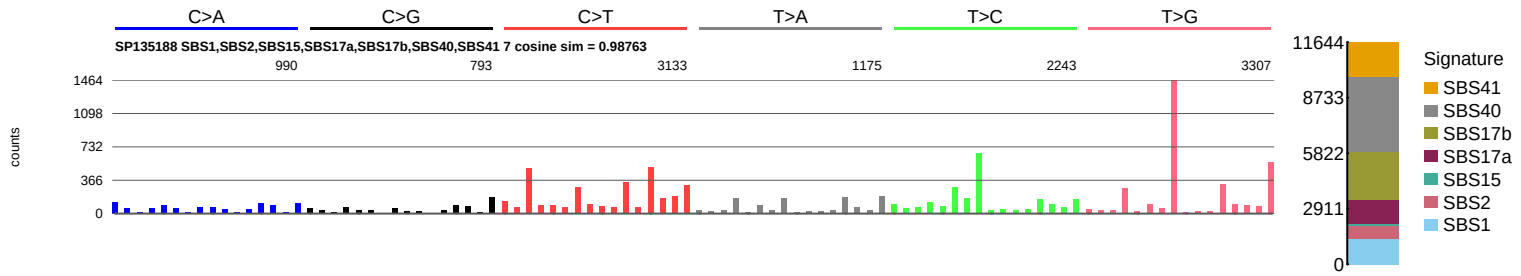

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.9877
871
1020
2948
1244
2388
3178
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

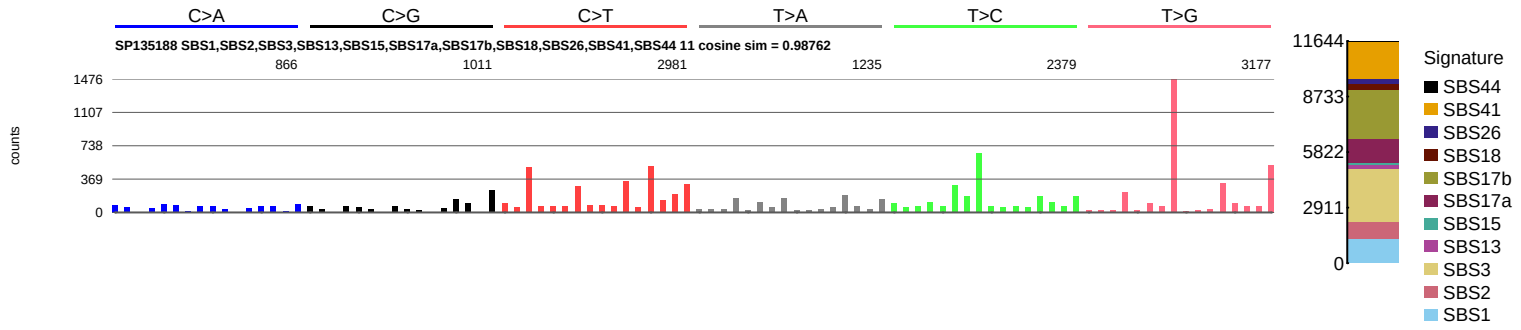

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98769
962
778
3091
1152
2367
3301
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

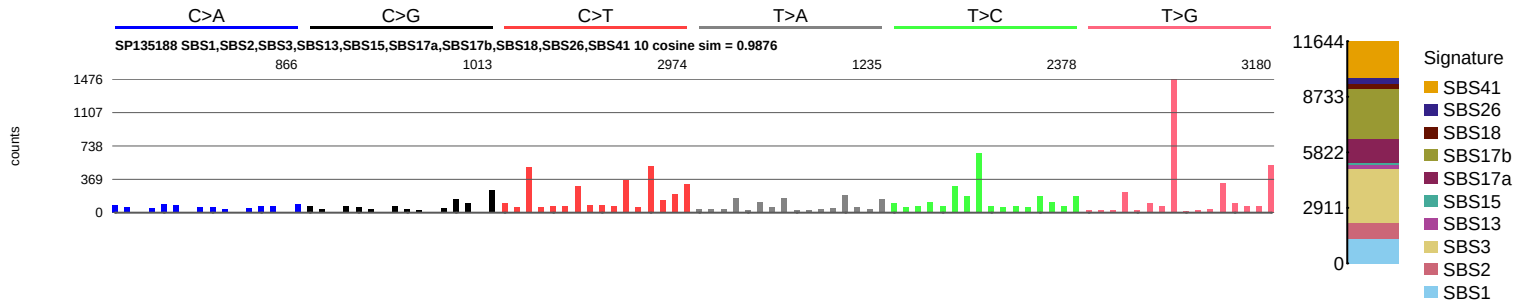

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98766
875
1023
2952
1251
2362
3178
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

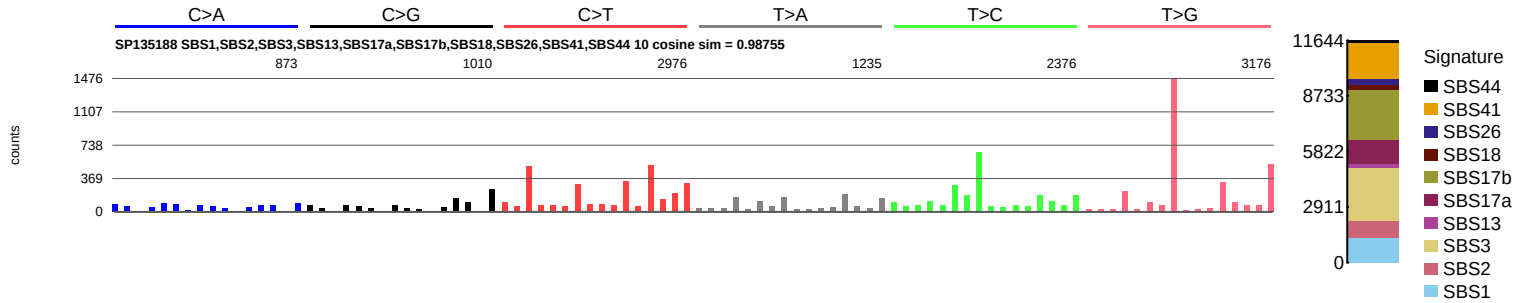

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98763
803
804
3176
1062
2655
3138
counts
1484
1113
742
371
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

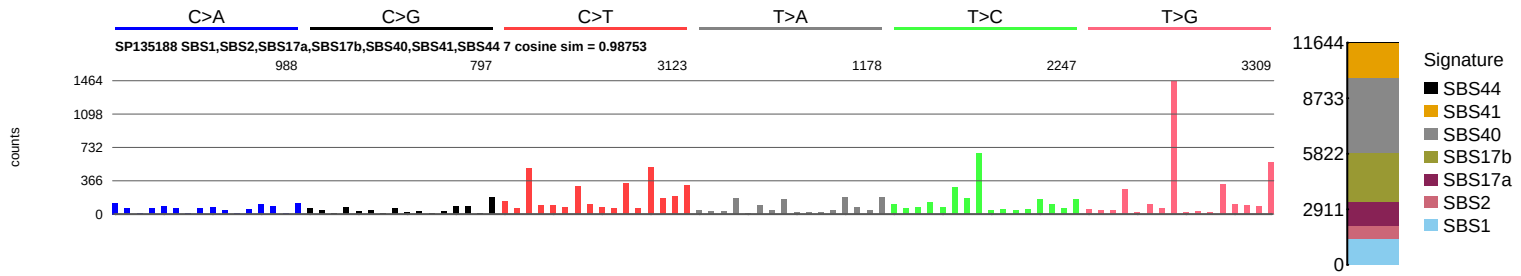

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98763
803
808
3166
1063
2661
3141
counts
1484
1113
742
371
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

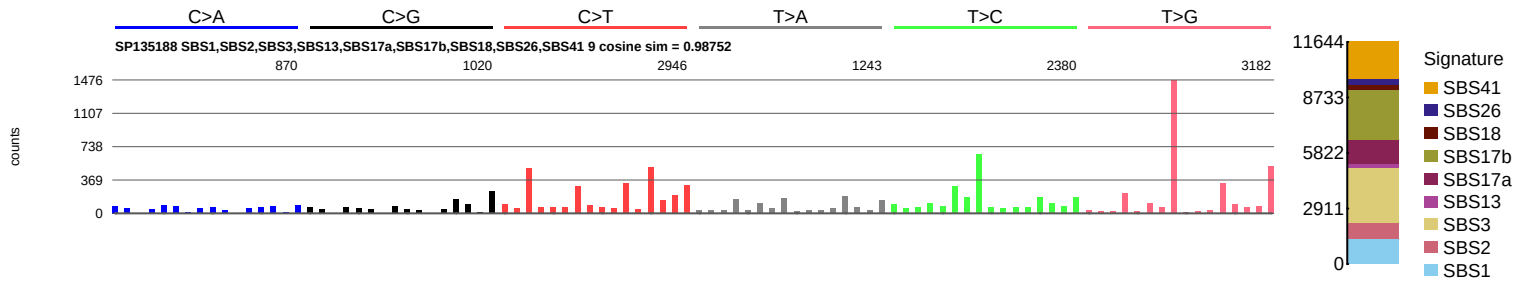

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98763
990
793
3133
1175
2243
3307
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

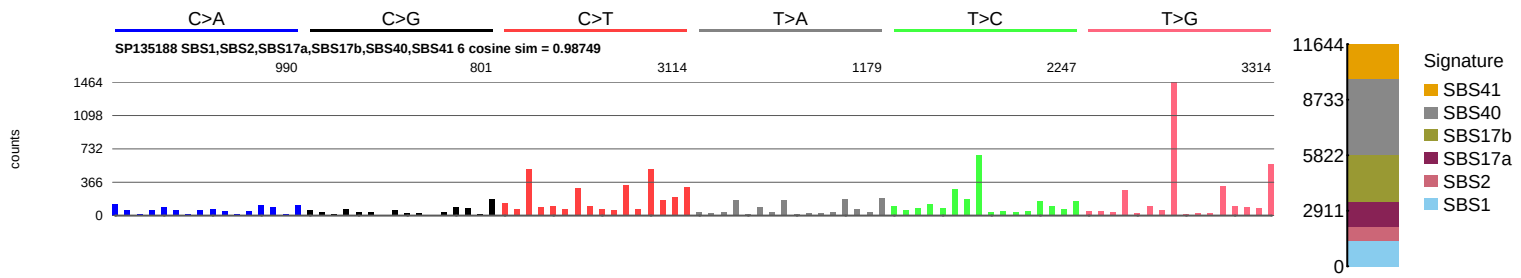

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98762
866
1011
2981
1235
2379
3177
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

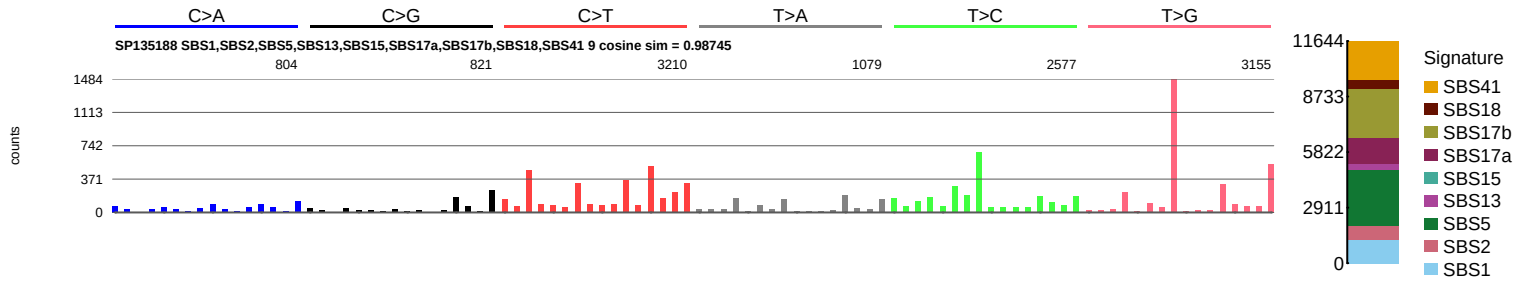

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.9876
866
1013
2974
1235
2378
3180
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

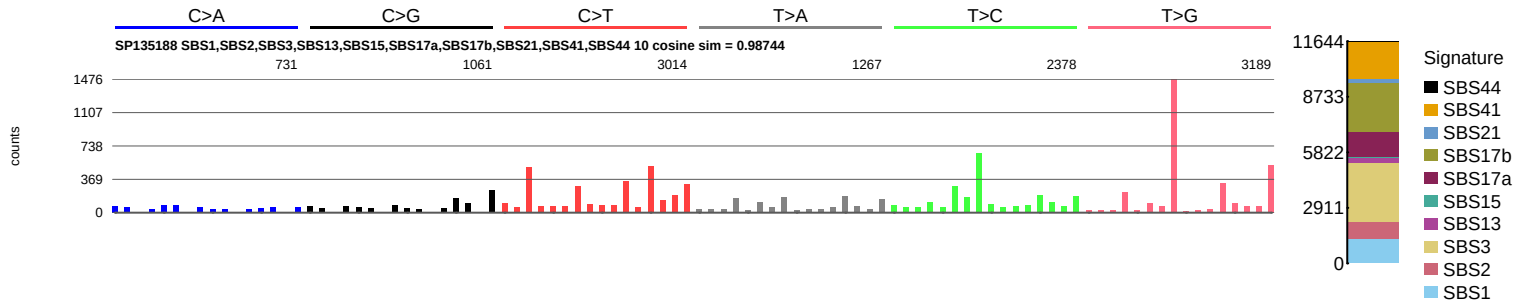

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98755
873
1010
2976
1235
2376
3176
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98753
988
797
3123
1178
2247
3309
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98752
870
1020
2946
1243
2380
3182
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98749
990
801
3114
1179
2247
3314
counts
1464
1098
732
366
0
11644
8733
5822
2911
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98745
804
821
3210
1079
2577
3155
counts
1484
1113
742
371
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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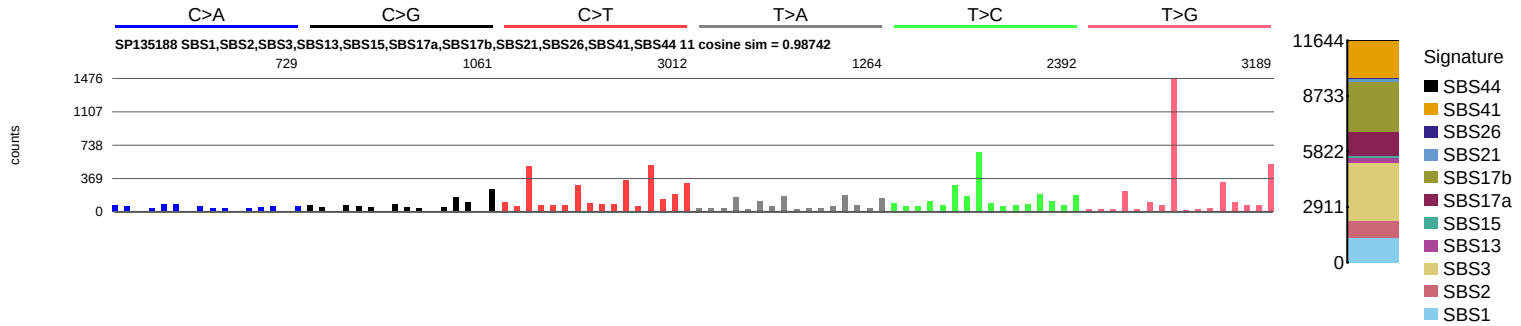
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.98744
731
1061
3014
1267
2378
3189
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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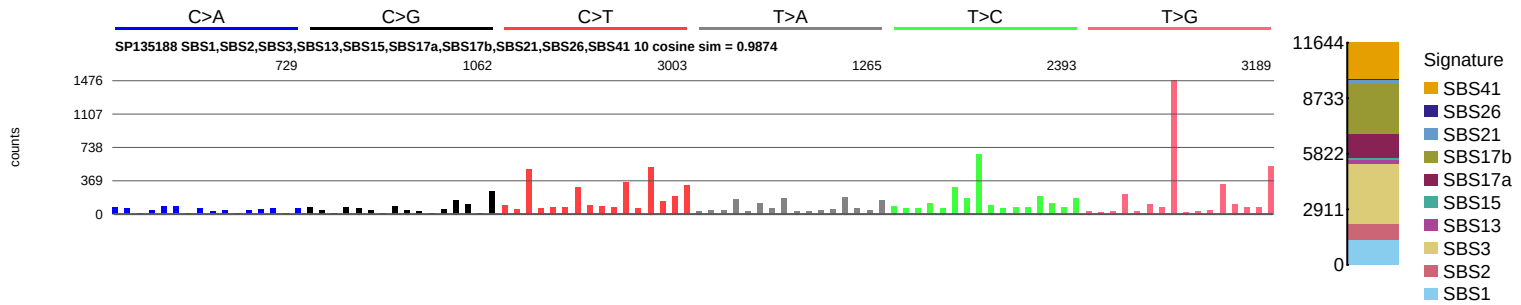
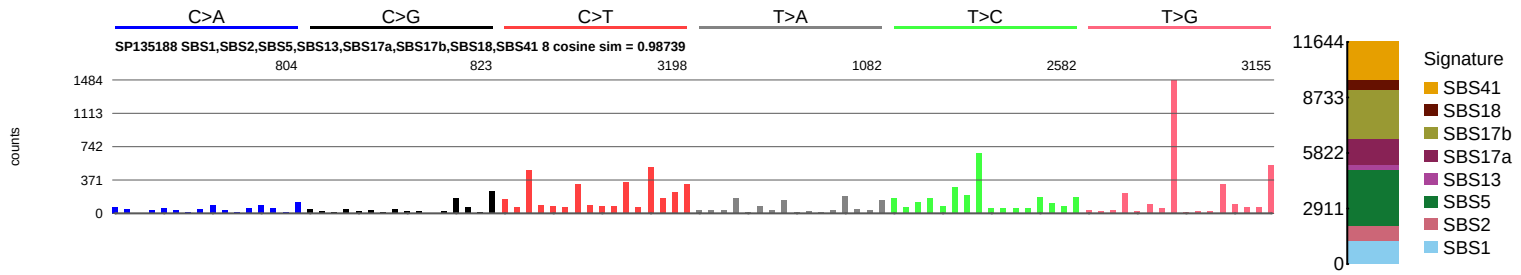
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98742
729
1061
3012
1264
2392
3189
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 10 cosine sim = 0.9874
729
1062
3003
1265
2393
3189
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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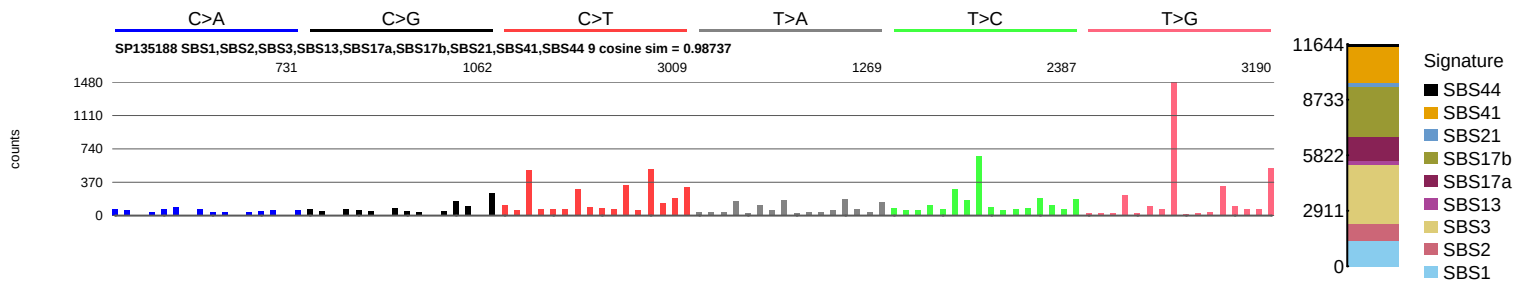
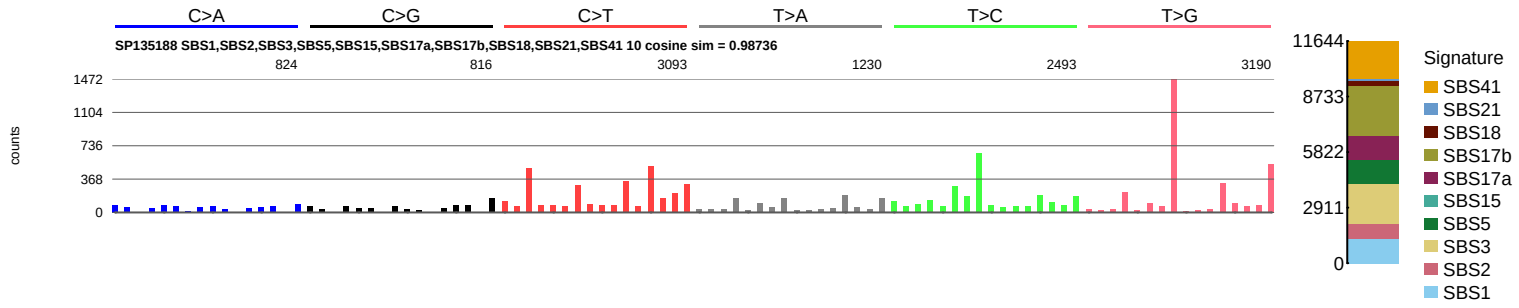
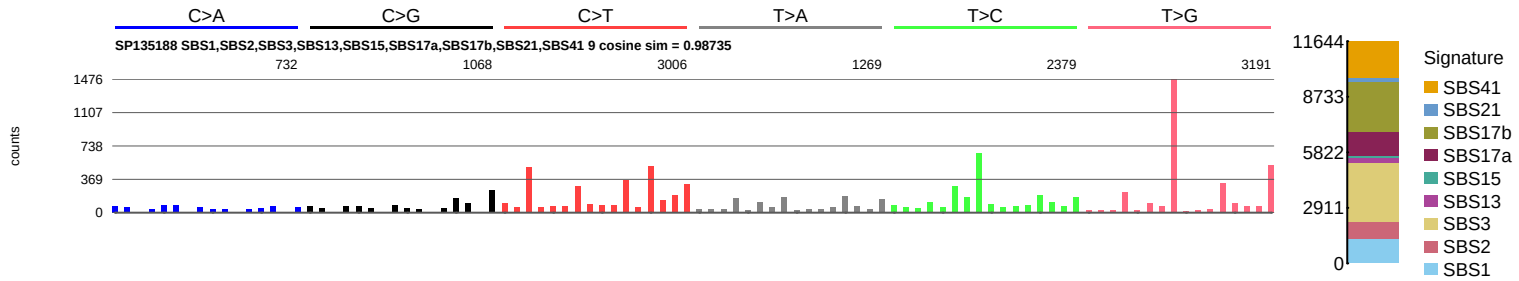
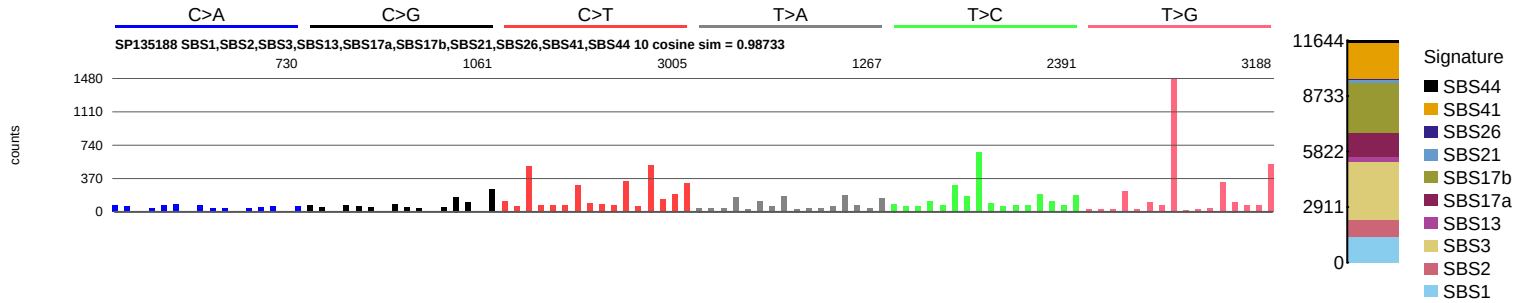
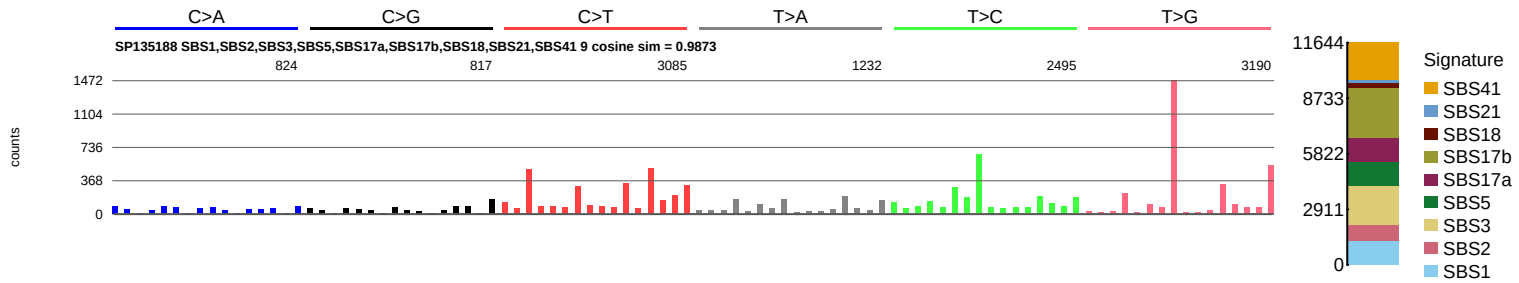
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98739
804
823
3198
1082
2582
3155
counts
1484
1113
742
371
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

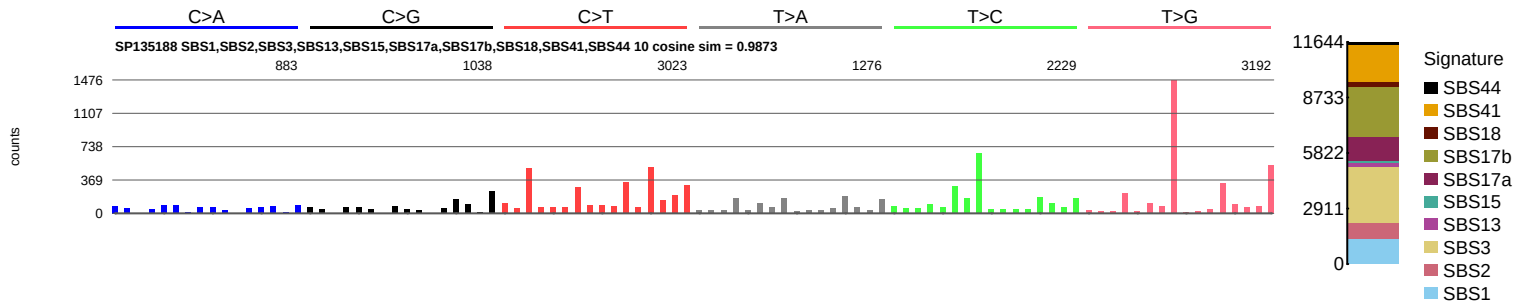

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98737
731
1062
3009
1269
2387
3190
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

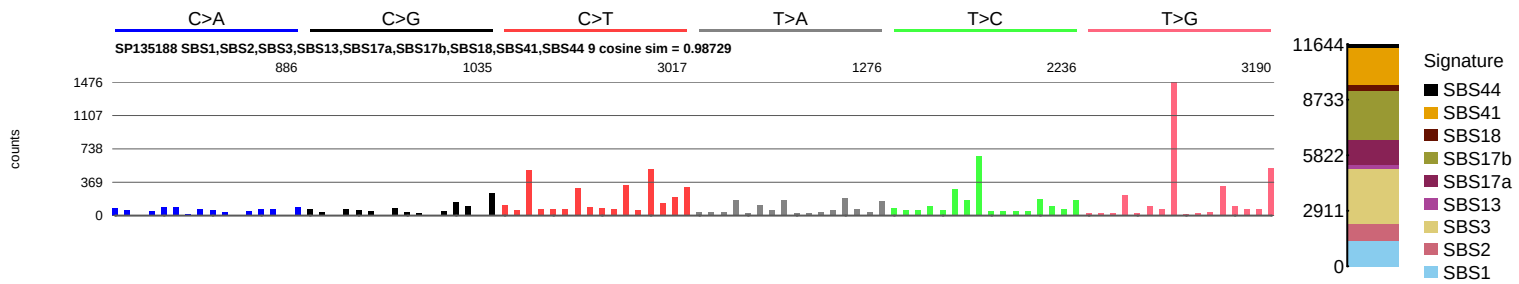

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98736
824
816
3093
1230
2493
3190
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

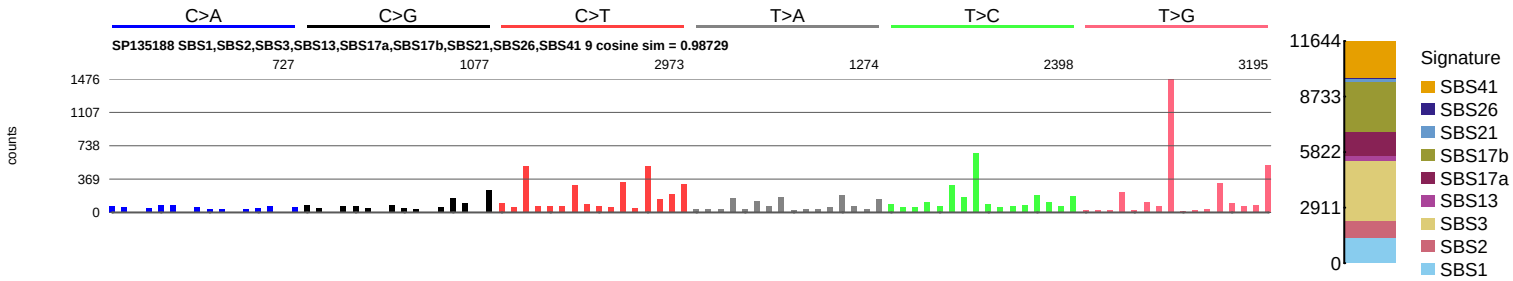

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98735
732
1068
3006
1269
2379
3191
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

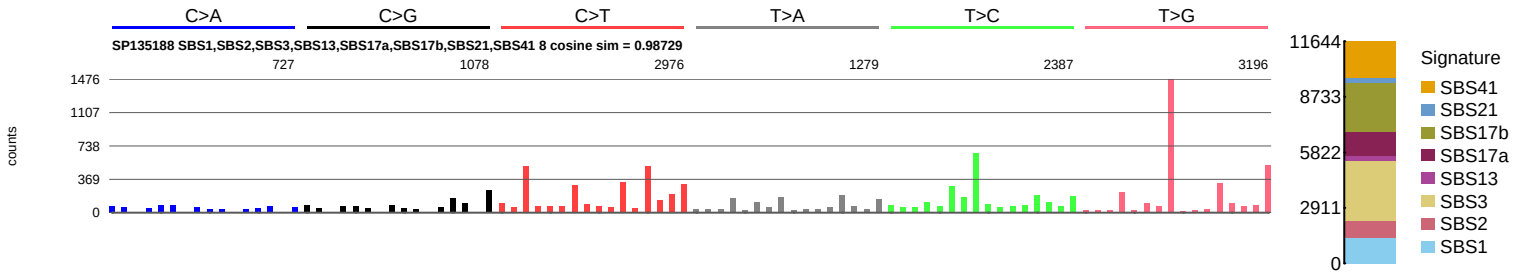

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98733
730
1061
3005
1267
2391
3188
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

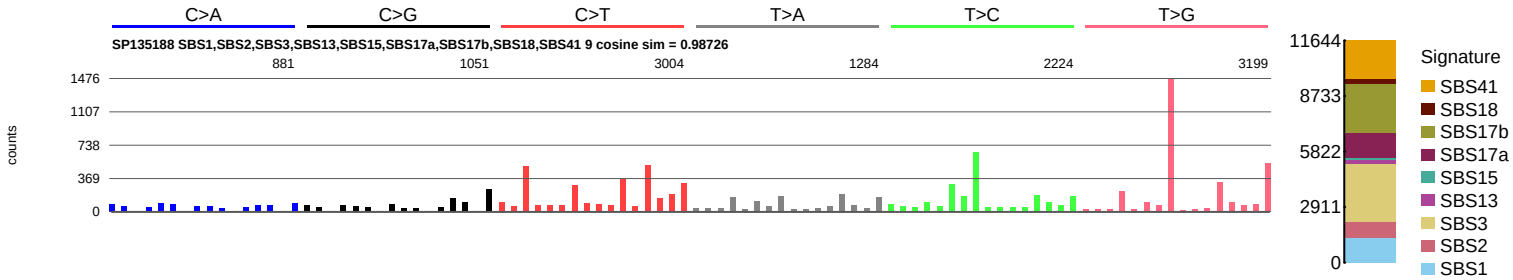

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9873
824
817
3085
1232
2495
3190
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

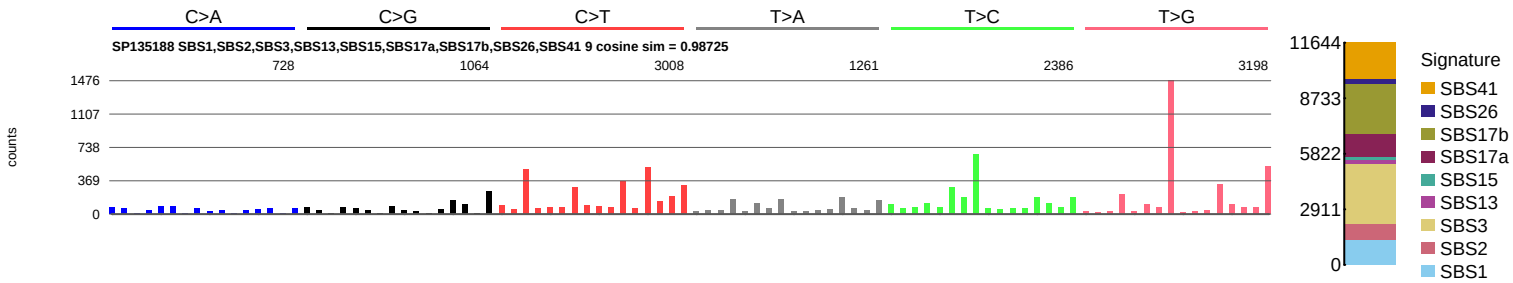

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.9873
883
1038
3023
1276
2229
3192
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98729
886
1035
3017
1276
2236
3190
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98729
727
1077
2973
1274
2398
3195
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98729
727
1078
2976
1279
2387
3196
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98726
881
1051
3004
1284
2224
3199
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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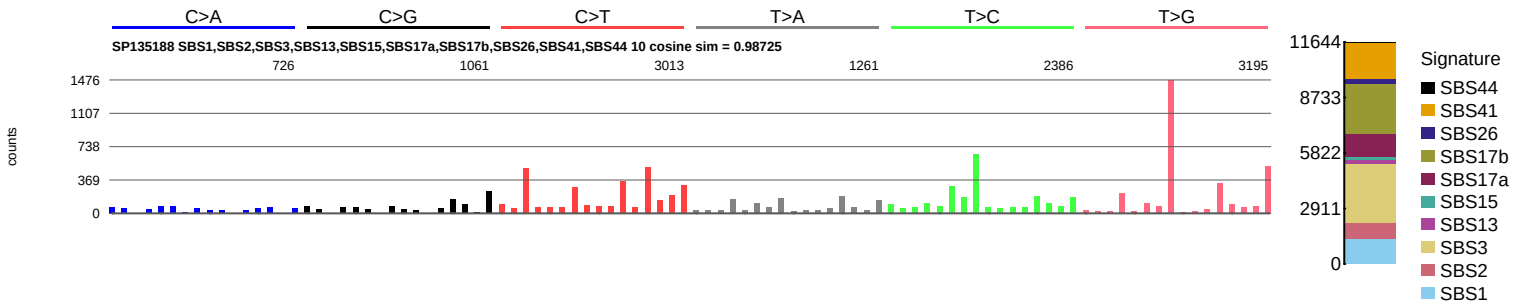
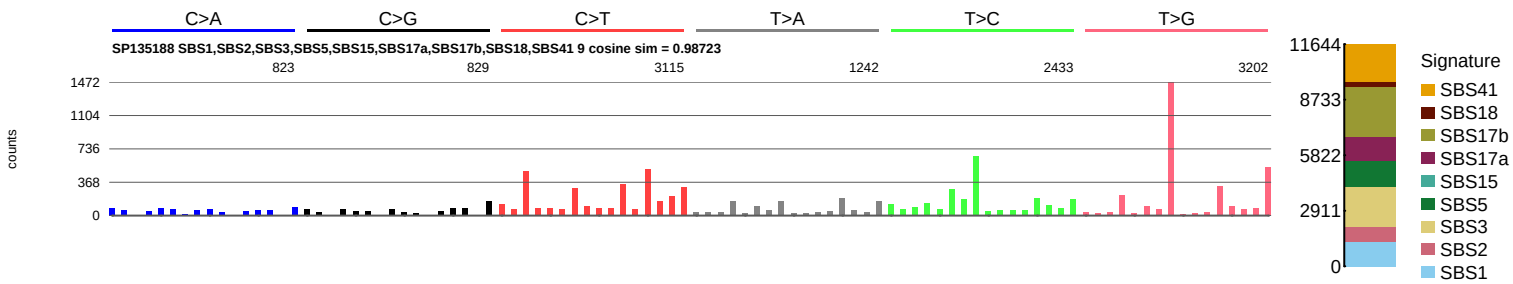
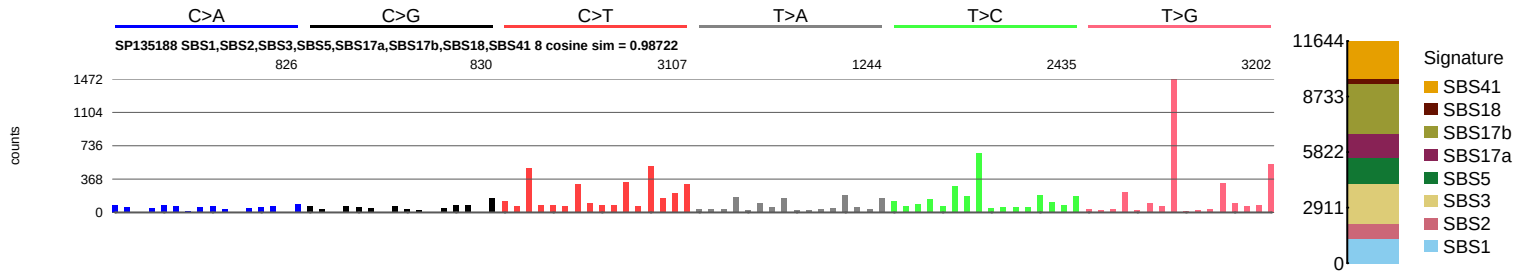
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.98725
728
1064
3008
1261
2386
3198
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 10 cosine sim = 0.98725
726
1061
3013
1261
2386
3195
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98723
823
829
3115
1242
2433
3202
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98722
826
830
3107
1244
2435
3202
counts
1472
1104
736
368
0
11644
8733
5822
2911
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Signature
SBS41
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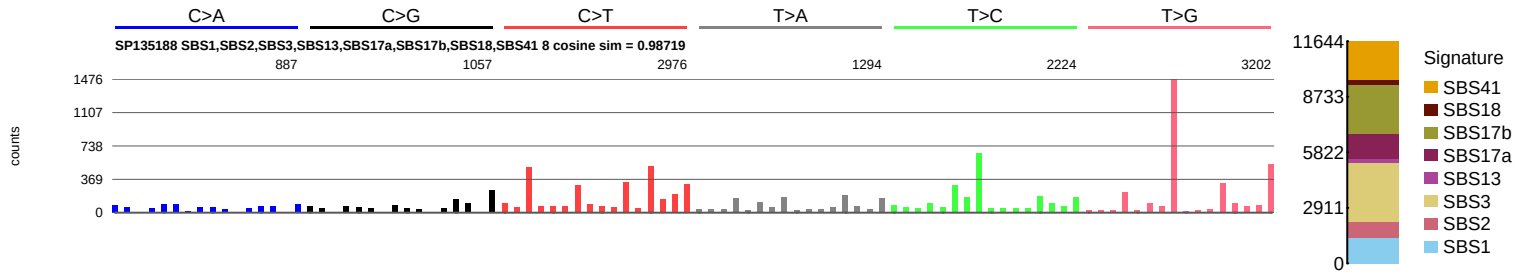
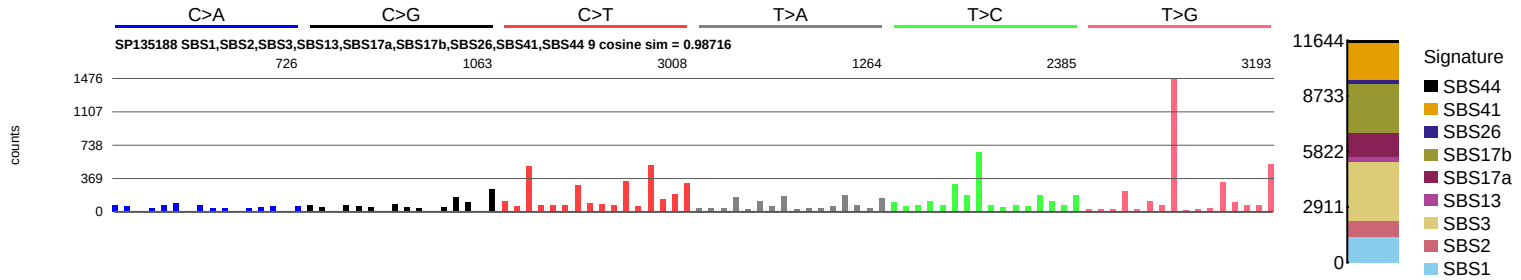
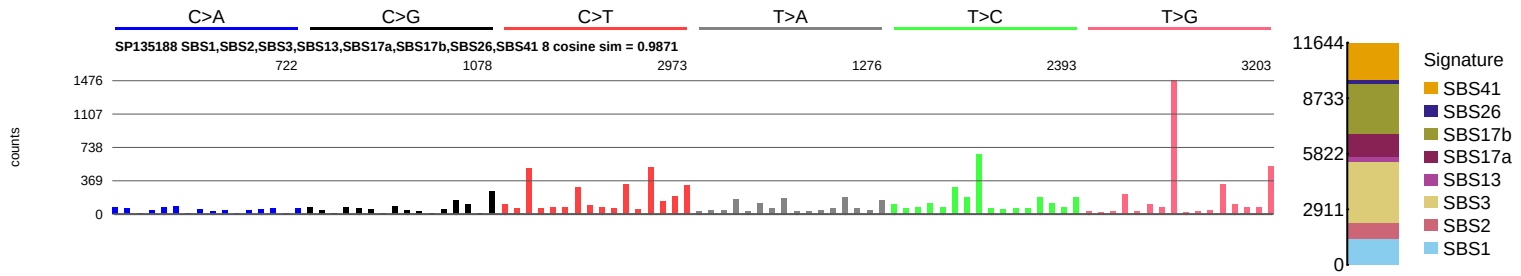
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98719
887
1057
2976
1294
2224
3202
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98716
726
1063
3008
1264
2385
3193
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.9871
722
1078
2973
1276
2393
3203
counts
1476
1107
738
369
0
11644
8733
5822
2911
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Signature
SBS41
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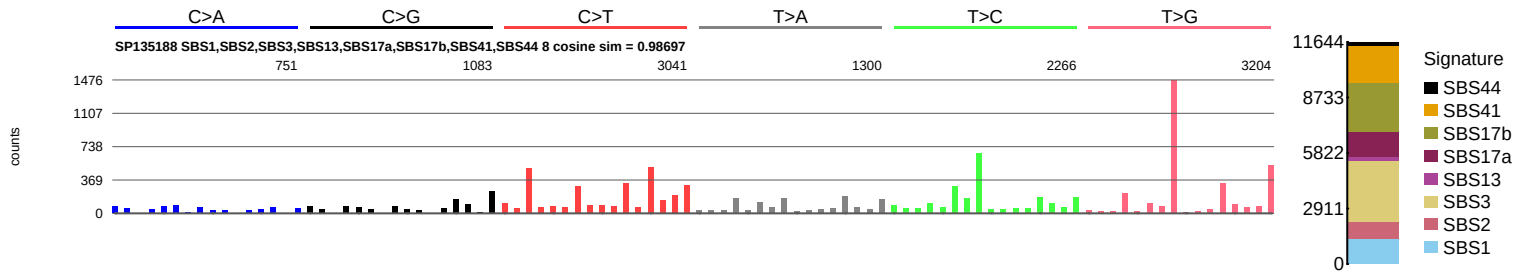
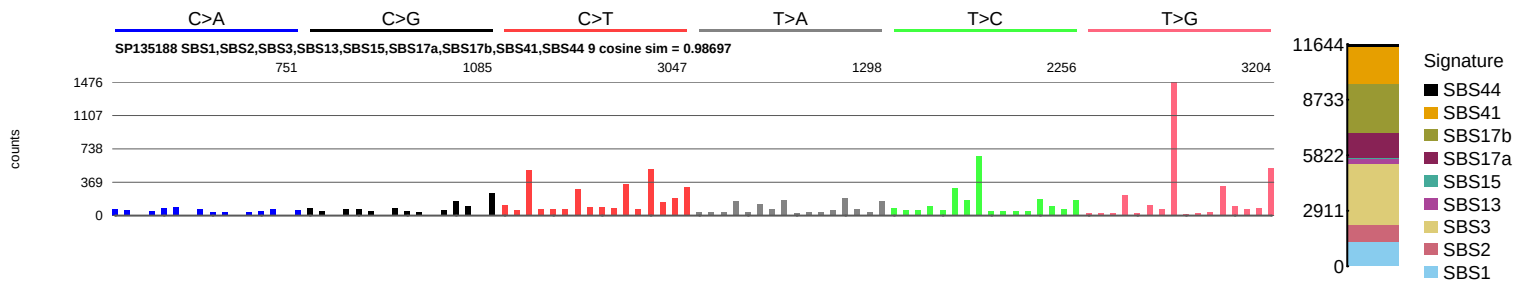
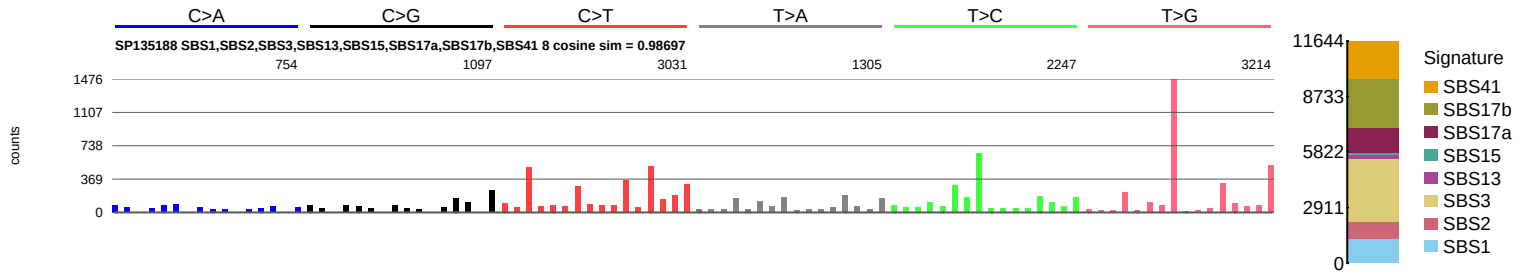
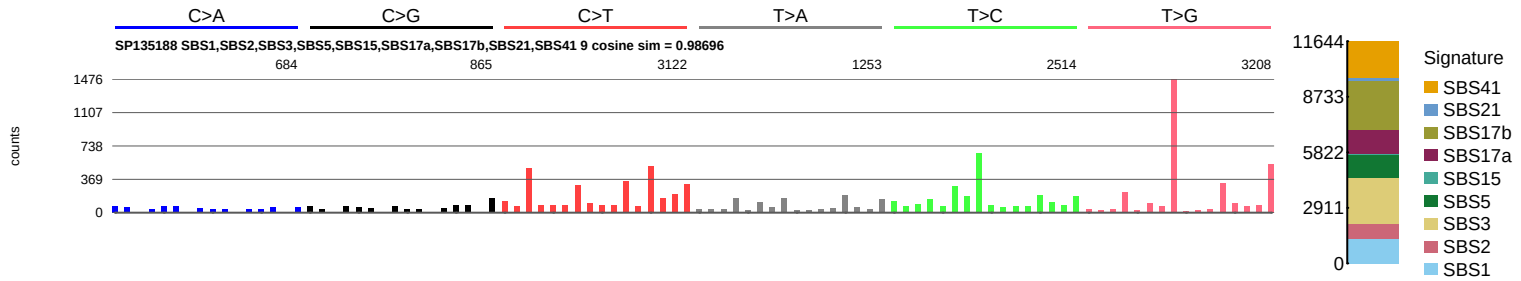
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98697
751
1083
3041
1300
2266
3204
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98697
751
1085
3047
1298
2256
3204
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98697
754
1097
3031
1305
2247
3214
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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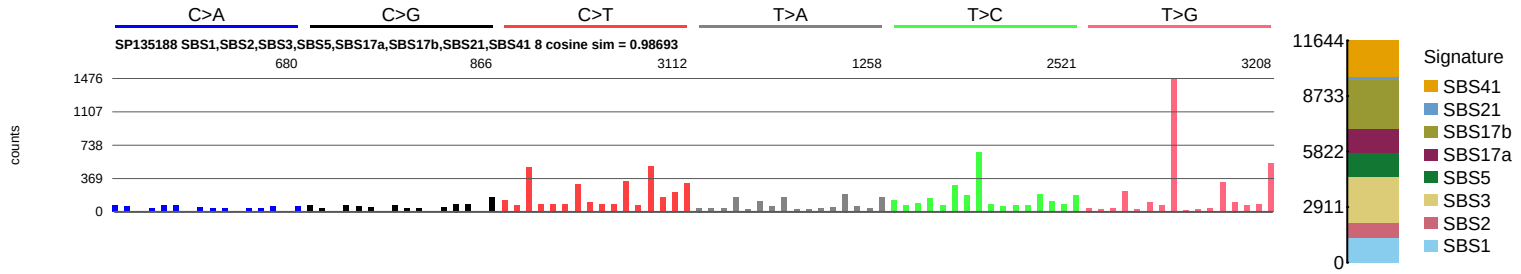
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98696
684
865
3122
1253
2514
3208
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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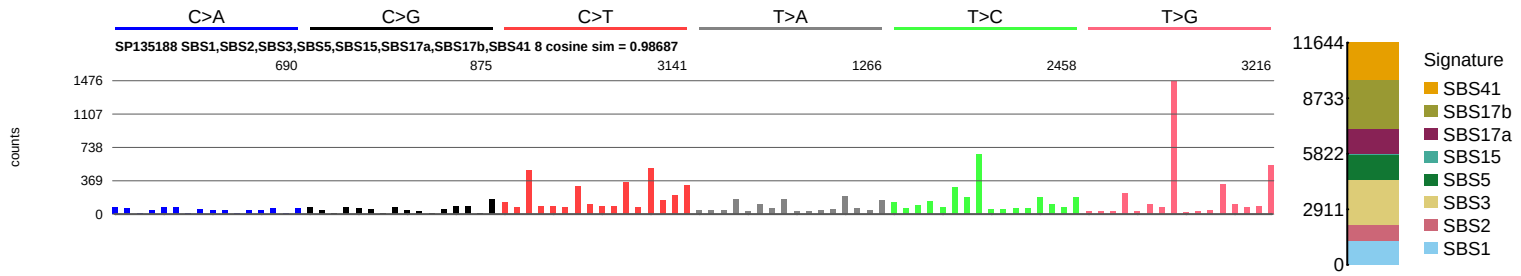
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98693
680
866
3112
1258
2521
3208
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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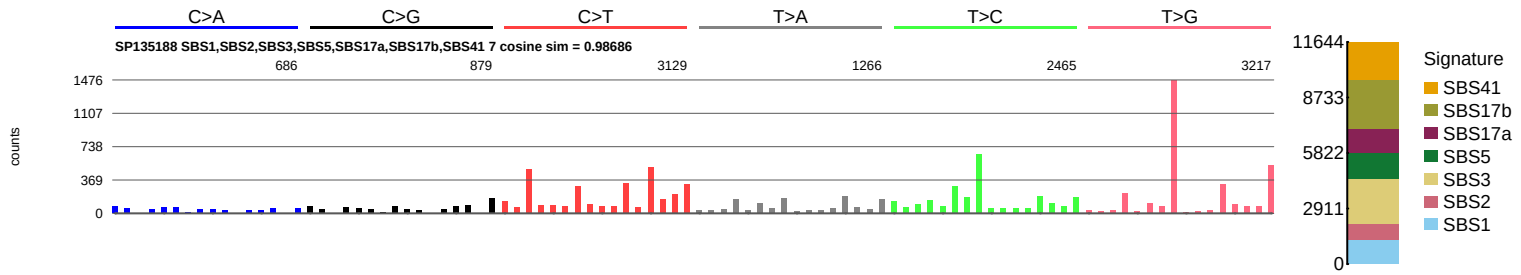
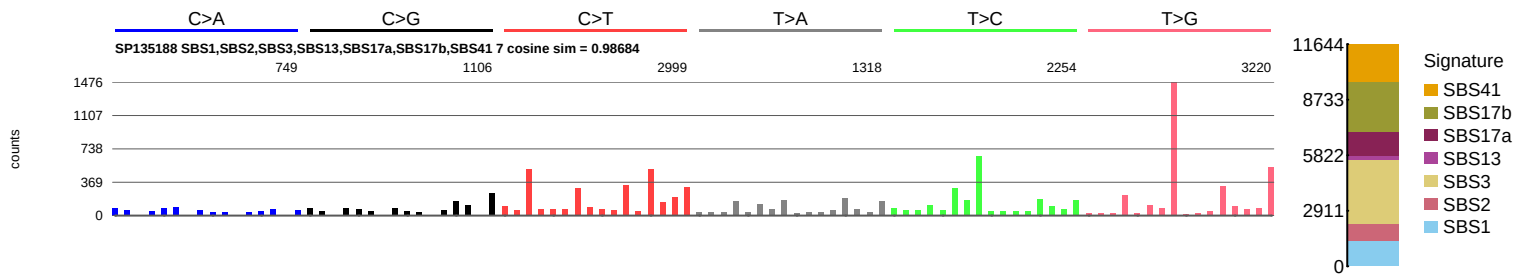
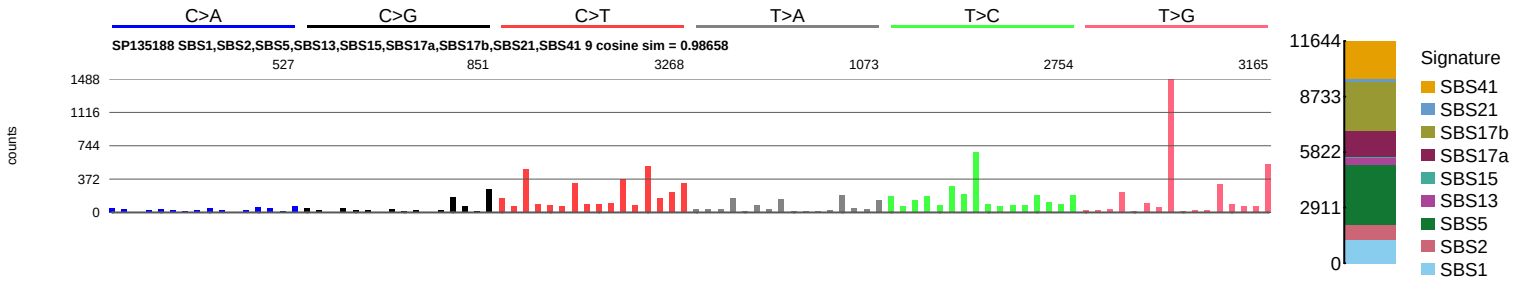
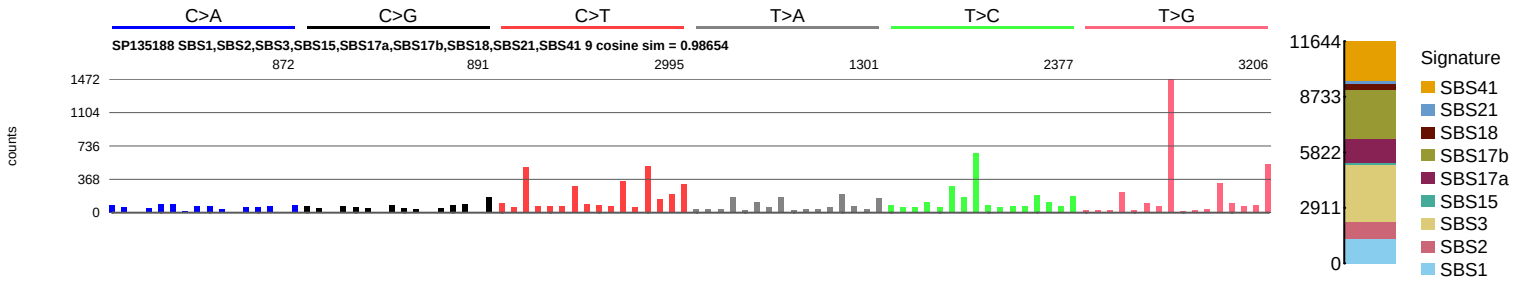
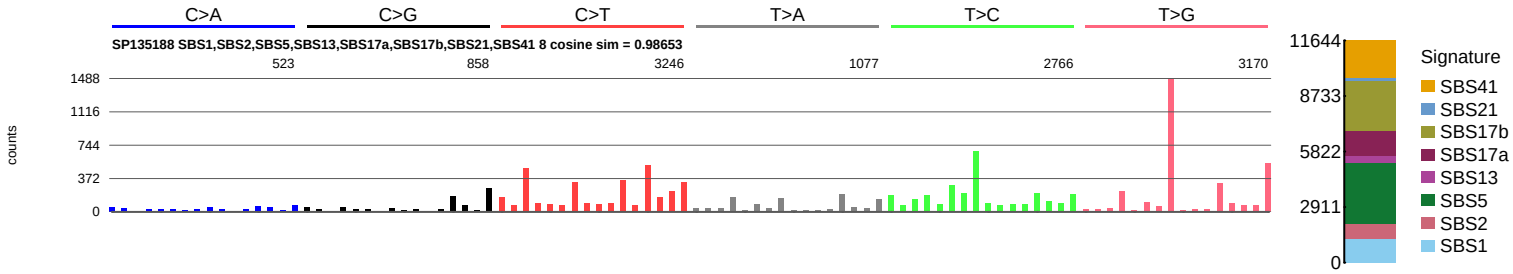
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98687
690
875
3141
1266
2458
3216
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98686
686
879
3129
1266
2465
3217
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98684
749
1106
2999
1318
2254
3220
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98658
527
851
3268
1073
2754
3165
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98654
872
891
2995
1301
2377
3206
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98653
523
858
3246
1077
2766
3170
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS41
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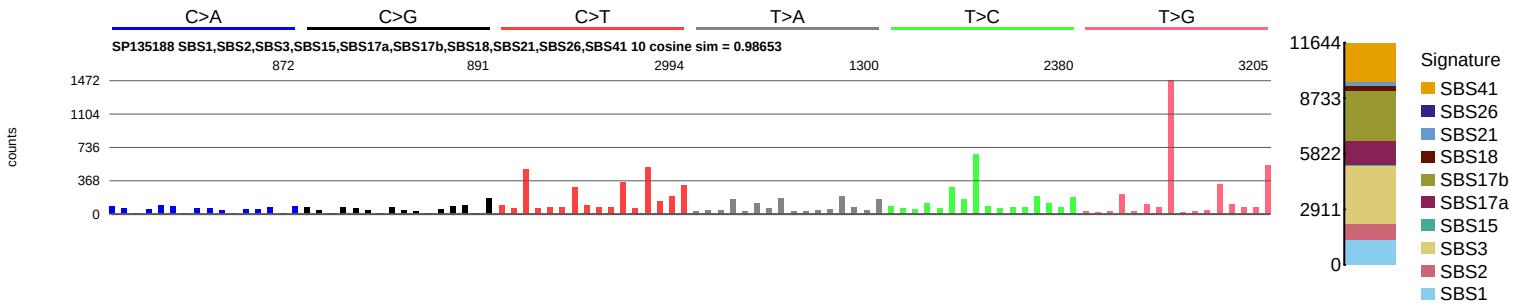
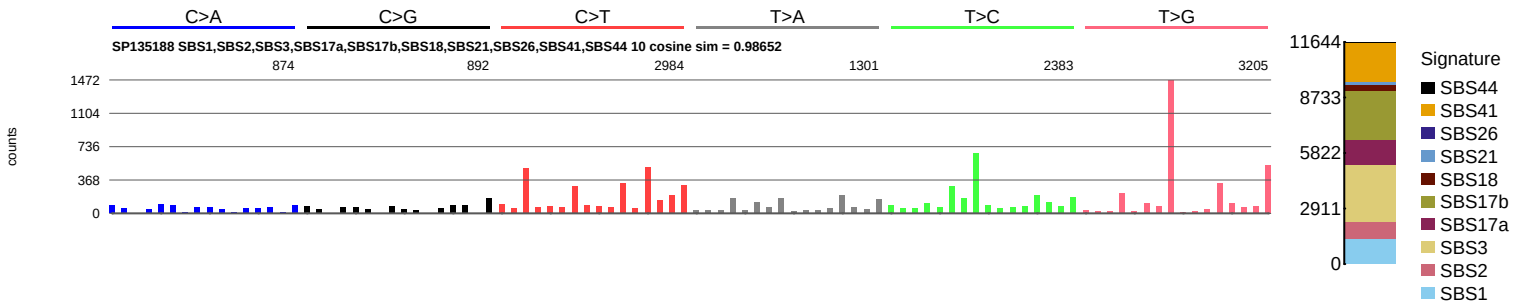
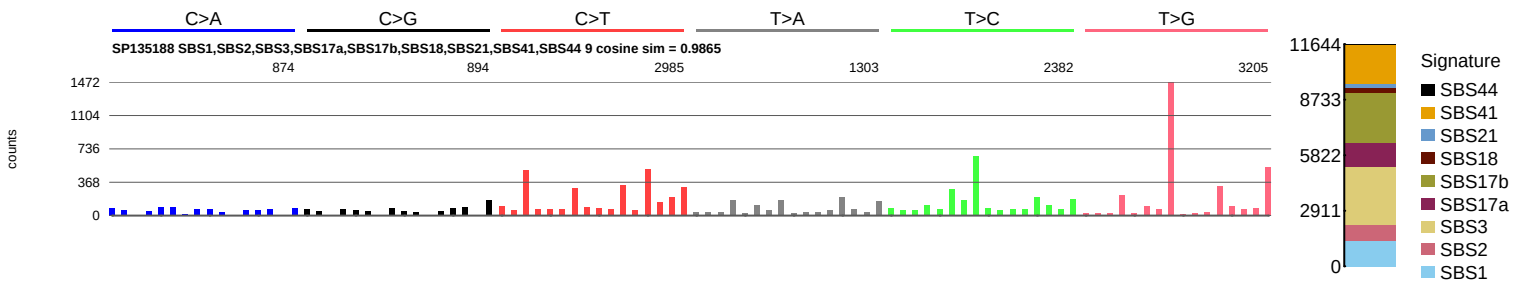
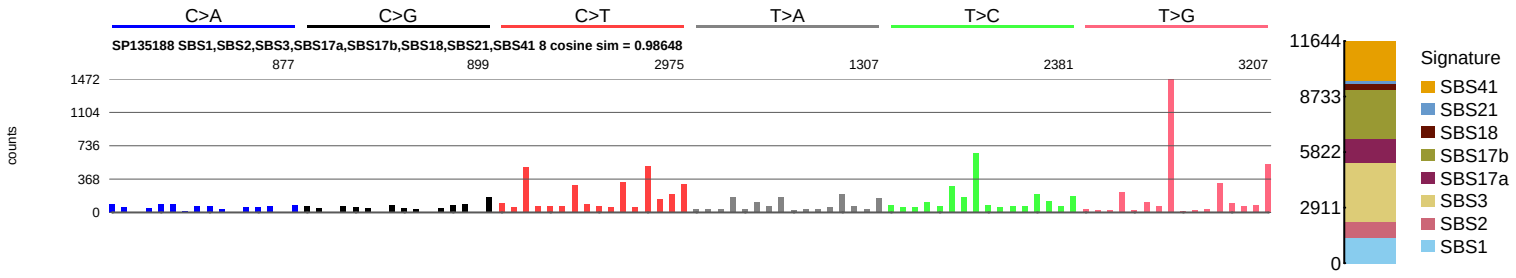
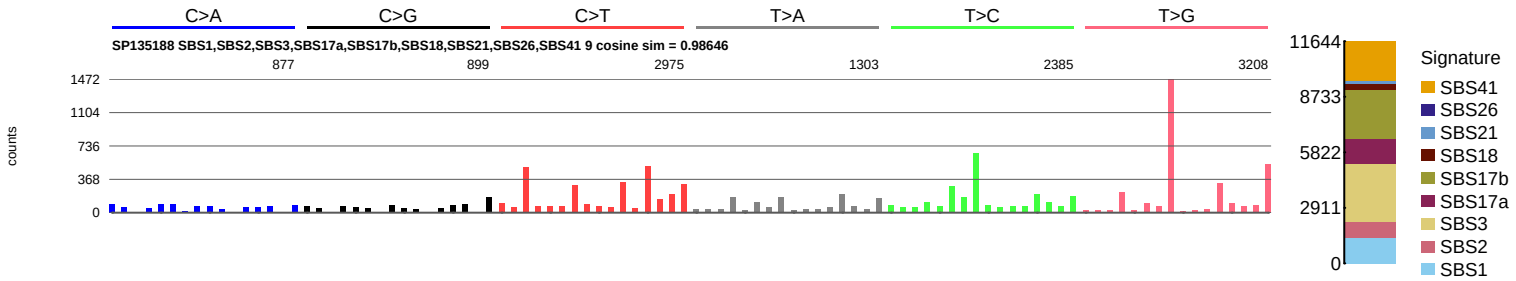
SBS2
SBS1

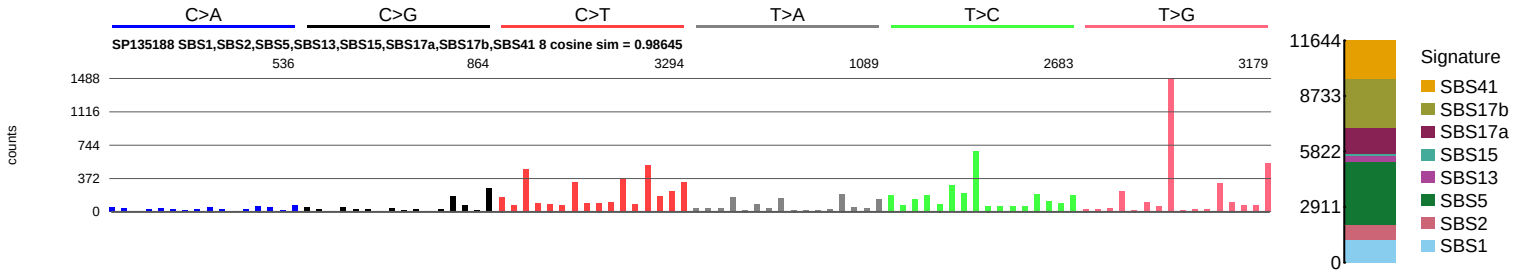

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98653
872
891
2994
1300
2380
3205
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

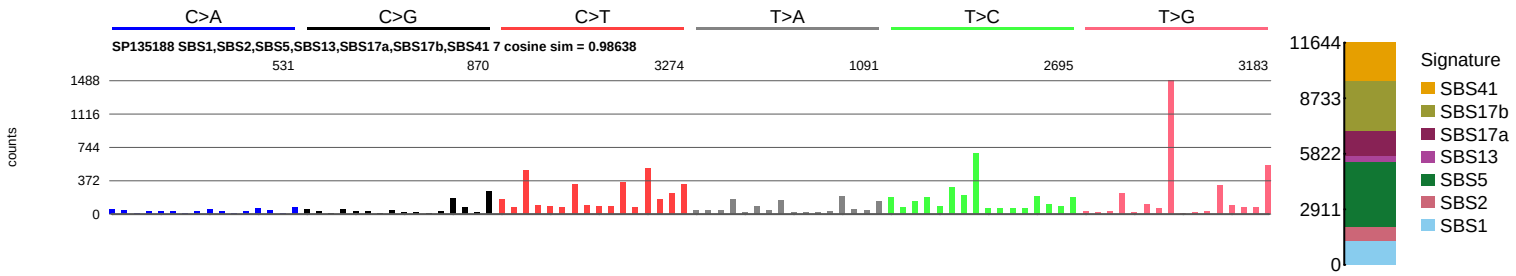

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98652
874
892
2984
1301
2383
3205
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

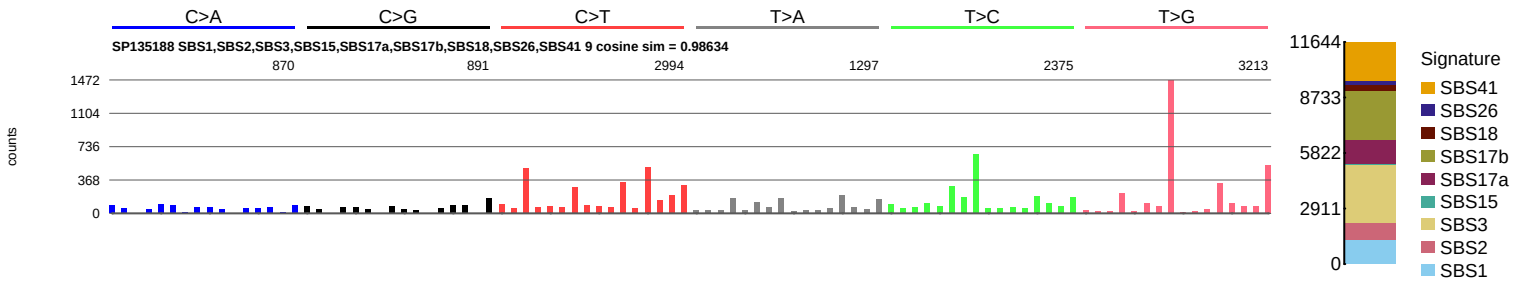

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9865
874
894
2985
1303
2382
3205
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

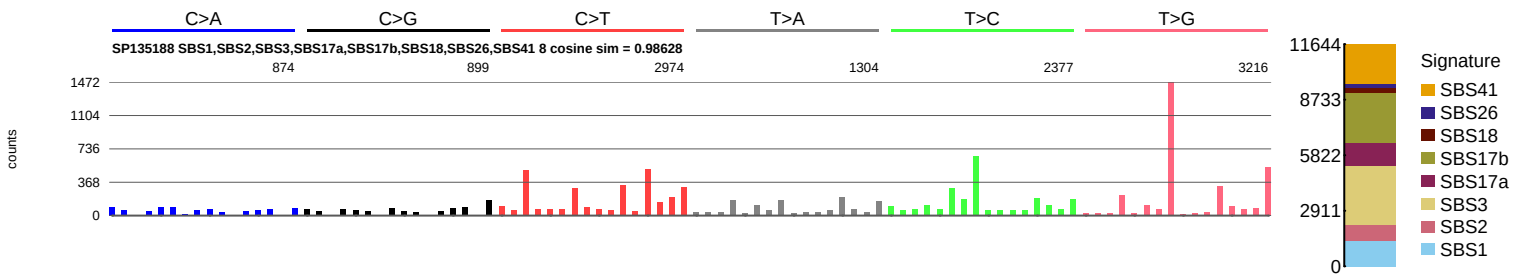

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98648
877
899
2975
1307
2381
3207
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

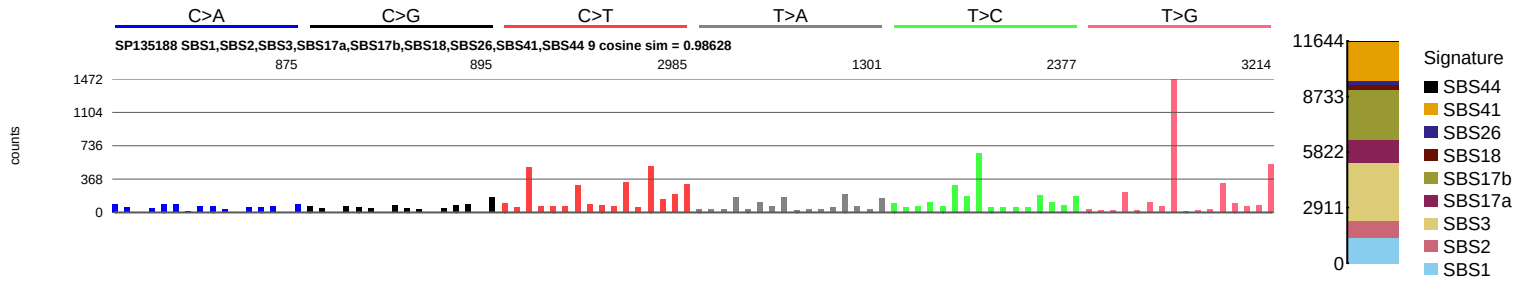

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98646
877
899
2975
1303
2385
3208
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

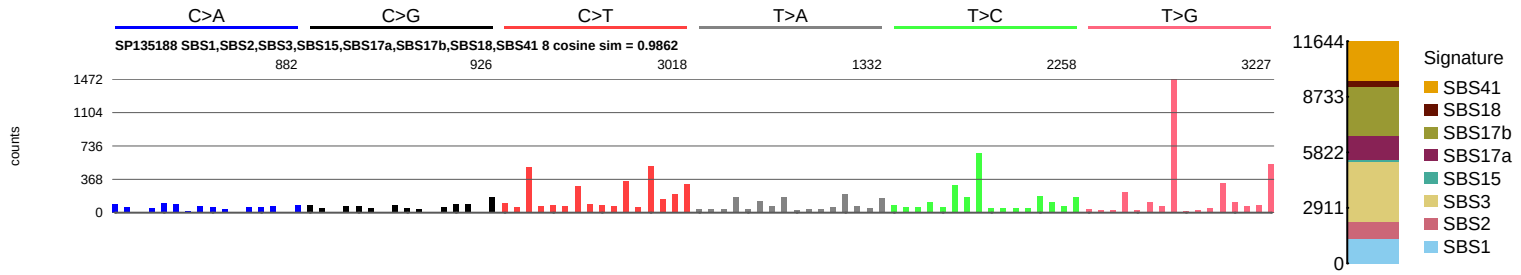

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98645
536
864
3294
1089
2683
3179
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98638
531
870
3274
1091
2695
3183
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98634
870
891
2994
1297
2375
3213
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98628
874
899
2974
1304
2377
3216
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98628
875
895
2985
1301
2377
3214
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9862
882
926
3018
1332
2258
3227
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
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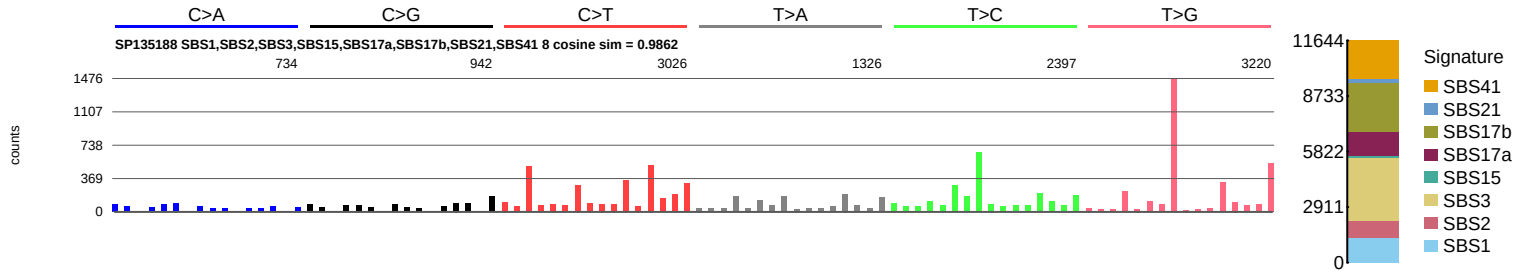
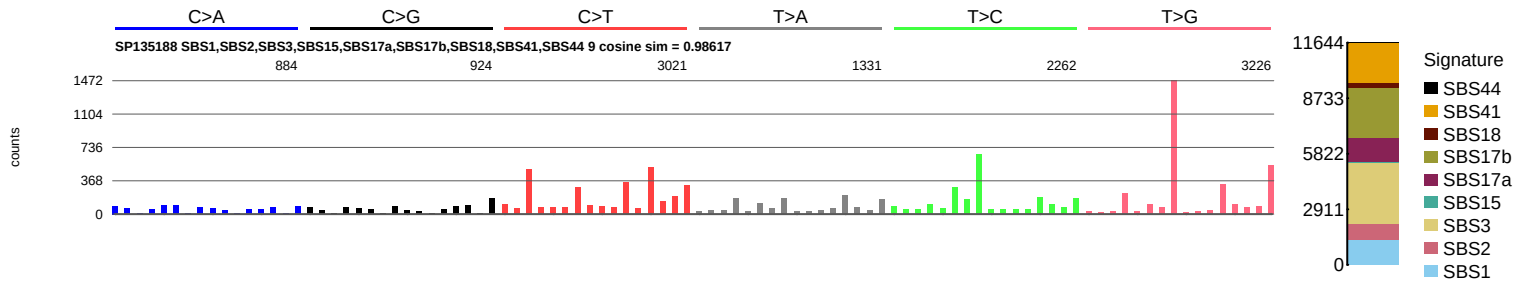
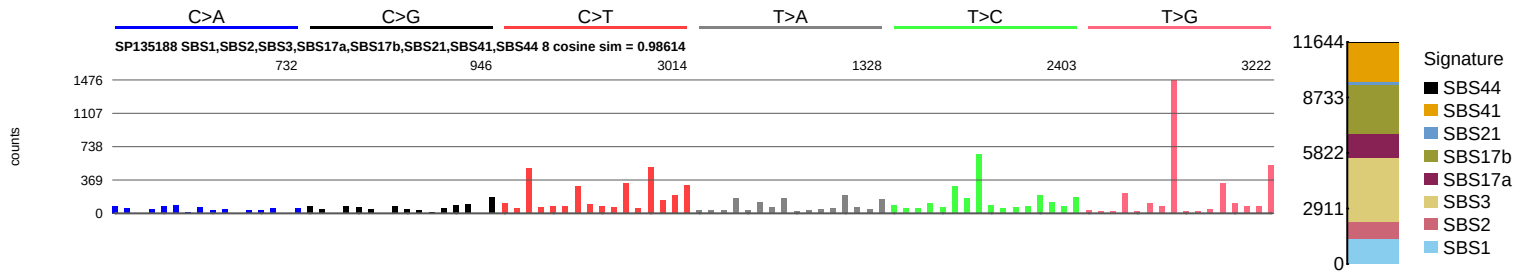
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9862
734
942
3026
1326
2397
3220
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98617
884
924
3021
1331
2262
3226
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98614
732
946
3014
1328
2403
3222
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
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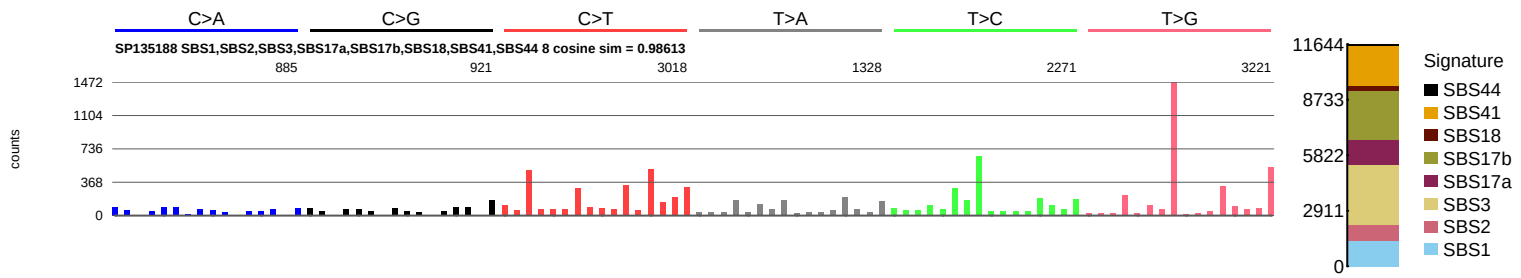
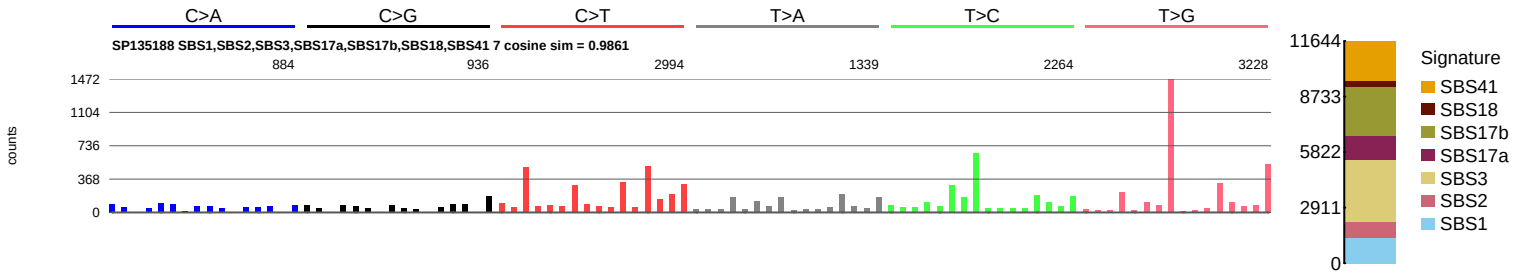
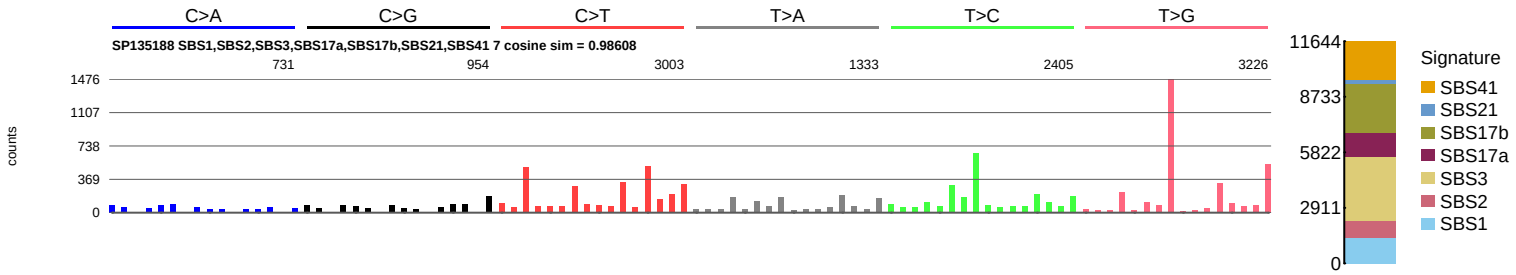
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98613
885
921
3018
1328
2271
3221
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9861
884
936
2994
1339
2264
3228
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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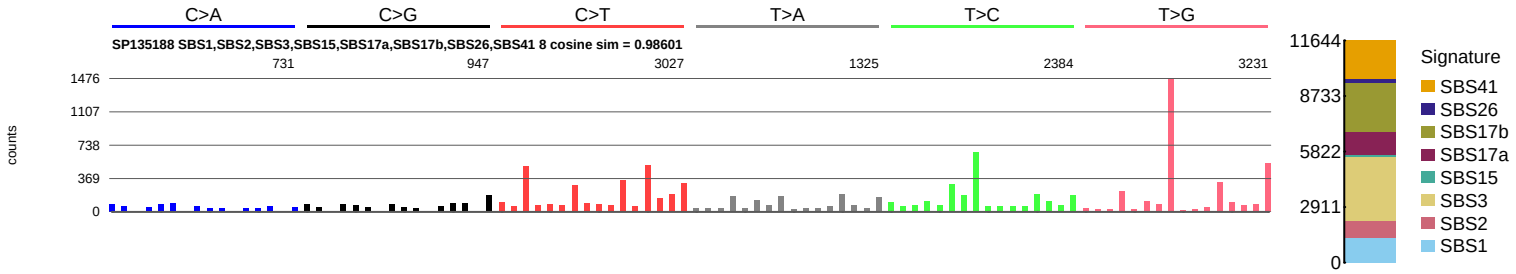
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98608
731
954
3003
1333
2405
3226
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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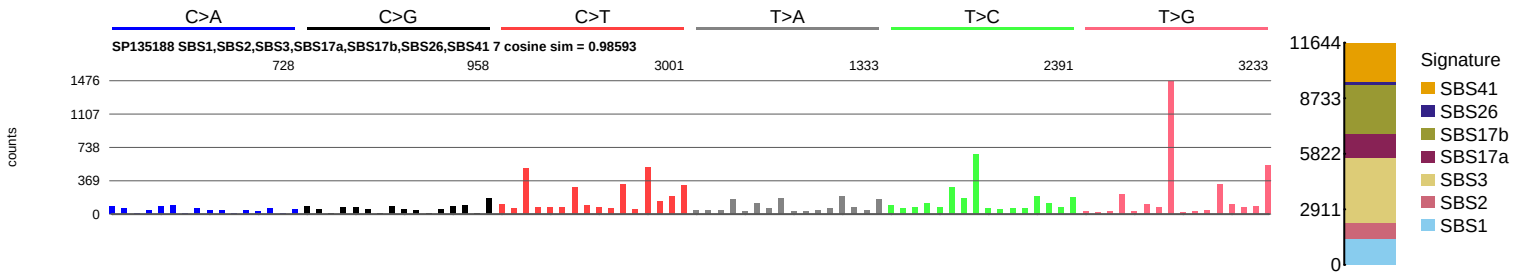
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98601
731
947
3027
1325
2384
3231
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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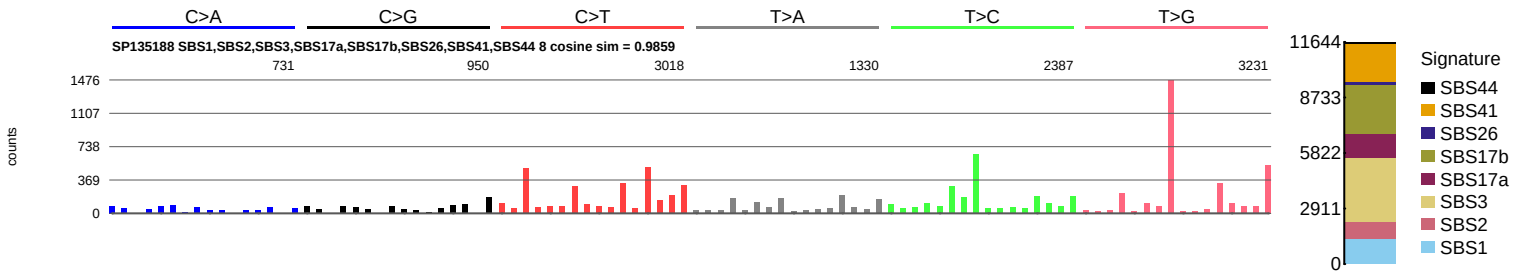
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98593
728
958
3001
1333
2391
3233
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1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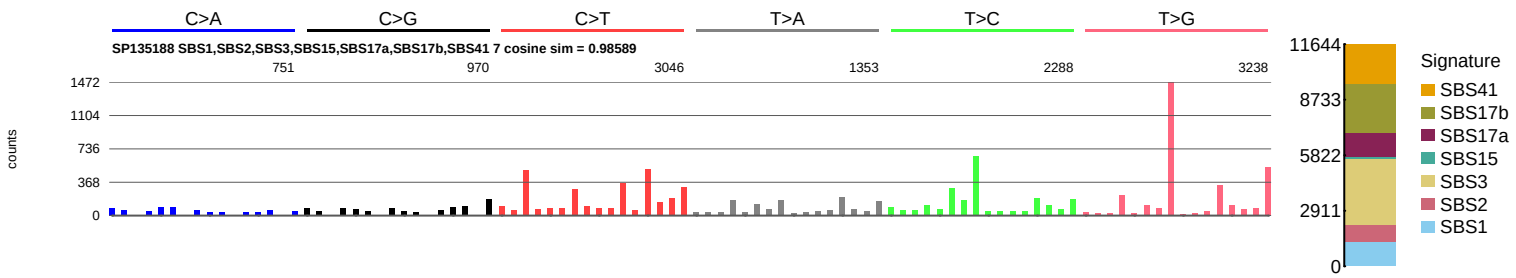
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.9859
731
950
3018
1330
2387
3231
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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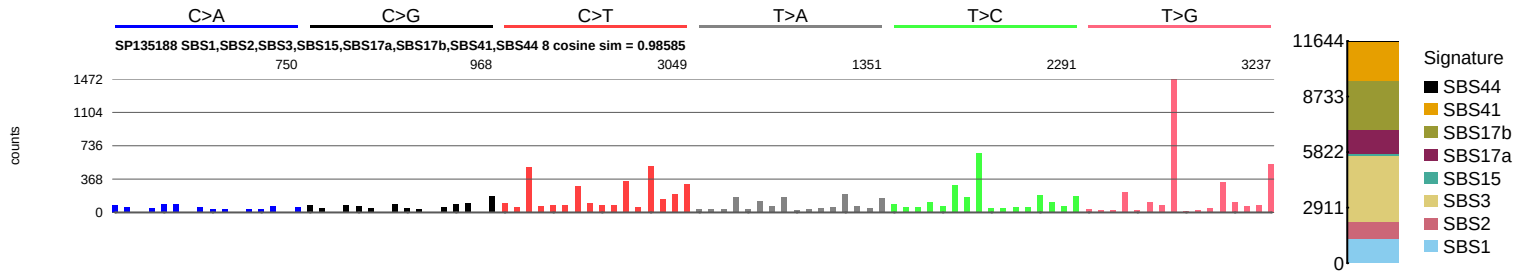
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98589
751
970
3046
1353
2288
3238
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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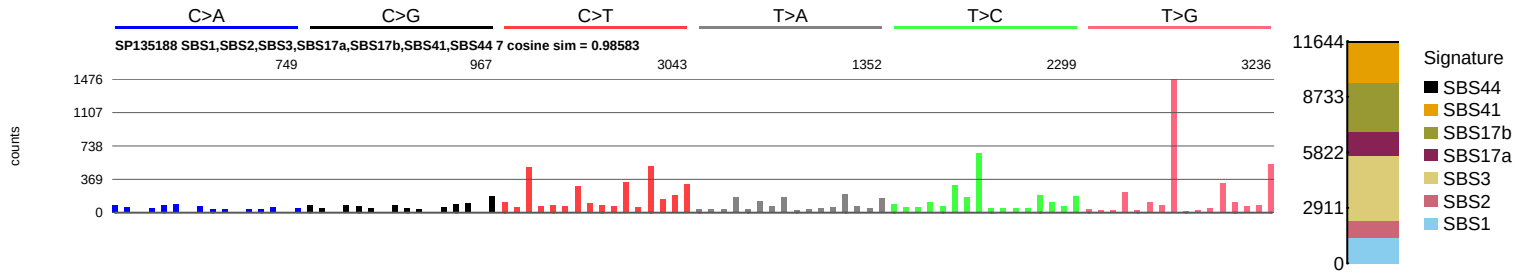
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98585
750
968
3049
1351
2291
3237
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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T>A
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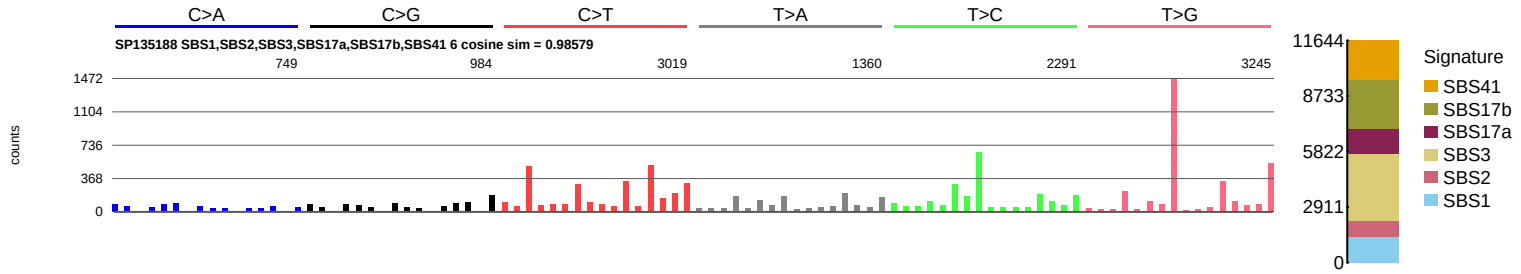
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98583
749
967
3043
1352
2299
3236
counts
1476
1107
738
369
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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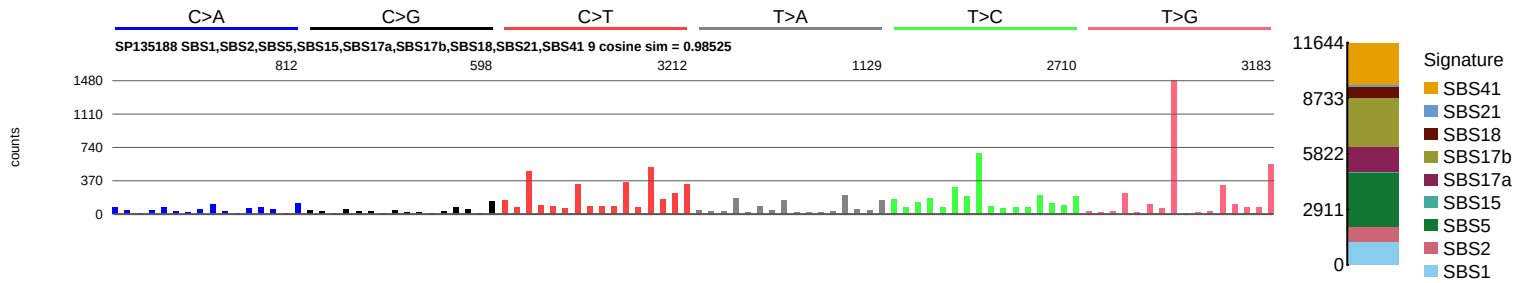
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98579
749
984
3019
1360
2291
3245
counts
1472
1104
736
368
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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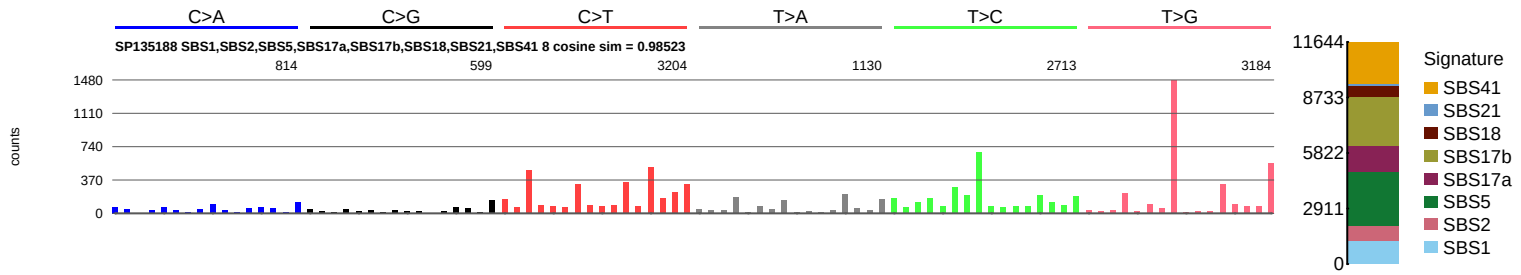
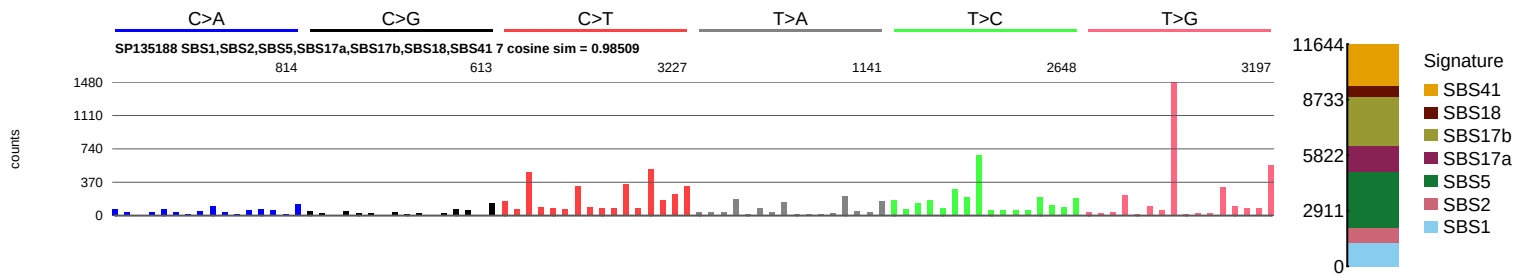
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98525
812
598
3212
1129
2710
3183
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98523
814
599
3204
1130
2713
3184
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98509
814
613
3227
1141
2648
3197
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
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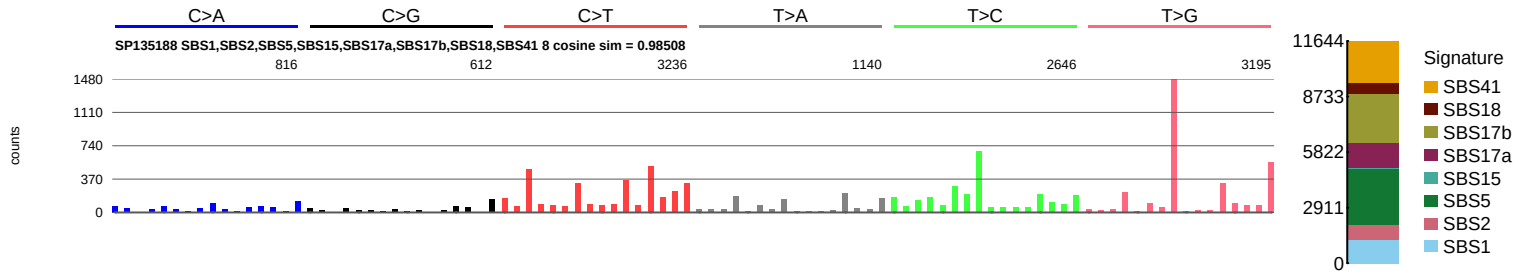
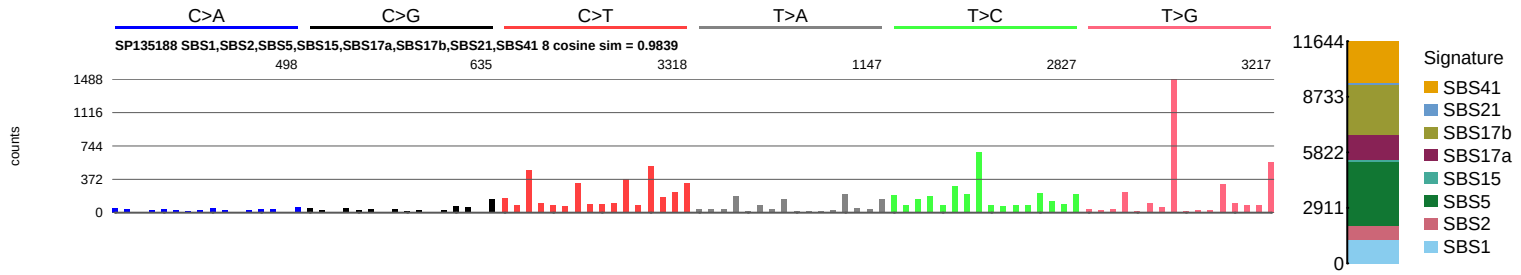
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98508
816
612
3236
1140
2646
3195
counts
1480
1110
740
370
0
11644
8733
5822
2911
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9839
498
635
3318
1147
2827
3217
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
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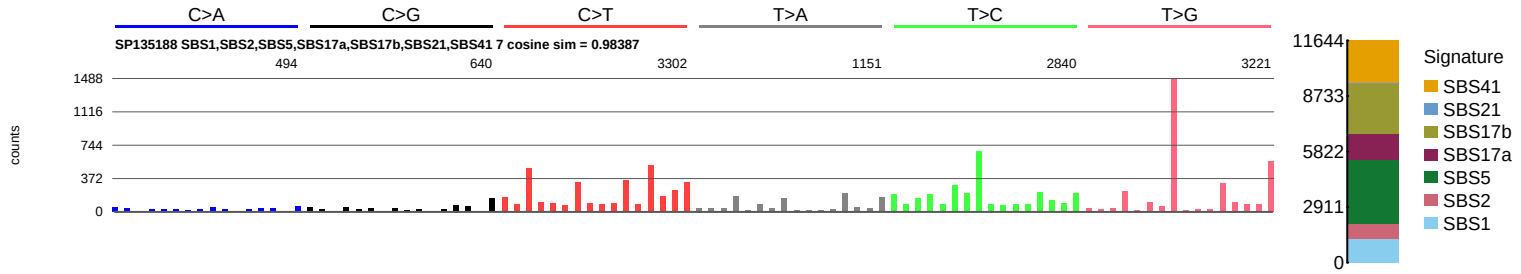
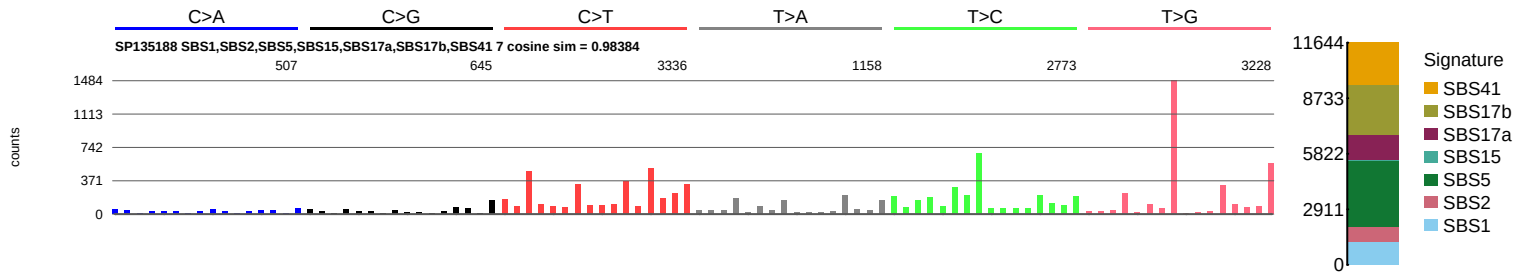
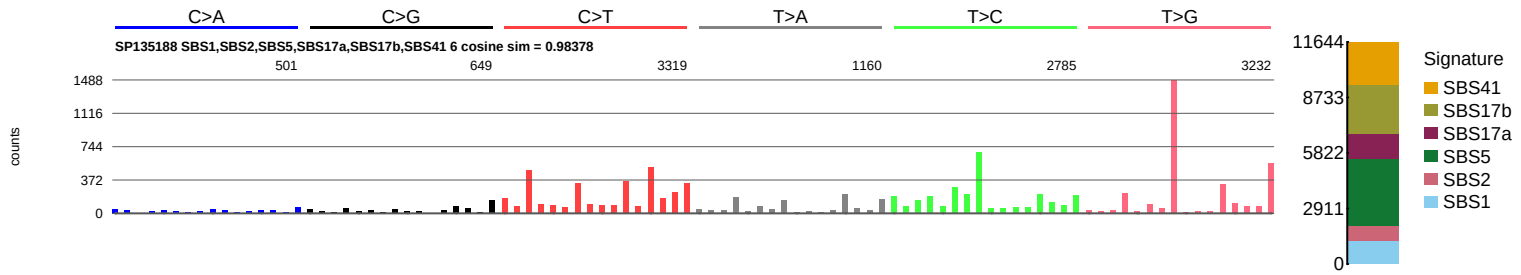
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98387
494
640
3302
1151
2840
3221
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98384
507
645
3336
1158
2773
3228
counts
1484
1113
742
371
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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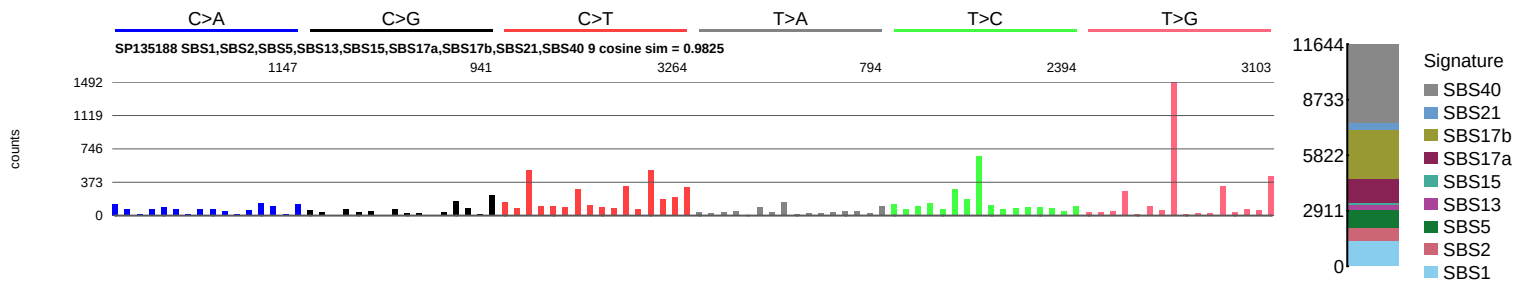
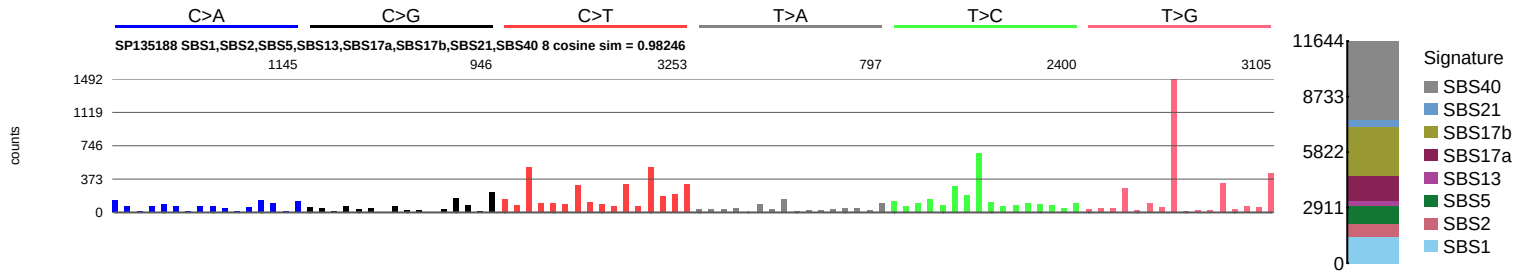
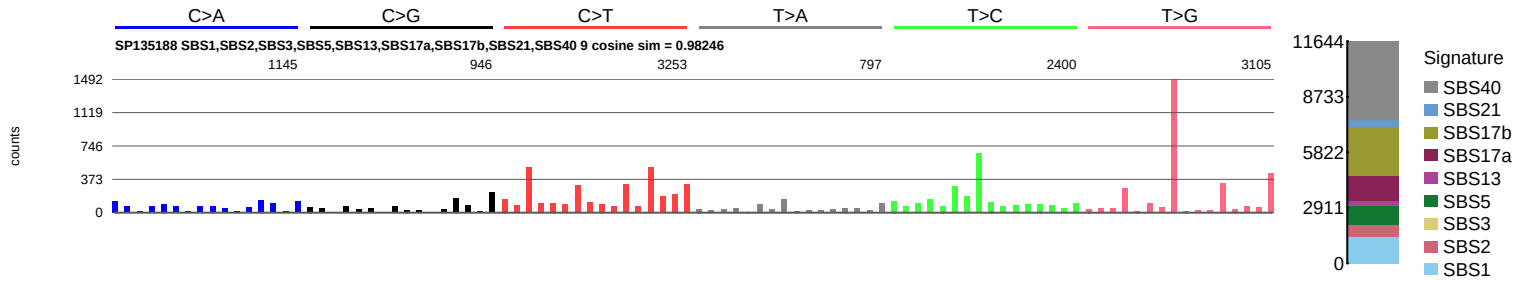
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98378
501
649
3319
1160
2785
3232
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.9825
1147
941
3264
794
2394
3103
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98246
1145
946
3253
797
2400
3105
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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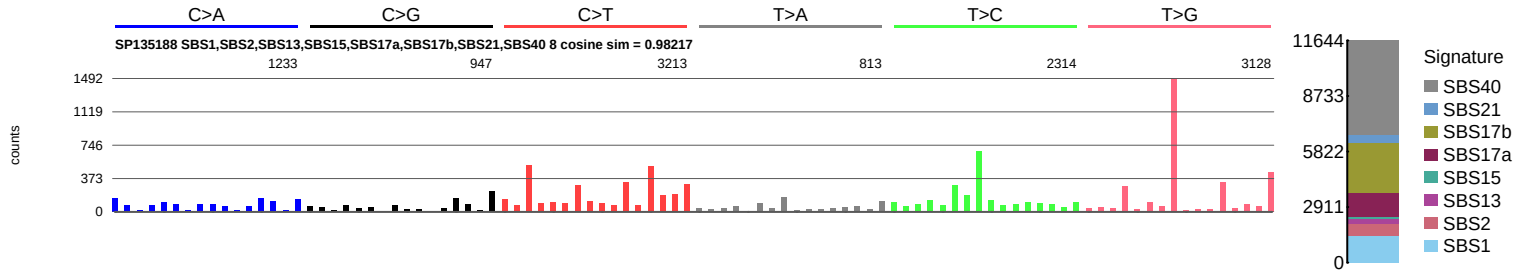
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98246
1145
946
3253
797
2400
3105
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98217
1233
947
3213
813
2314
3128
counts
1492
1119
746
373
0
11644
8733
5822
2911
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Signature
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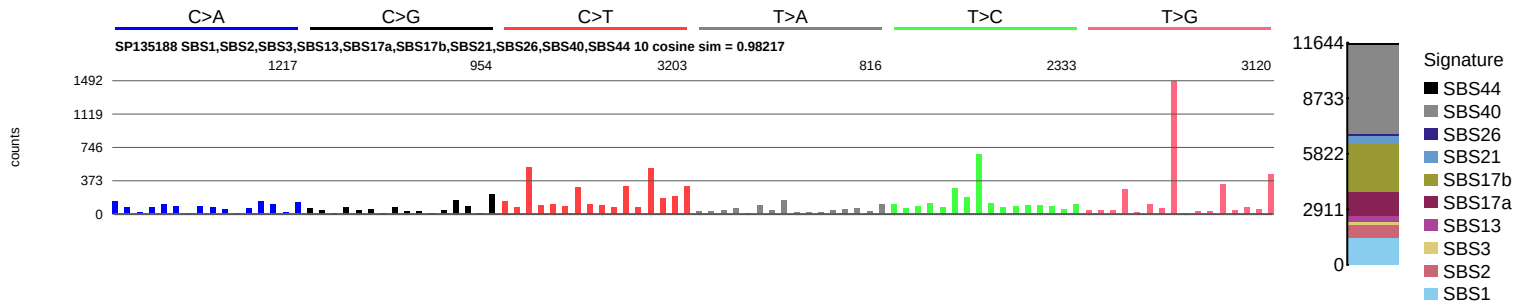
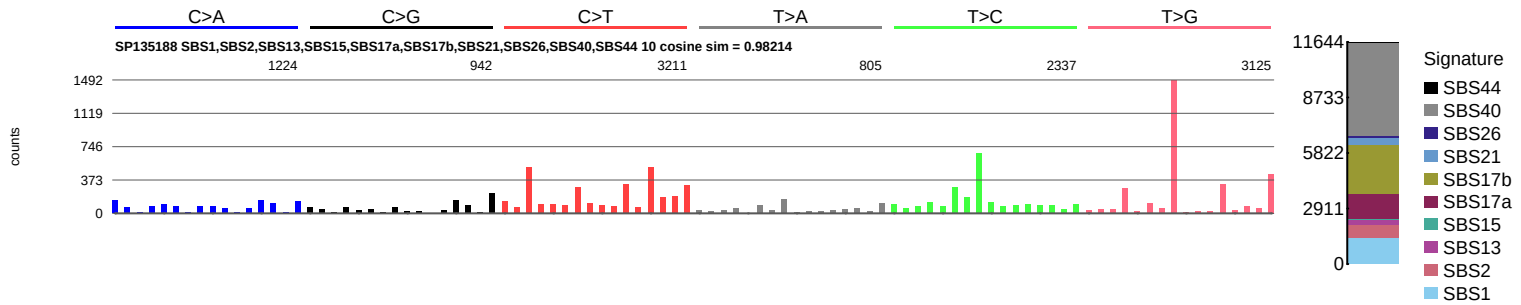
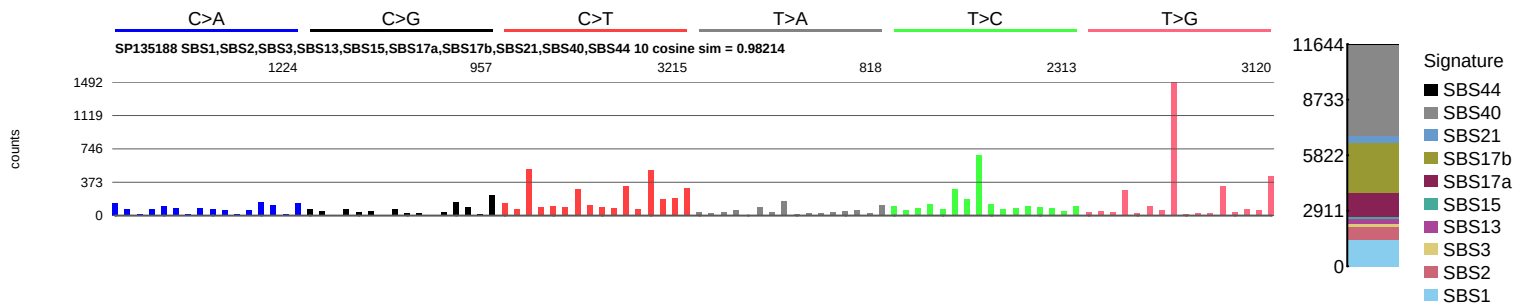
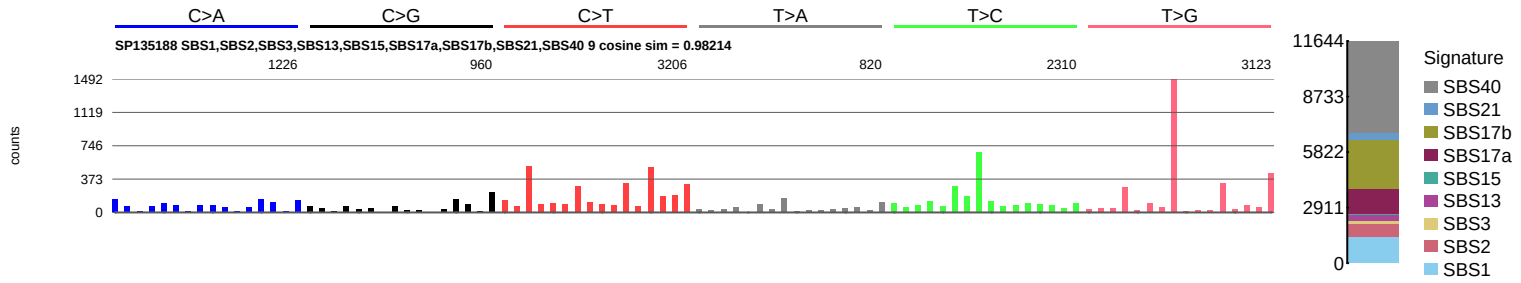
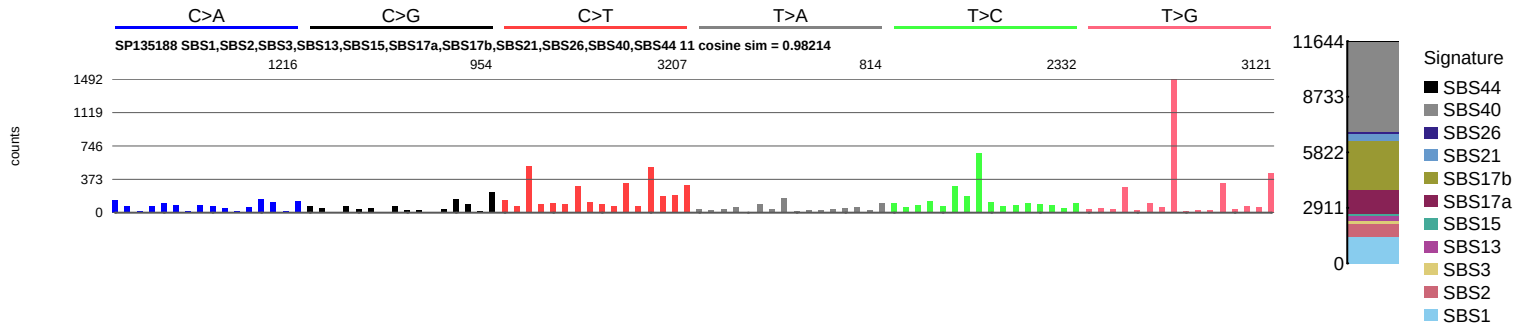
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98217
1217
954
3203
816
2333
3120
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98214
1224
942
3211
805
2337
3125
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98214
1224
957
3215
818
2313
3120
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98214
1226
960
3206
820
2310
3123
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98214
1216
954
3207
814
2332
3121
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
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SBS17b
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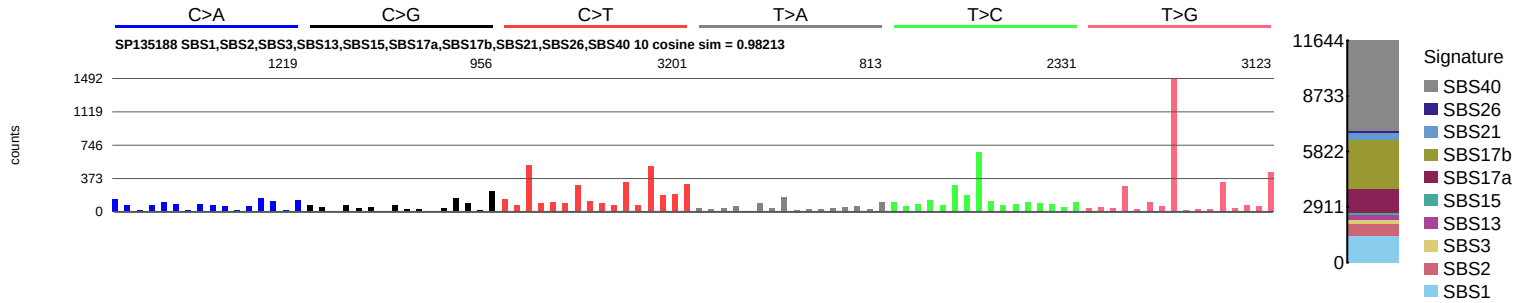
SBS1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T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 10 cosine sim = 0.98213
1219
956
3201
813
2331
3123
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
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SBS17b
SBS17a
SBS15
SBS13
SBS3
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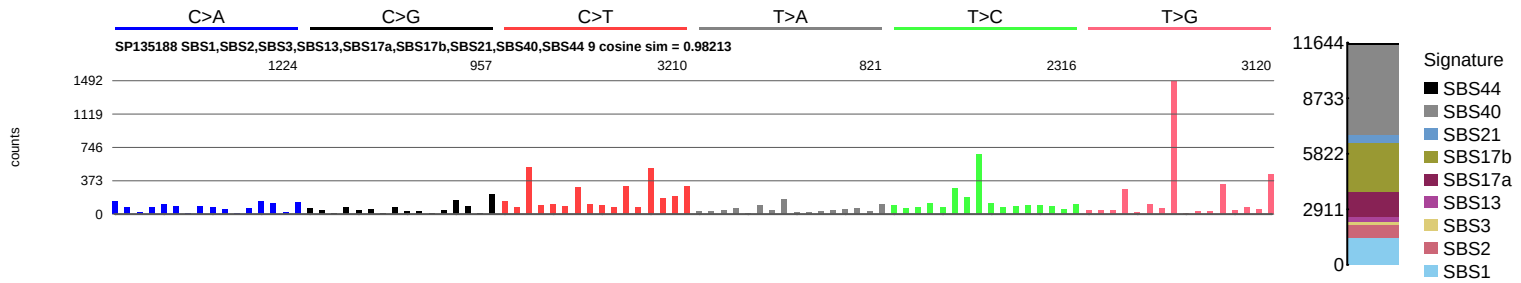
SBS1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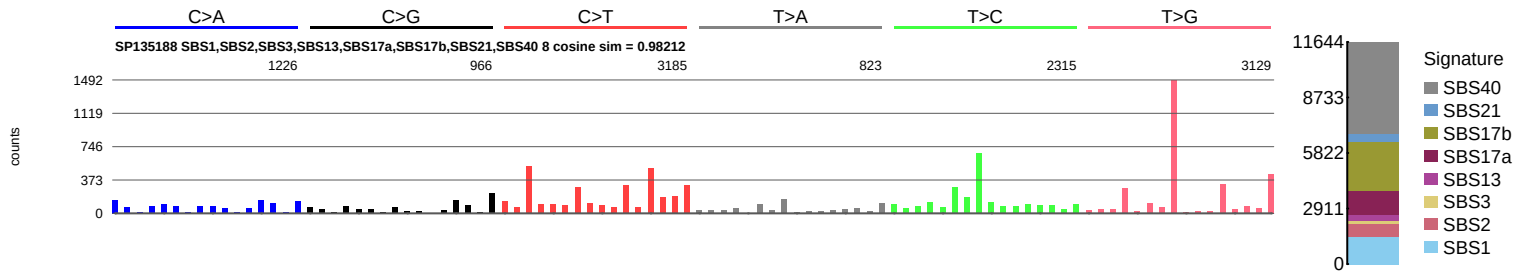
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98213
1224
957
3210
821
2316
3120
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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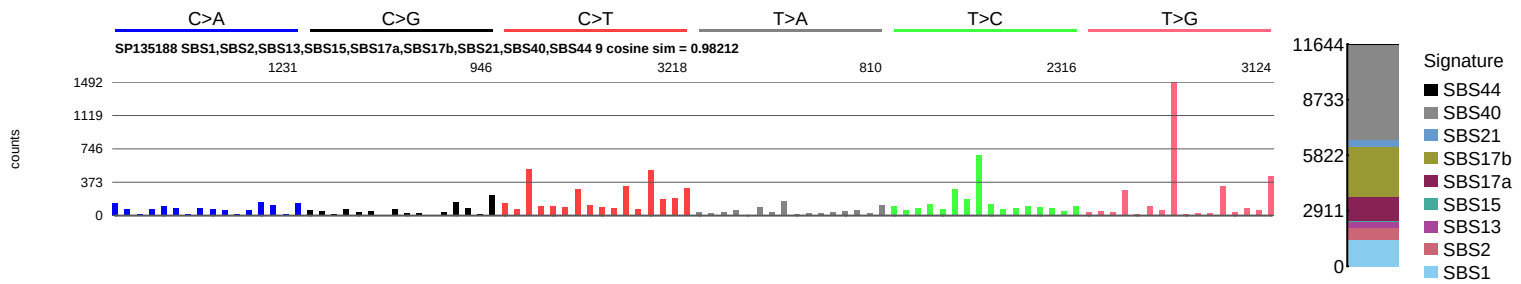
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98212
1226
966
3185
823
2315
3129
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98212
1231
946
3218
810
2316
3124
counts
1492
1119
746
373
0
11644
8733
5822
2911
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SBS44
SBS40
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SBS17b
SBS17a
SBS15
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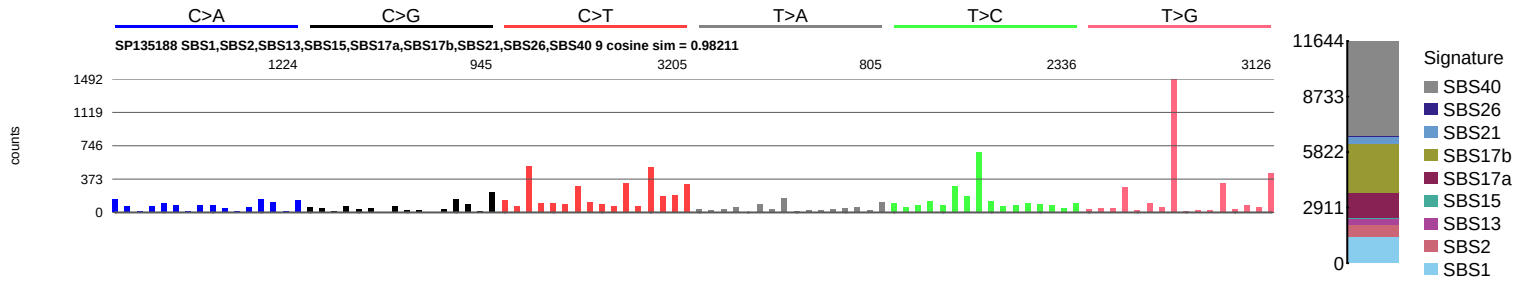
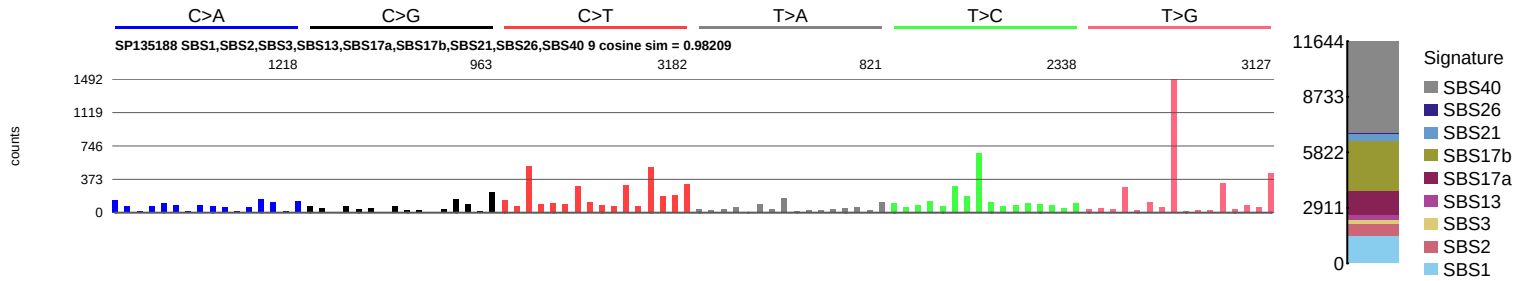
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98211
1224
945
3205
805
2336
3126
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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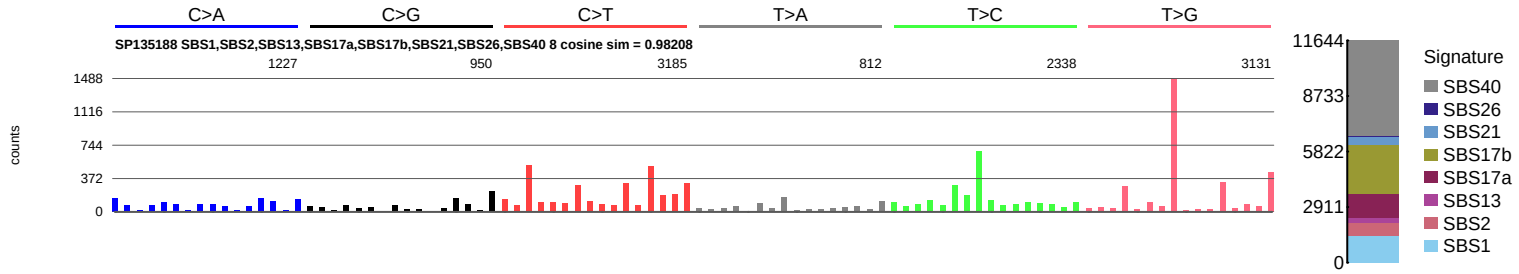
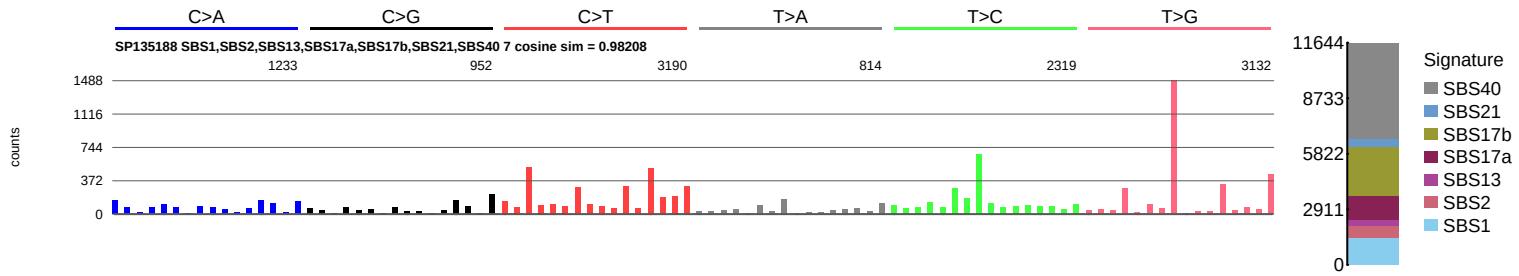
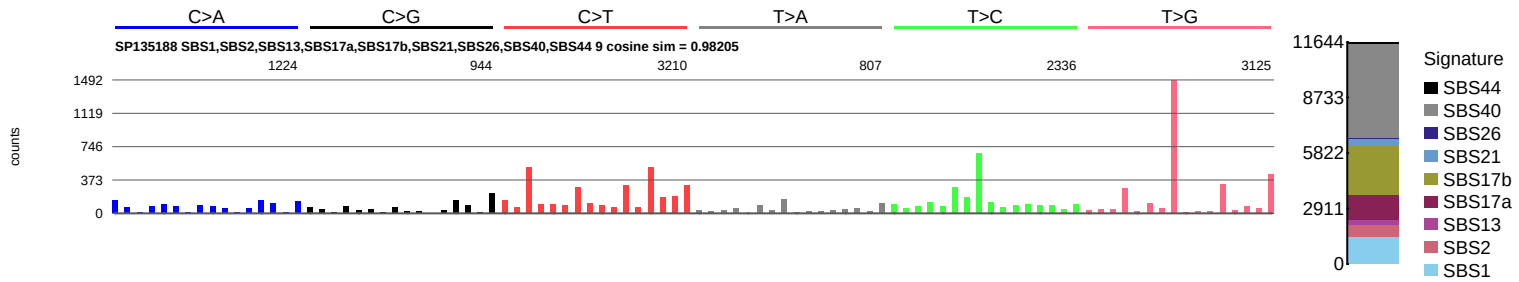
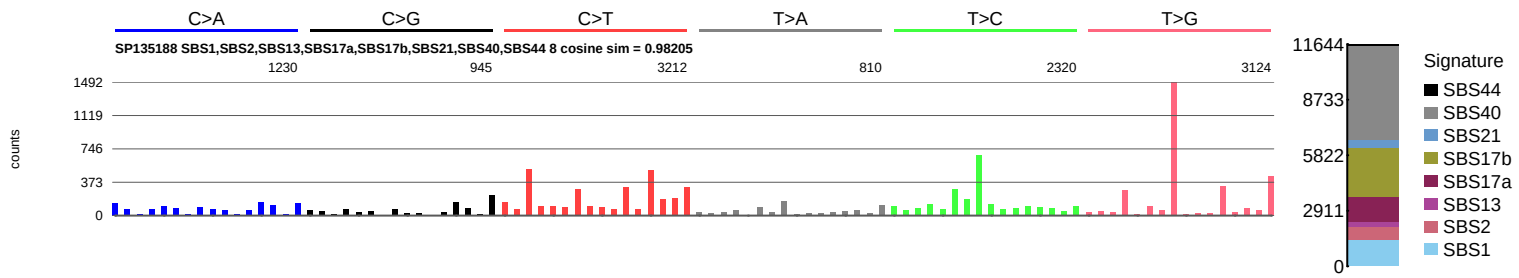
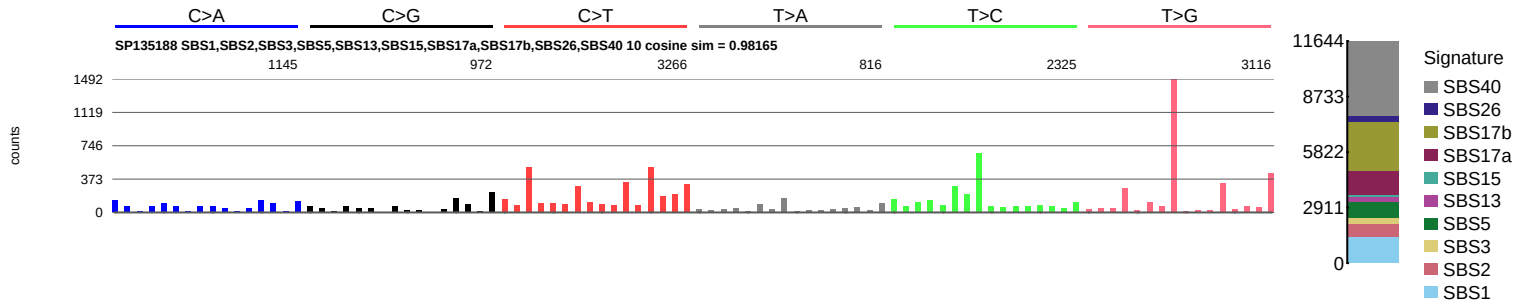
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98209
1218
963
3182
821
2338
3127
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

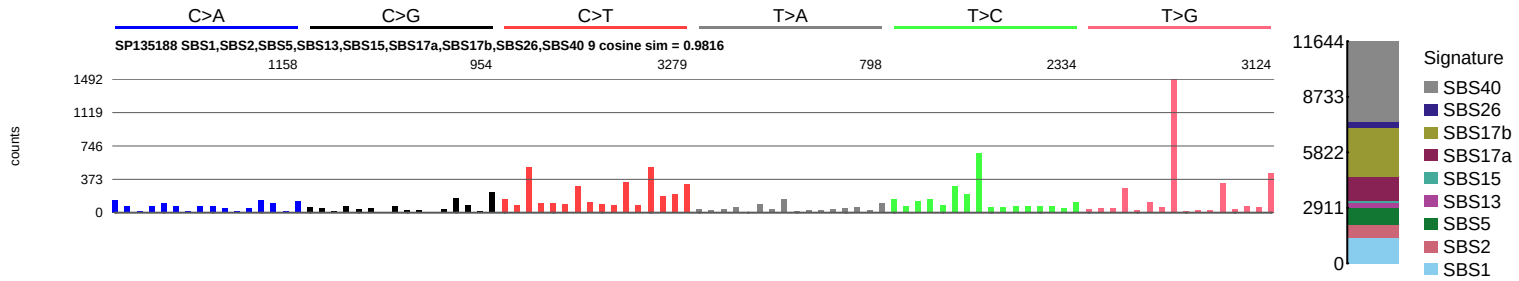

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98208
1227
950
3185
812
2338
3131
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

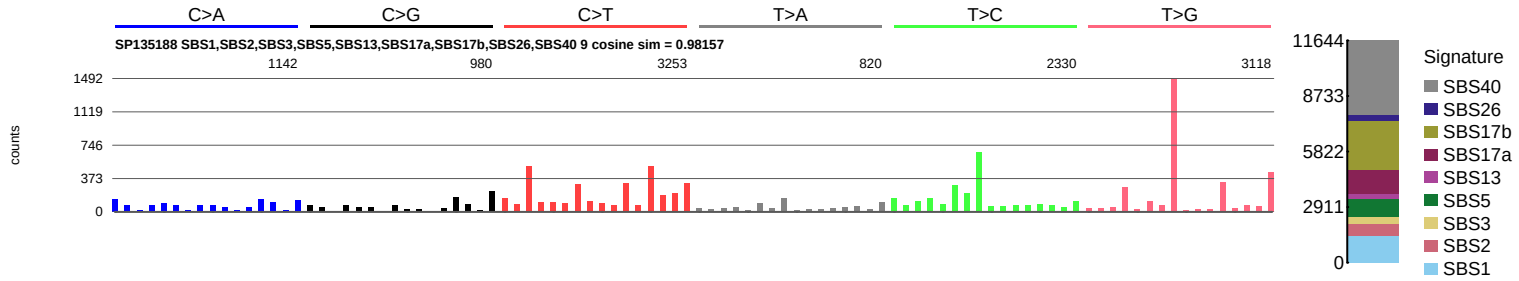

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98208
1233
952
3190
814
2319
3132
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

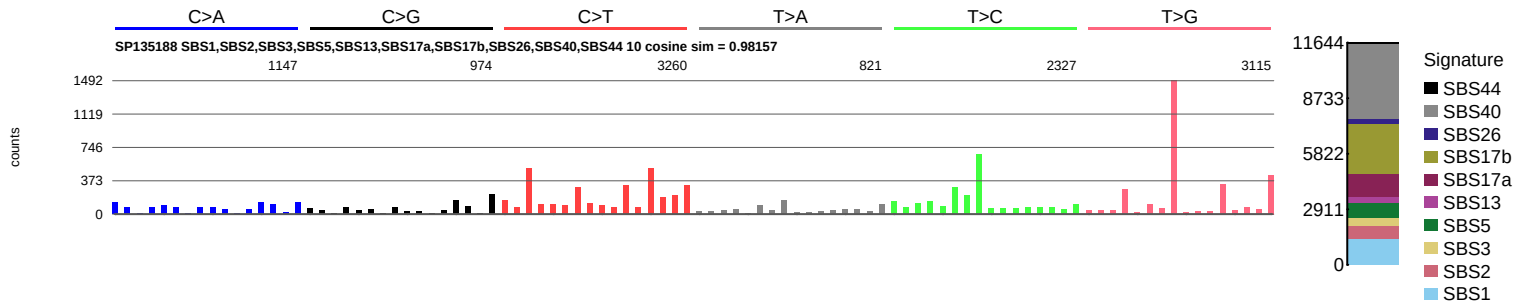

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98205
1224
944
3210
807
2336
3125
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

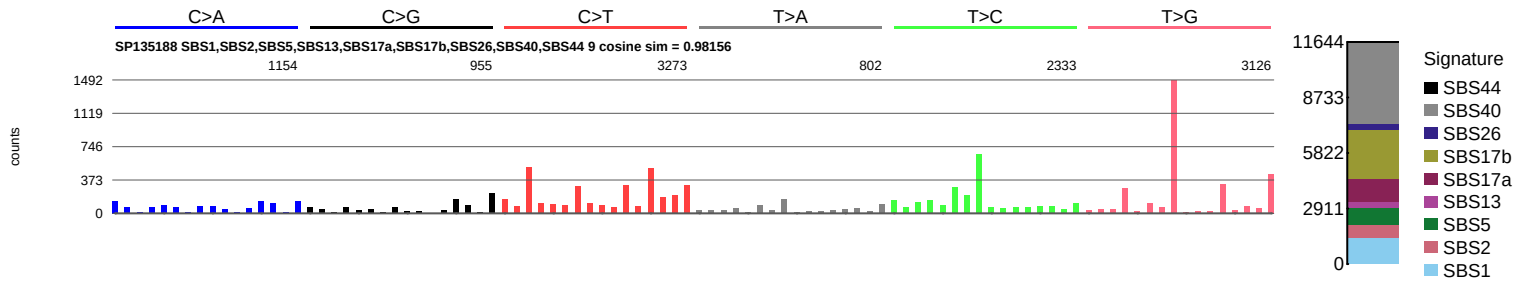

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98205
1230
945
3212
810
2320
3124
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

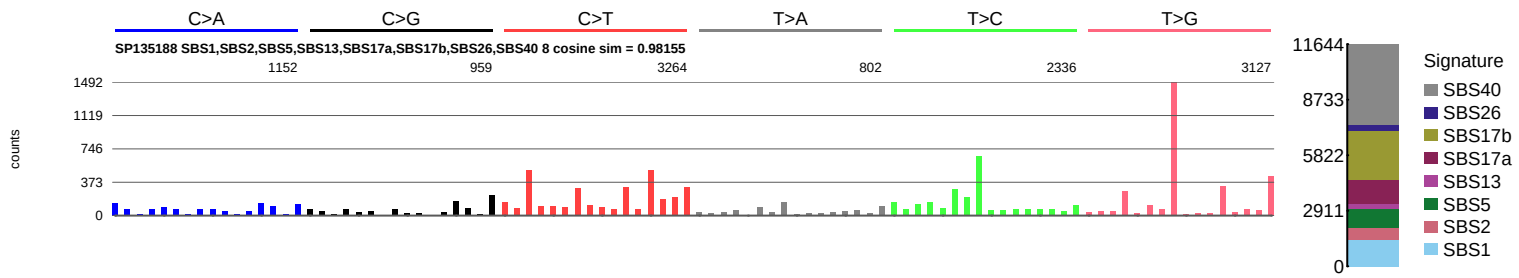

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 10 cosine sim = 0.98165
1145
972
3266
816
2325
3116
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

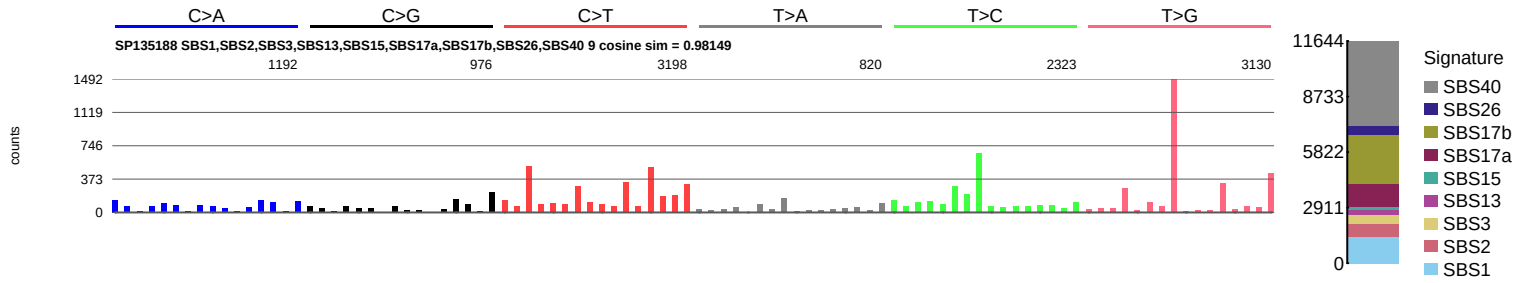

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.9816
1158
954
3279
798
2334
3124
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

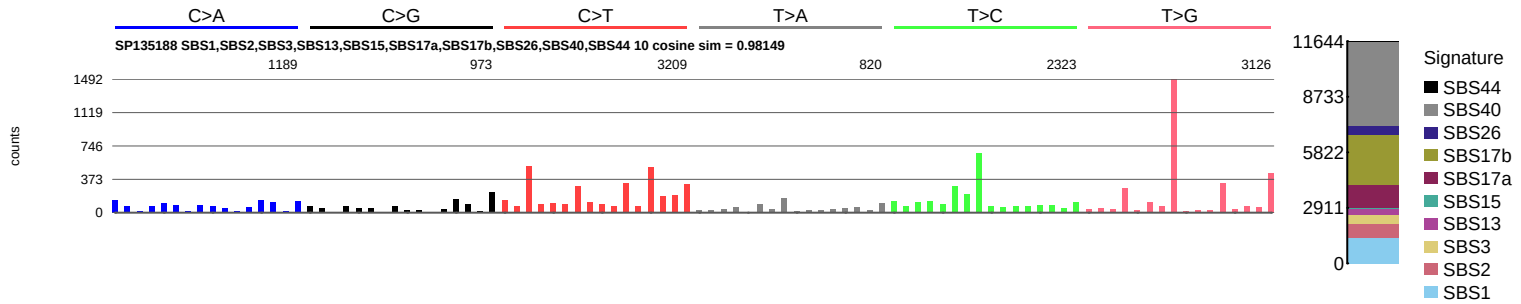

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98157
1142
980
3253
820
2330
3118
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

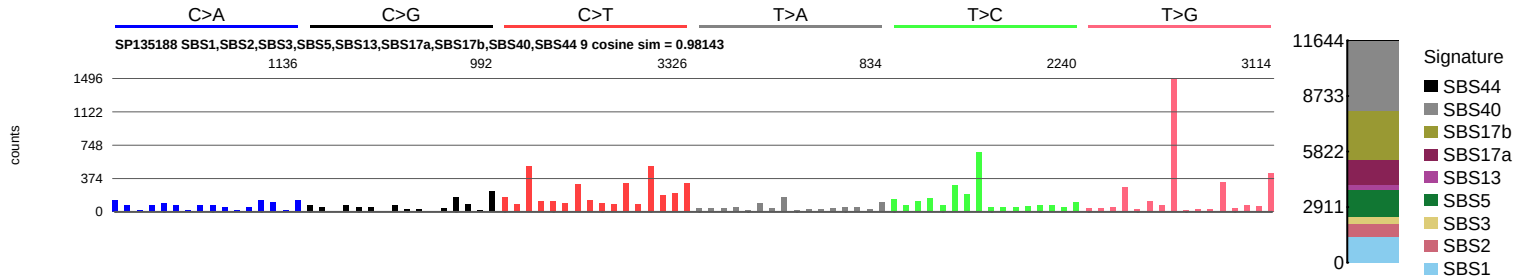

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98157
1147
974
3260
821
2327
3115
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

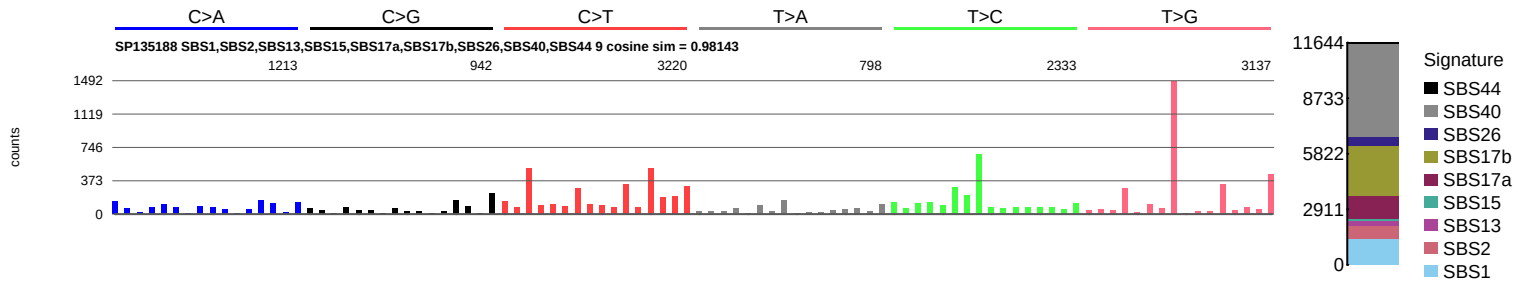

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98156
1154
955
3273
802
2333
3126
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

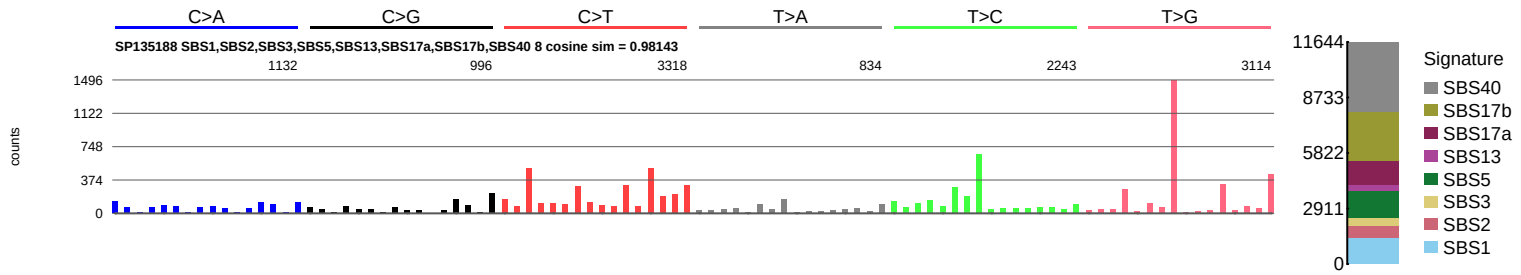

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98155
1152
959
3264
802
2336
3127
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98149
1192
976
3198
820
2323
3130
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98149
1189
973
3209
820
2323
3126
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98143
1136
992
3326
834
2240
3114
counts
1496
1122
748
374
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98143
1213
942
3220
798
2333
3137
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98143
1132
996
3318
834
2243
3114
counts
1496
1122
748
374
0
11644
8733
5822
2911
0
Signature
SBS40
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SBS17a
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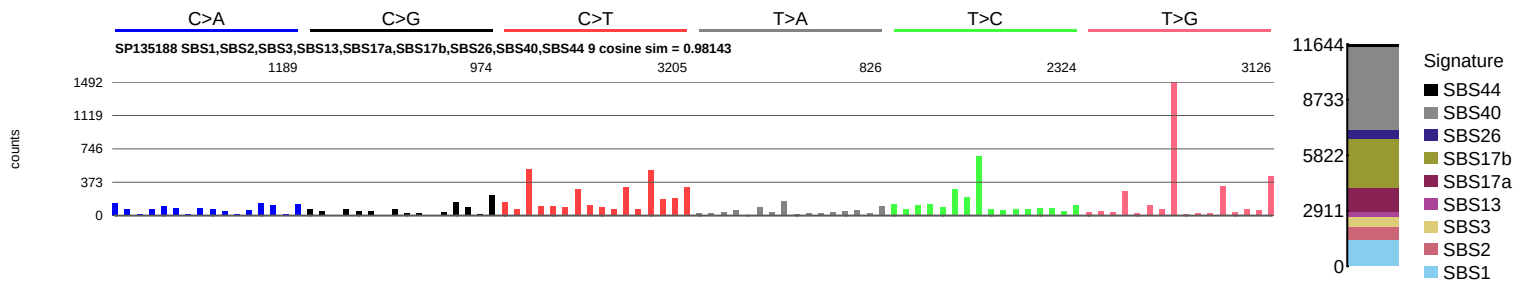
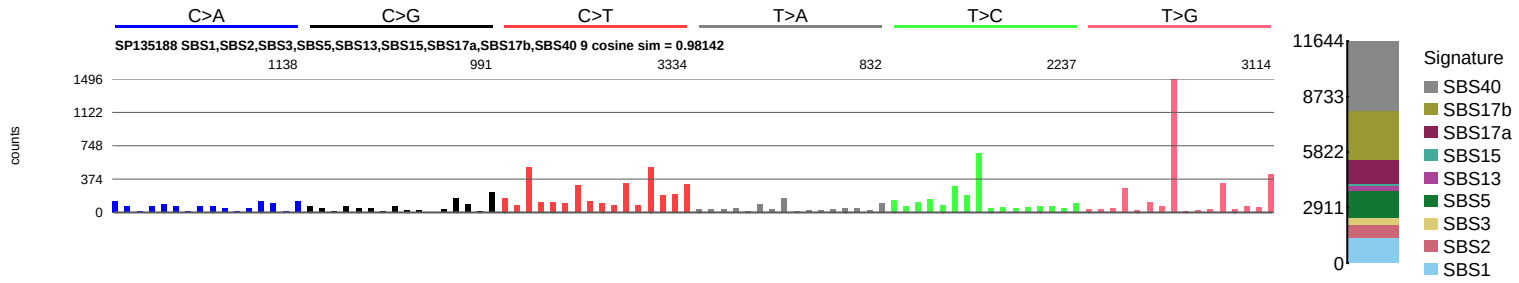
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98143
1189
974
3205
826
2324
3126
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 9 cosine sim = 0.98142
1138
991
3334
832
2237
3114
counts
1496
1122
748
374
0
11644
8733
5822
2911
0
Signature
SBS40
SBS17b
SBS17a
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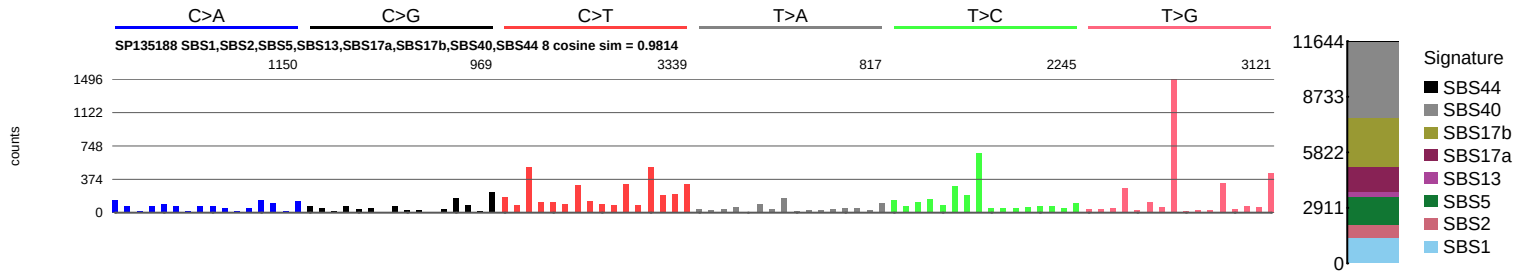
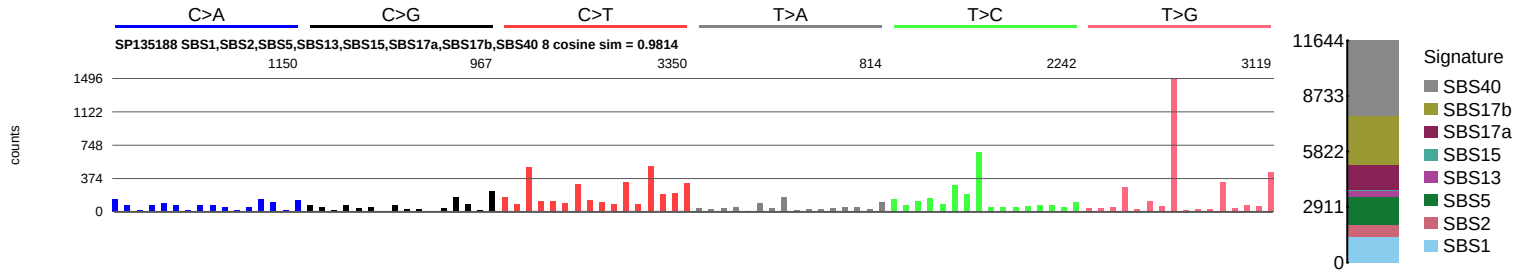
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.9814
1150
969
3339
817
2245
3121
counts
1496
1122
748
374
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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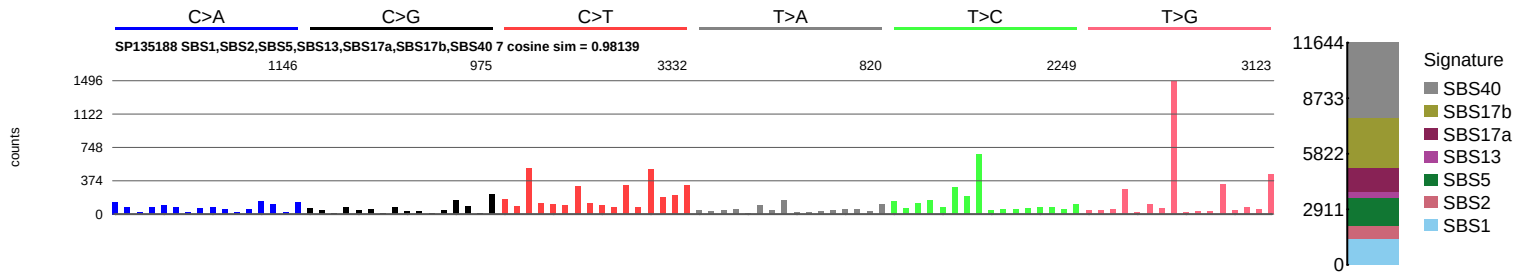
SP135188 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.9814
1150
967
3350
814
2242
3119
counts
1496
1122
748
374
0
11644
8733
5822
2911
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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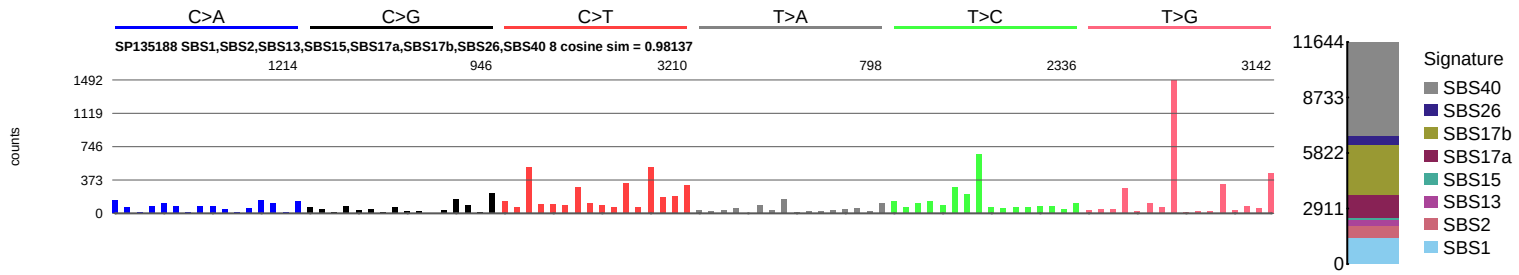
SP135188 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98139
1146
975
3332
820
2249
3123
counts
1496
1122
748
374
0
11644
8733
5822
2911
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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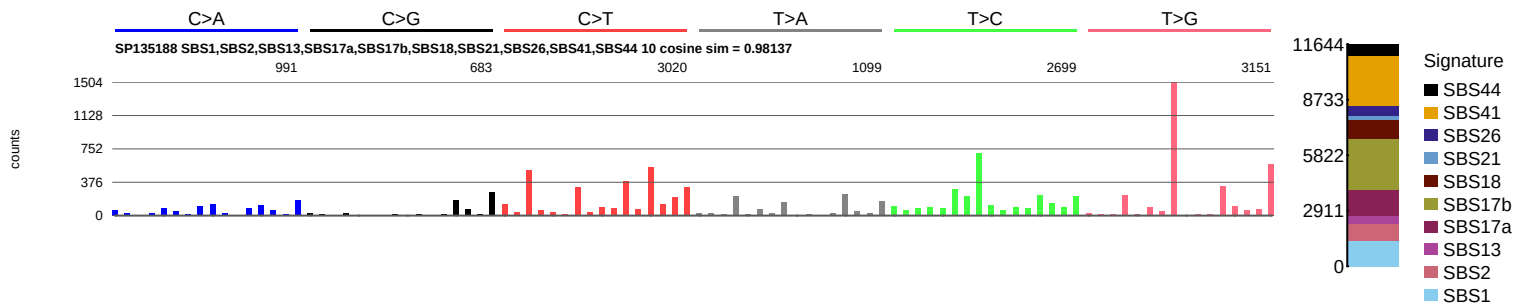
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98137
1214
946
3210
798
2336
3142
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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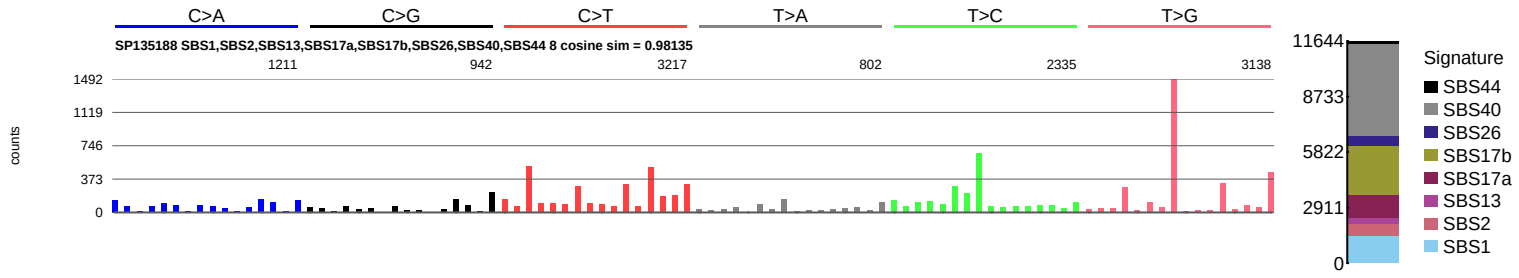
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98137
991
683
3020
1099
2699
3151
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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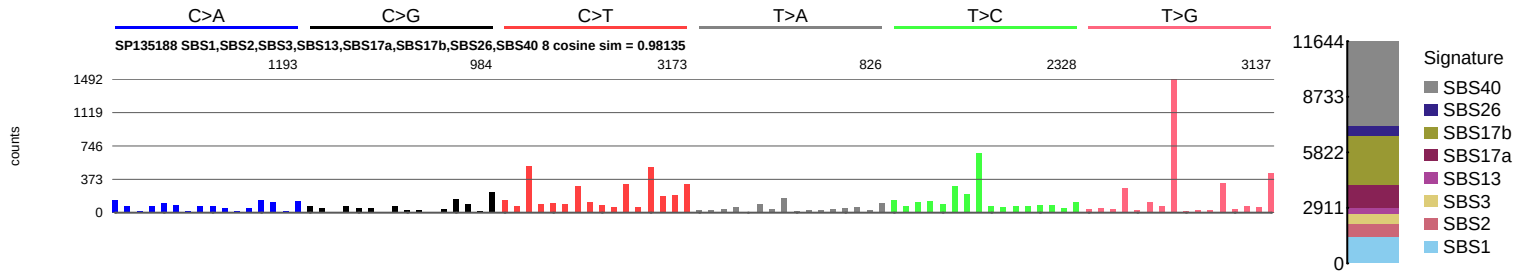
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98135
1211
942
3217
802
2335
3138
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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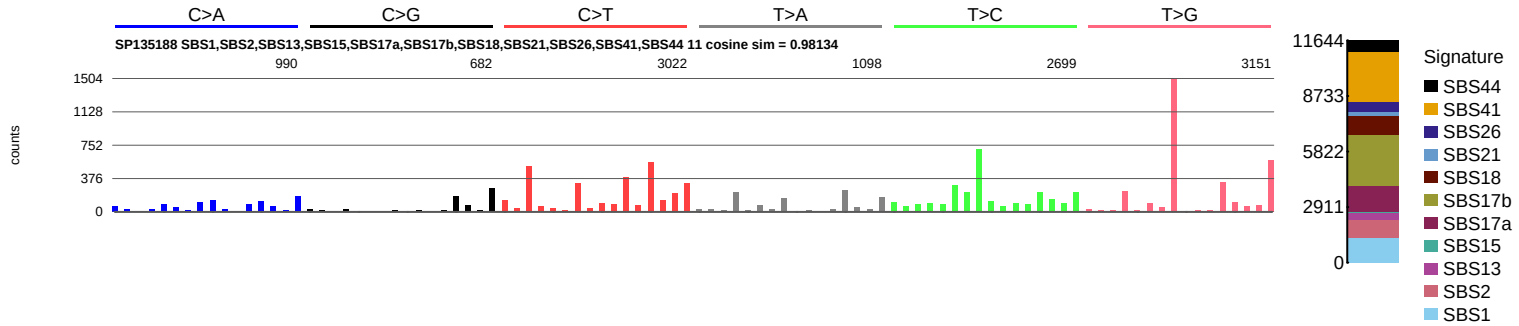
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98135
1193
984
3173
826
2328
3137
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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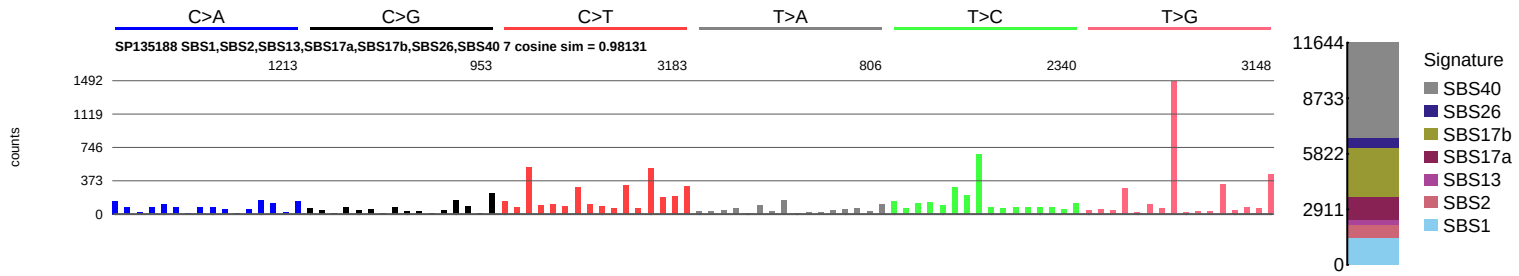
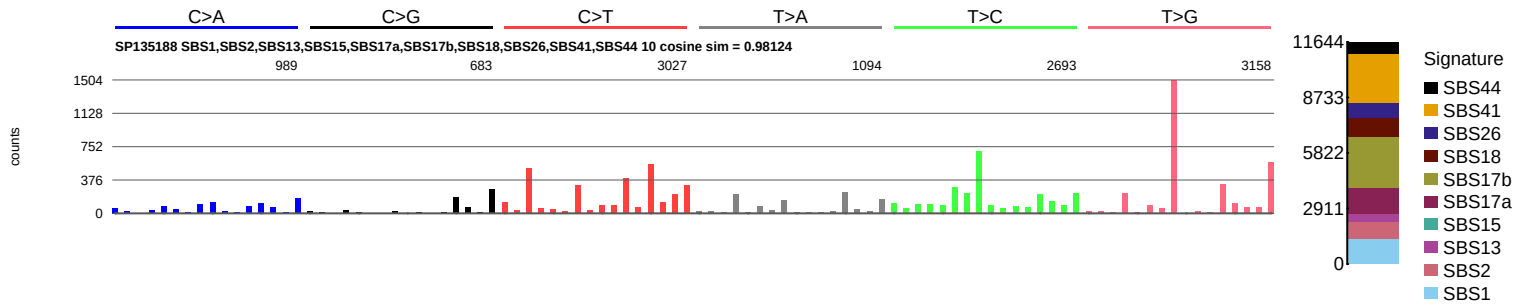
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98134
990
682
3022
1098
2699
3151
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98131
1213
953
3183
806
2340
3148
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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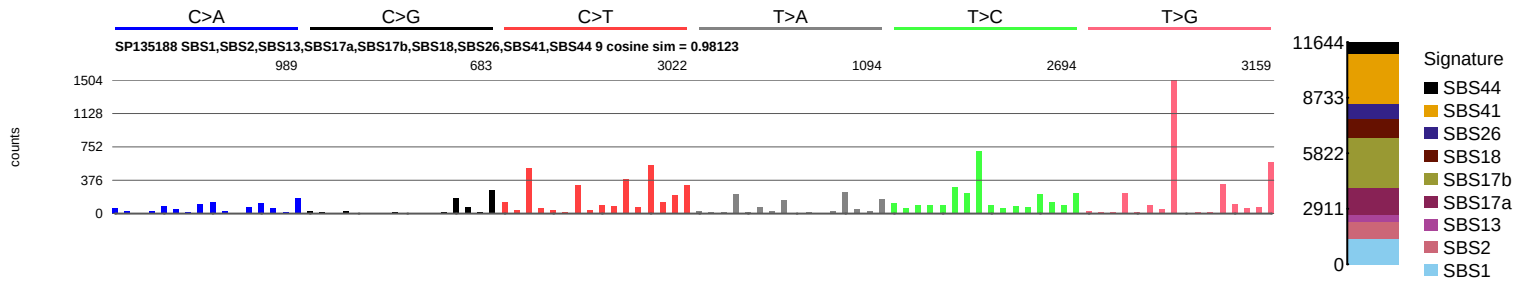
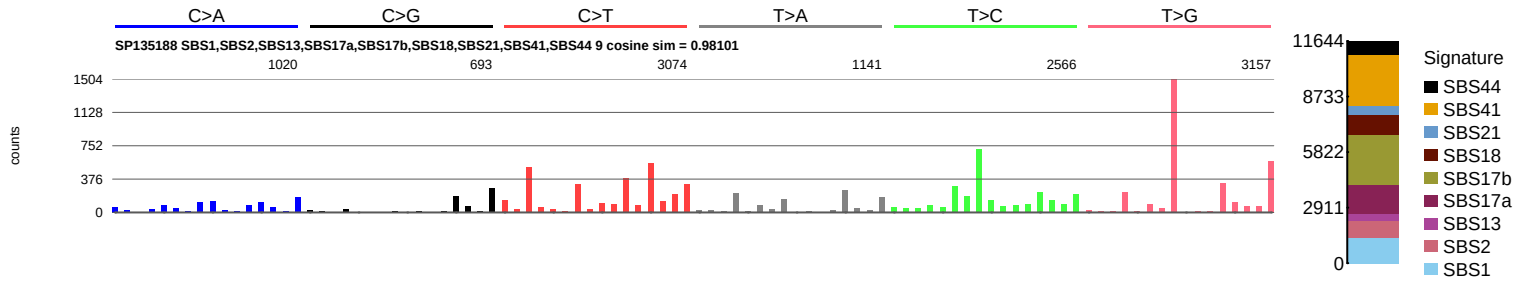
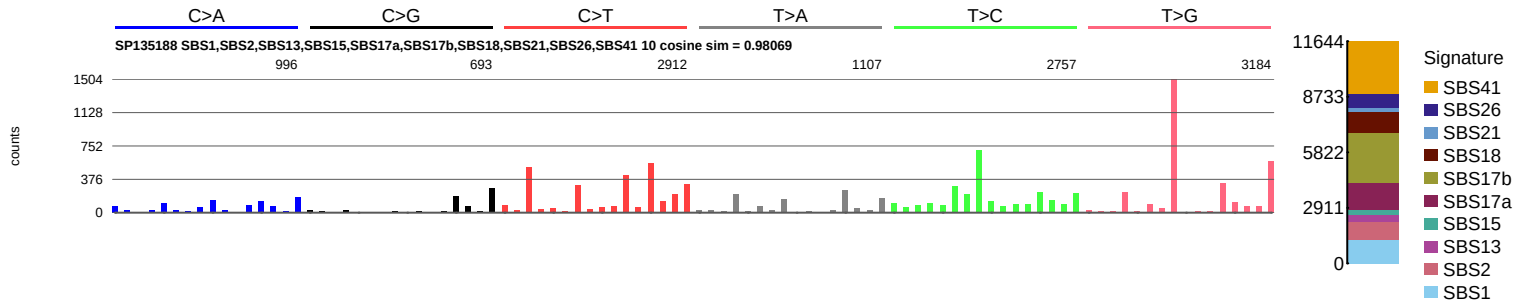
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98124
989
683
3027
1094
2693
3158
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98123
989
683
3022
1094
2694
3159
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98101
1020
693
3074
1141
2566
3157
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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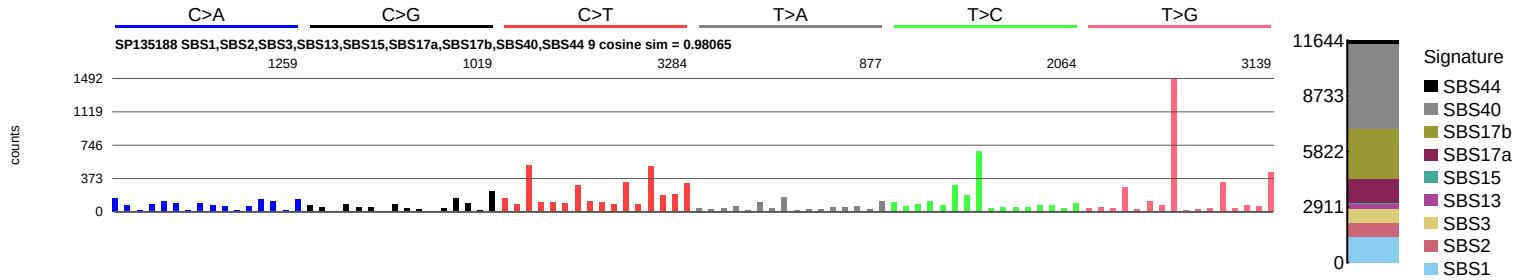
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98069
996
693
2912
1107
2757
3184
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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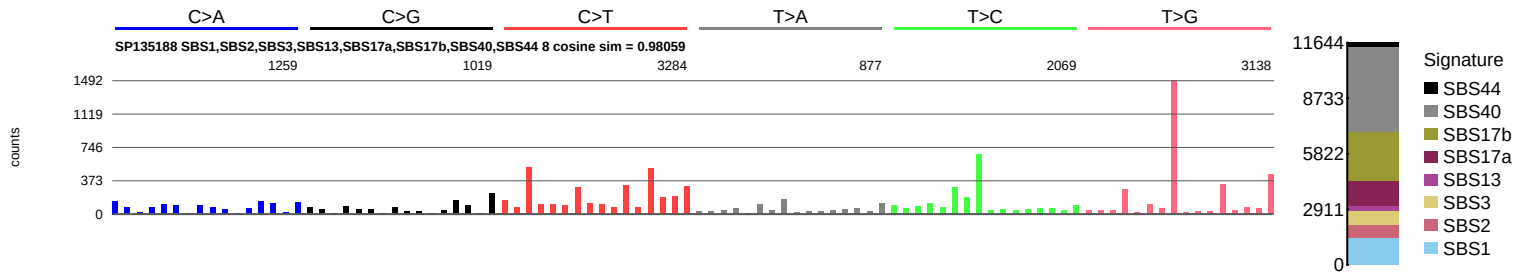
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98065
1259
1019
3284
877
2064
3139
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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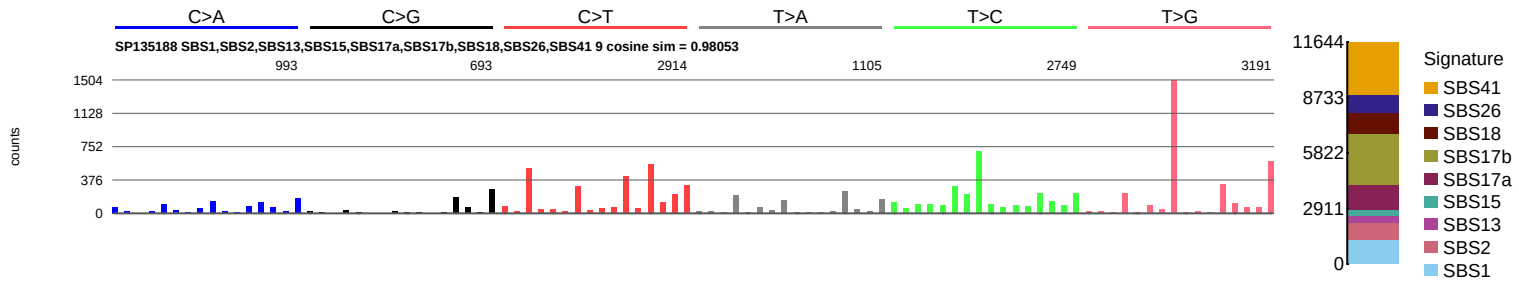
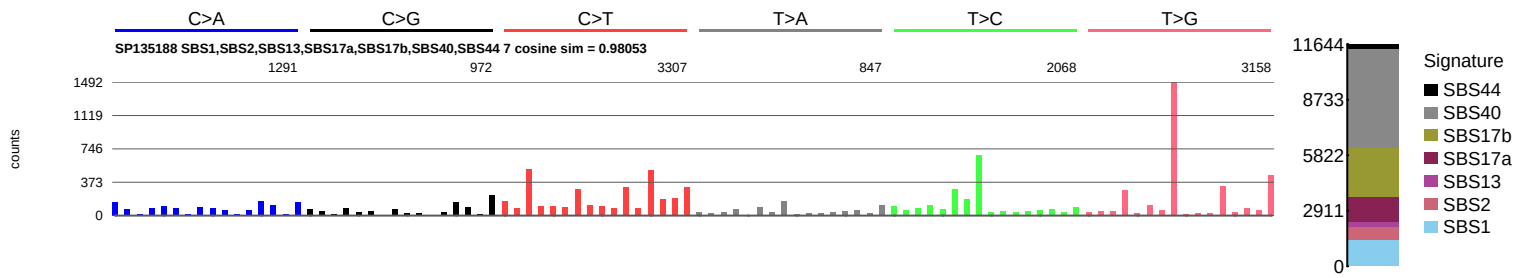
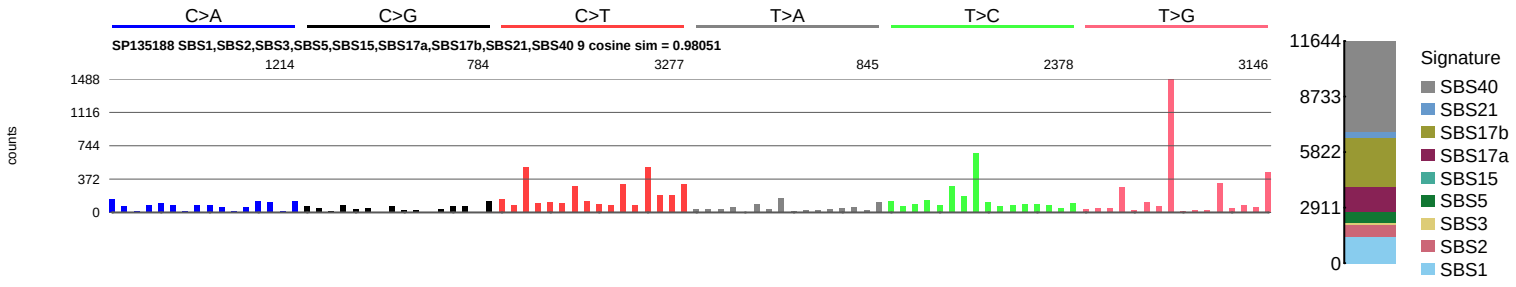
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98059
1259
1019
3284
877
2069
3138
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98053
993
693
2914
1105
2749
3191
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98053
1291
972
3307
847
2068
3158
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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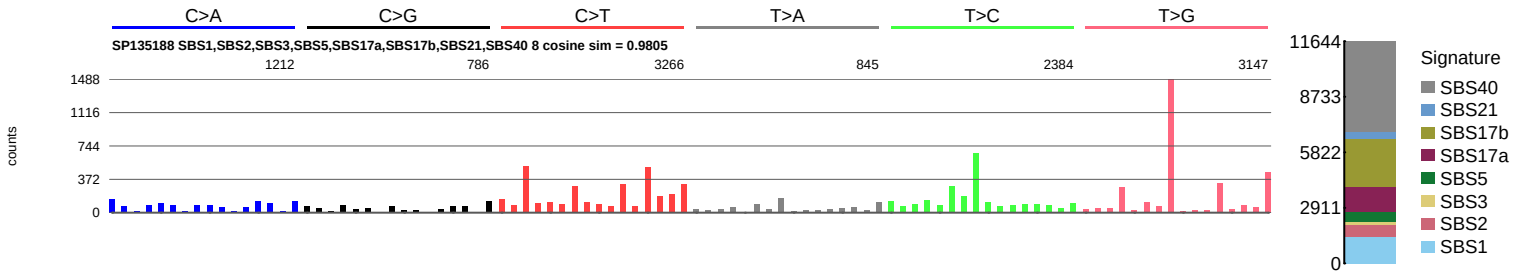
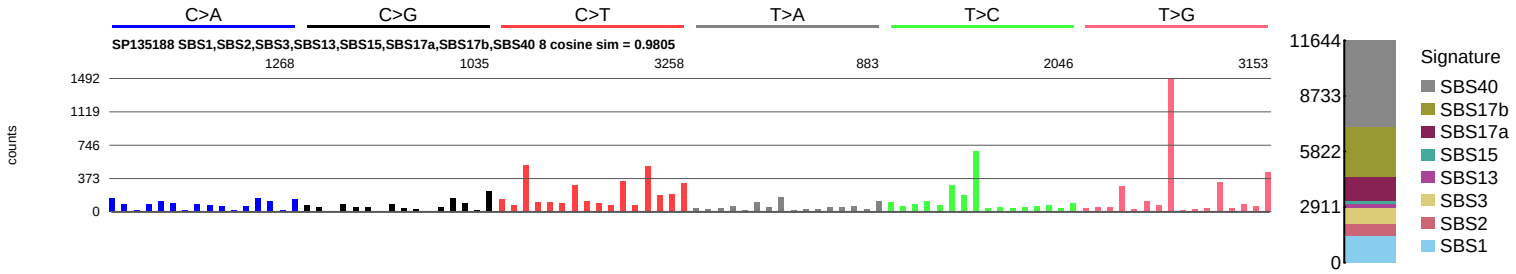
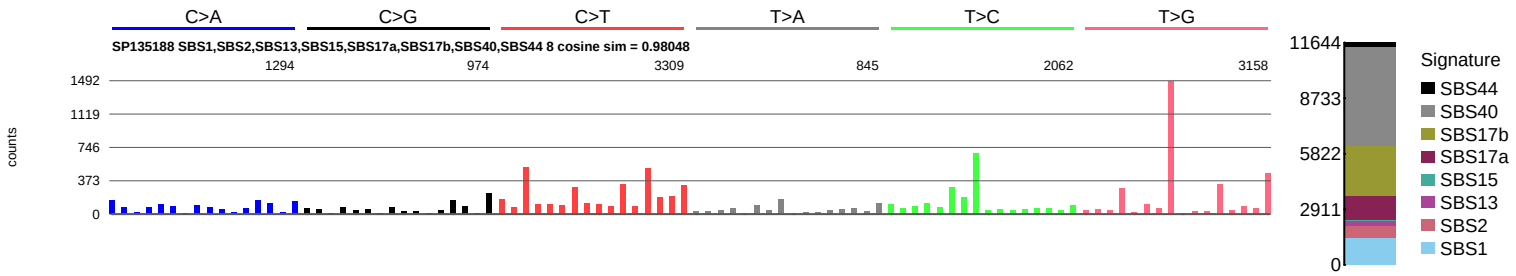
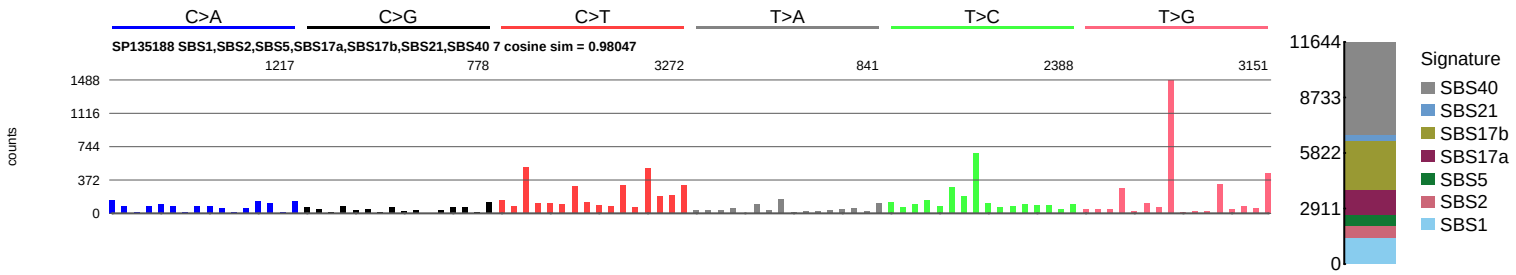
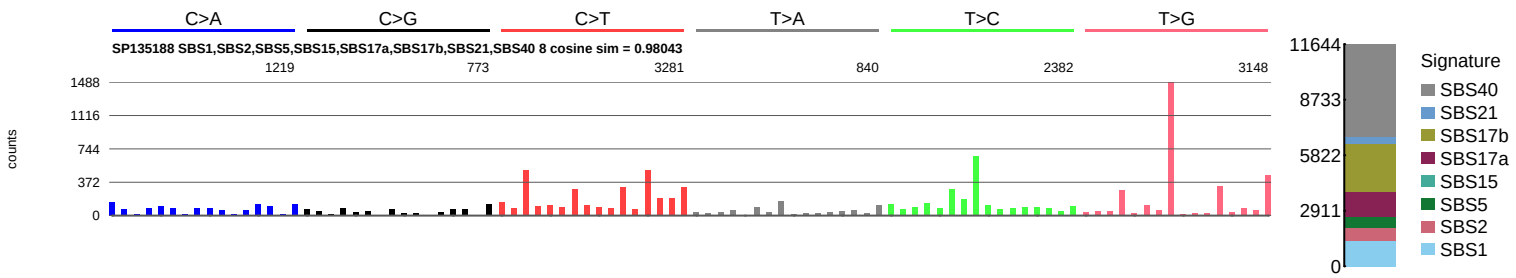
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98051
1214
784
3277
845
2378
3146
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

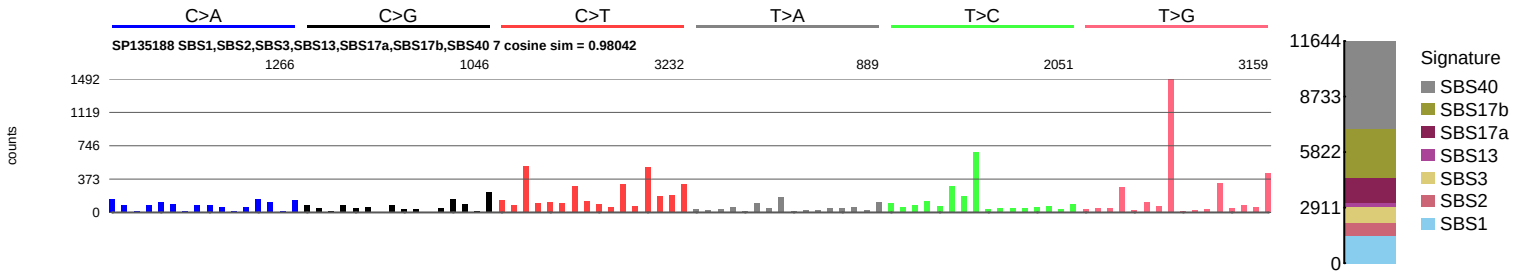

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.9805
1212
786
3266
845
2384
3147
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

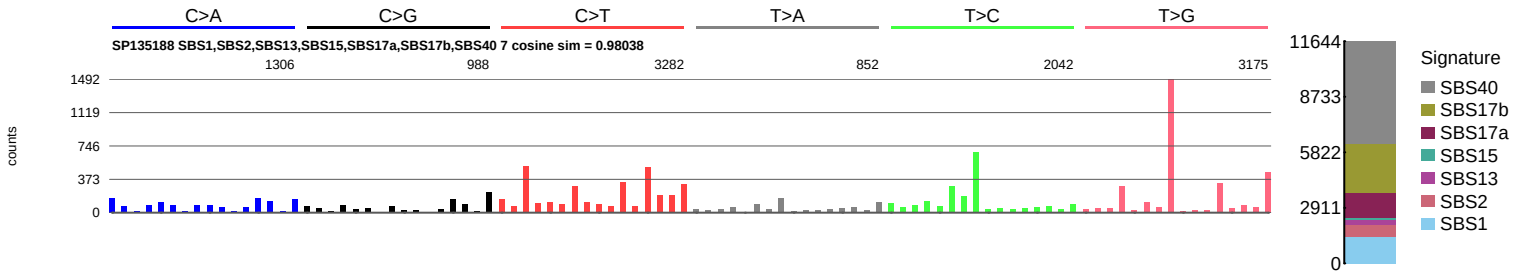

C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.9805
1268
1035
3258
883
2046
3153
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98048
1294
974
3309
845
2062
3158
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98047
1217
778
3272
841
2388
3151
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98043
1219
773
3281
840
2382
3148
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98042
1266
1046
3232
889
2051
3159
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98038
1306
988
3282
852
2042
3175
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
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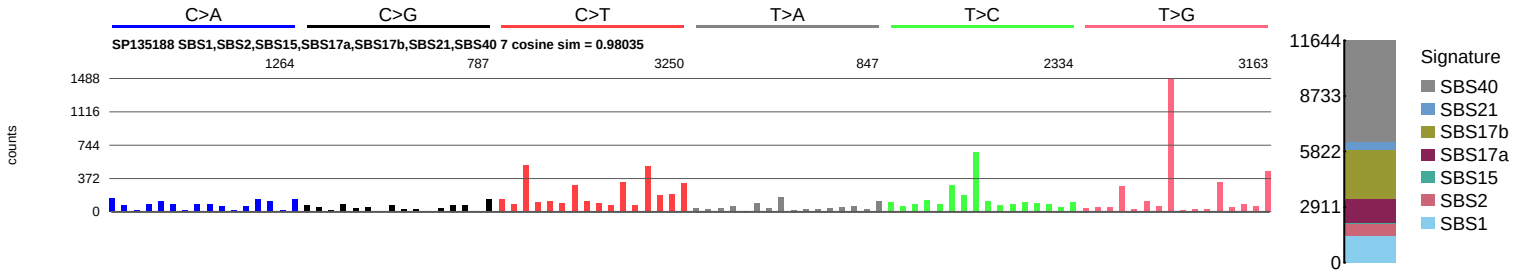
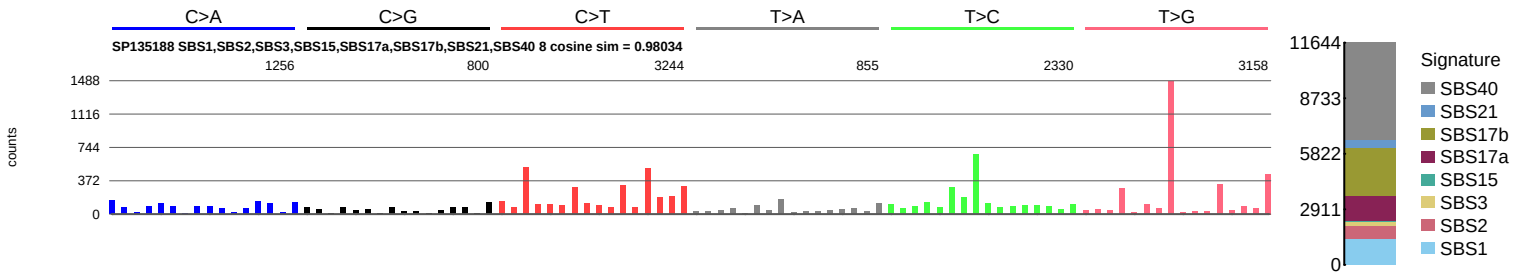
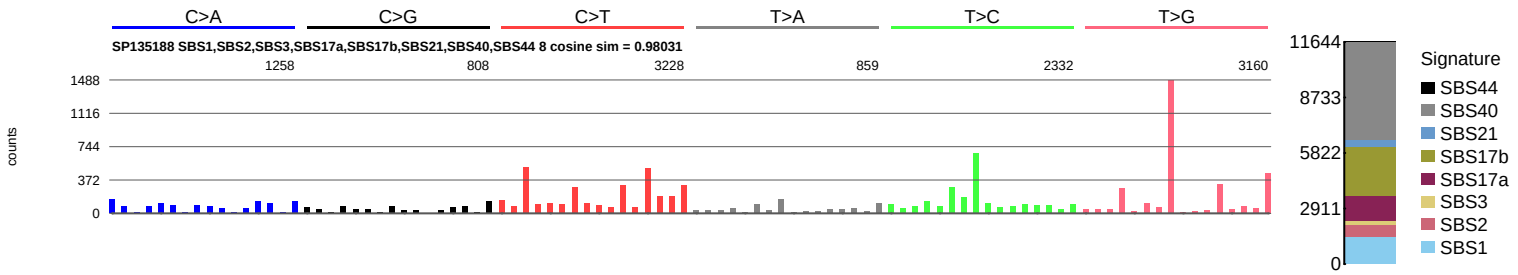
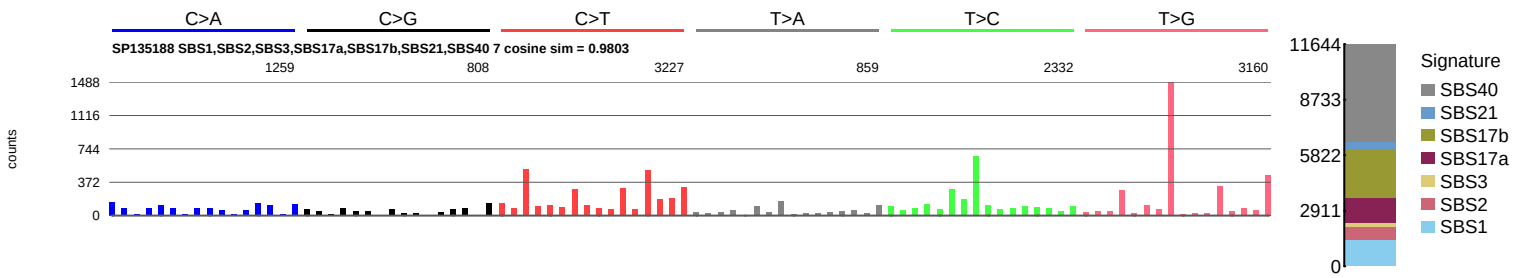
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98035
1264
787
3250
847
2334
3163
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98034
1256
800
3244
855
2330
3158
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98031
1258
808
3228
859
2332
3160
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.9803
1259
808
3227
859
2332
3160
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
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SBS21
SBS17b
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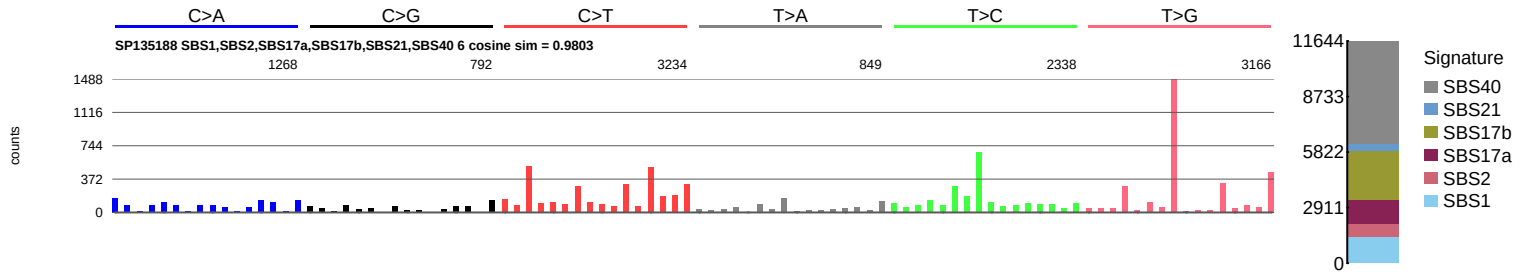
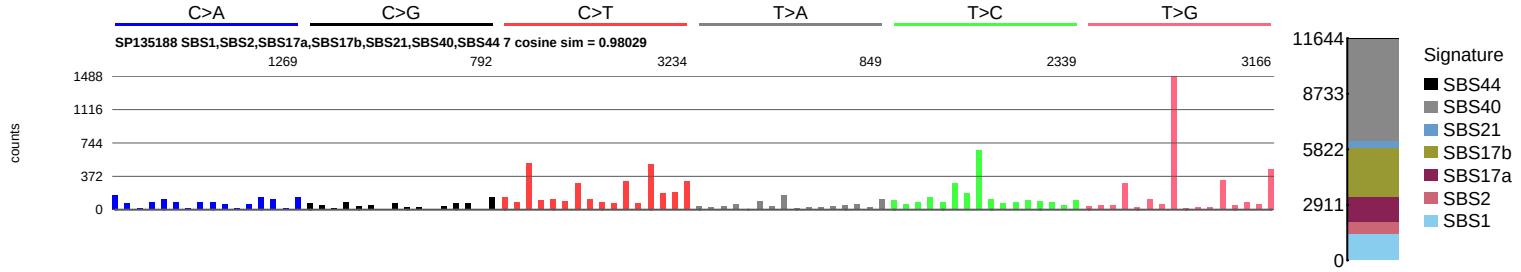
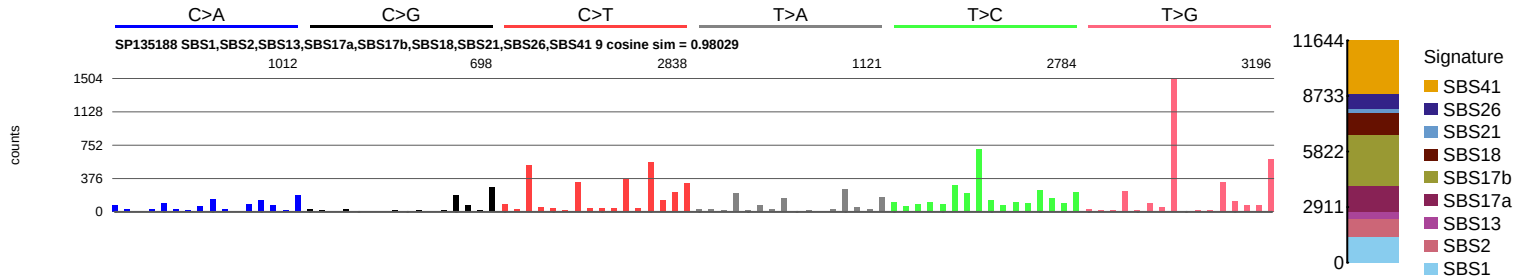
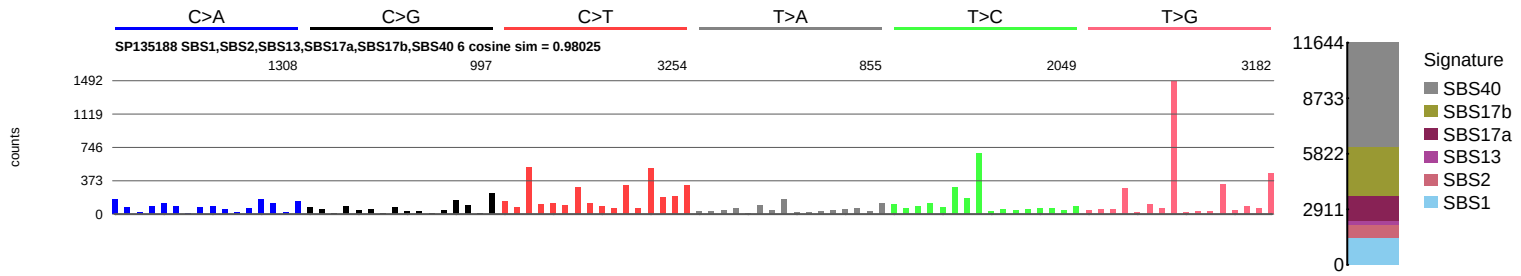
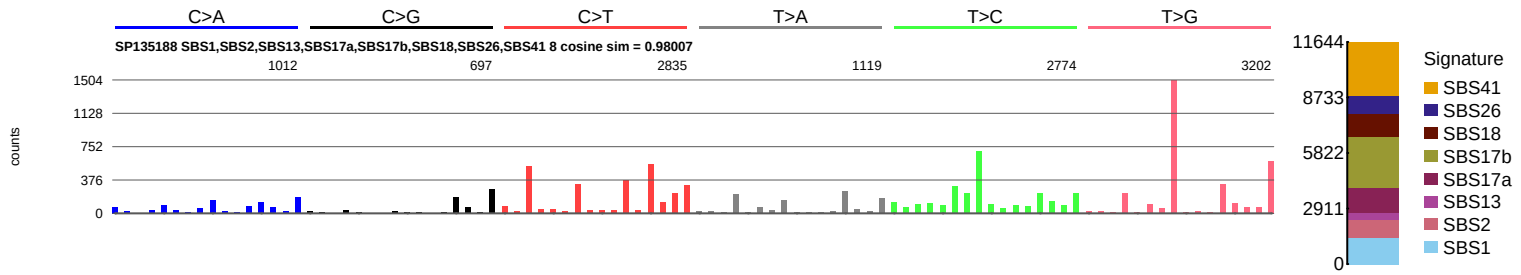
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.9803
1268
792
3234
849
2338
3166
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98029
1269
792
3234
849
2339
3166
counts
1488
1116
744
372
0
11644
8733
5822
2911
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98029
1012
698
2838
1121
2784
3196
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98025
1308
997
3254
855
2049
3182
counts
1492
1119
746
373
0
11644
8733
5822
2911
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135188 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98007
1012
697
2835
1119
2774
3202
counts
1504
1128
752
376
0
11644
8733
5822
2911
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1